

Global Fueter Primitives

Affine monodromy, spherical residues, and the energy of a singular sphere
with a self-contained account of slice-regular and Cauchy–Fueter analysis

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1 Part 0. The question, the status ledger, and the conventions

1.1 The single question

Two notions of regularity coexist over the quaternions, and a second-order differential operator — the Euclidean Laplacian of $\mathbb{R}^4$ — maps one into the other. Fueter’s mapping theorem says that if f is slice regular then $\Delta_4 f$ is Cauchy–Fueter regular. The whole of this document is organized around the inverse of that map, and around one question about it.

The problem. Let U be a connected open subset of the upper half-plane $\mathbb{C}^+ = \{x + iy : y > 0\}$, let

$$\Omega_U = \{x + Iy : x + iy \in U, I \in \mathbb{S}\}$$

be its circularization, and let g be a single-valued axially monogenic function on Ω_U , that is, a smooth $g(x + Iy) = P(x, y) + IQ(x, y)$ with $\mathcal{D}g = 0$. *When does there exist a single-valued slice-regular f on the same domain Ω_U with $\Delta_4 f = g$, and when does not, what measures the failure?* The same question is asked for a connected conjugation-invariant planar set $D \subseteq \mathbb{C}$ that meets $\mathbb{R}$, whose circularization Ω_D is a genuine slice domain rather than a product domain.

The emphasis on *the same domain* is the entire content. Locally the answer is classical and affirmative: on a simply connected base one integrates two closed one-forms in succession and obtains a primitive, unique up to $q \mapsto qa + b$ (Theorem 2.54 below, which is the last theorem of Part I and the seam at which the rest of the document begins). What the local construction leaves open is exactly what happens when a local primitive is continued around a hole of U : it can return changed by an affine function, and the Laplacian cannot see that change. Local surjectivity is therefore compatible with a nonzero global cokernel, and an integral formula — however explicit — does not by itself remove monodromy.

The answers, proved in Parts II–IX, are these. Two closed quaternion-valued one-forms ω_g and θ_g , both computed directly from the target g with no primitive chosen, have periods $(a_\gamma, b_\gamma) \in \mathbb{H}^2$ that vanish on every loop precisely when a single-valued primitive exists; they are precisely the coefficients of the affine monodromy $f \mapsto f + qa_\gamma + b_\gamma$. On a base with a finite winding basis of size m the cokernel of Δ_4 has real dimension exactly $8m$, realized by explicit logarithmic modes. All of this data is carried by a single holomorphic $\mathbb{H}_{\mathbb{C}}$ -valued function $\mathcal{H}_g$. At a missing nonreal sphere the same eight real parameters reappear as an affine spherical residue pair, as the density of a distributional surface source, as two conserved three-dimensional fluxes, and as the exact coefficient of a logarithmic divergence of the L^2 energy.

1.2 What is background and what is proposed

This document is assembled from ten separately written manuscripts: one expository synthesis of the two regularity theories, and nine independent attacks on the problem just stated. Part I is the expository trunk; Parts II through IX are the extension. One status paragraph is stated here and is not repeated anywhere else.

Established background, claimed by nobody here as new: the quaternionic algebra and the stem formalism of Ghiloni–Perotti [3]; slice regularity and the classical function theory built on it; Cauchy–Fueter regularity, the Green identity and the Cauchy–Fueter formula; Fueter’s mapping theorem; the affine kernel $\text{Aff}_{\mathbb{H}}$; the axial Vekua system; the local inverse Fueter theory of Colombo–Sabadini–Sommen [2] and Colombo–Peña Peña–Sabadini–Sommen [1], including the arctangent primitives and the sphere-integrated Cauchy kernels; Perotti’s surjectivity theorem under the hypothesis that every component of the planar generating set is simply connected; the classical scalar Runge [11] and Mittag–Leffler theorems; Weyl’s lemma; and the standard logarithmic-cutoff capacity argument for L^2 removability. Every one of these is used, and several are reproved in Part I to keep the document self-contained; none is advertised as a contribution.

Proposed as contributions: the two simultaneously defined, target-computable closed forms and the resulting same-domain solvability criterion with no topological hypothesis; its identification with an additive affine monodromy $H_1(U; \mathbb{Z}) \rightarrow \mathbb{H} \oplus \mathbb{H}$; the reflection-sensitive split range theorem and the $8m$ cokernel with the correct index; the holomorphic encoding $\mathcal{H}_g$ with its exact range, its exact kernel, and the reality constraint on its Laurent coefficients; the singularity classification along a nonreal sphere under the weakest hypothesis (local integrability) together with the growth filtration above it and the exact leading constants in both norms; the distributional surface source and the two conserved fluxes; the exact energy law with its sharp growth-free lower bound; and the Mittag–Leffler realization theorem.

Established, used here	Proposed, proved here
Stems, slice regularity, the classical slice apparatus (Part I)	Two target-computable closed forms; same-domain criterion; affine monodromy (Part II)
Cauchy–Fueter theory: Green, Cauchy–Pompeiu, $o(r^{-3})$ removability, flux residues (Part I)	The holomorphic invariant $\mathcal{H}_g$, its exact range $\mathcal{O}_{\text{mom}}$, its kernel, and the residue reality constraint (Part III)
Fueter’s mapping theorem; $\ker(\Delta_4 _{S\mathcal{R}}) = \text{Aff}_{\mathbb{H}}$ (Part I)	Logarithmic modes, split exact sequence, $\dim_{\mathbb{R}} \text{coker } \Delta_4 = 8m$ (Parts IV–V)
Local inverse on a simply connected base (Part I)	L^1 and growth classifications at a nonreal sphere with exact constants (Part VII)
Arctangent and spherical Cauchy kernels; Runge; Mittag–Leffler; Weyl; the L^2 cutoff method	Surface source, conserved fluxes, exact energy law, realization theorem (Parts VIII–IX)

The novelty claim is qualified and is not certified. The literature audit behind it was carried out on 21 September 2026 and is targeted, not exhaustive; it is described once, in Part XI. Every mathematical claim below rests on its proof and not on that audit. One editorial position is taken once and held throughout: this document states a necessary and sufficient period criterion, exhibits explicit obstructed axially monogenic fields on $\mathbb{H} \setminus \mathbb{S}$, and observes that an *unqualified* same-domain surjectivity assertion is therefore false as stated; it declines to adjudicate whether any particular published theorem, read with its own domain, locality and branch hypotheses, is incorrect. The topologically restricted surjectivity theorems cited above are correct as hypothesized, and the contribution here is the removal of the hypothesis, not the correction of the theorem.

1.3 Conventions, and the collisions they resolve

The ten sources disagreed on nearly every symbol, and several of the disagreements are dangerous rather than cosmetic: the same letter was used for two quantities obeying laws with genuinely different constants. The conventions below are fixed for the whole document and are never varied.

Quaternions. $q = x_0 + \mathbf{i}x_1 + \mathbf{j}x_2 + \mathbf{k}x_3$ with $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$; $\bar{q}$ is quaternionic conjugation, which reverses products; $|q|^2 = q\bar{q}$; $\text{Re } q = \frac{1}{2}(q + \bar{q})$ and $\text{Im } q = q - \text{Re } q$; $\text{Re}_{\mathbb{H}}$ and $\text{Im}_{\mathbb{H}}$ denote the same operations when the ambient algebra could be misread. The sphere of imaginary units and its planes are

$$\mathbb{S} = \{I \in \mathbb{H} : I^2 = -1\}, \quad \mathbb{C}_I = \mathbb{R} + I\mathbb{R}.$$

Every nonreal quaternion is uniquely $x + Iy$ with $y > 0$ and $I \in \mathbb{S}$, and its conjugacy class is the two-sphere $[q] = x + y\mathbb{S}$.

The auxiliary complex unit. $\mathbb{H}_{\mathbb{C}} = \mathbb{H} \oplus \iota\mathbb{H}$ with ι *central*, $\iota^2 = -1$; the complex conjugation of this algebra is $\kappa(A + \iota B) = A - \iota B$ and is *not* quaternionic conjugation; $\operatorname{Re}_{\mathbb{C}}(A + \iota B) = A$ and $\operatorname{Im}_{\mathbb{C}}(A + \iota B) = B$ take values in $\mathbb{H}$, and $\|A + \iota B\|_{\mathbb{H}_{\mathbb{C}}}^2 = |A|^2 + |B|^2$. **Warning.** ι is never an element of $\mathbb{S}$. Confusing the two is the single most common way to destroy a statement in this subject: the kernel of Δ_4 on slice-regular functions consists of $qa + b$ with $a, b \in \mathbb{H}$ and would be twice as large if a, b were allowed in $\mathbb{H}_{\mathbb{C}}$ (Theorem 2.52 and Remark 2.53).

Sides. All slice-regular functions are *left* slice regular and all quaternionic vector spaces are *right* vector spaces; every quaternionic coefficient multiplies on the right, in power series, in period pairs, in Laurent coefficients and in every integral formula. A coefficient moved to the left changes the problem.

The operator and the map.

$$\mathcal{D} = \partial_{x_0} + \mathbf{i}\partial_{x_1} + \mathbf{j}\partial_{x_2} + \mathbf{k}\partial_{x_3}, \quad \overline{\mathcal{D}} = \partial_{x_0} - \mathbf{i}\partial_{x_1} - \mathbf{j}\partial_{x_2} - \mathbf{k}\partial_{x_3}, \quad \overline{\mathcal{D}}\mathcal{D} = \mathcal{D}\overline{\mathcal{D}} = \Delta_4 = \sum_{\mu=0}^3 \partial_{x_\mu}^2.$$

There is *no* factor $\frac{1}{2}$ anywhere in $\mathcal{D}$ or in Δ_4 ; consequently $\mathcal{D}q = -2$, $\Delta_4 q^2 = -4$, and every constant in the energy law of Part VIII is normalized against these. The Fueter map on stems is written out as $F \mapsto \Delta_4 \mathcal{I}(F)$ throughout and is **never** abbreviated by a single letter. Three of the sources called it T , one called T the holomorphic invariant, and one called T the normalized inverse — three incompatible arrows of the same commuting diagram. Likewise the letter J is not used: it named the splitting section in two sources and the invariant in four. The section is S , the projection is $\Pi = \operatorname{id} - S\operatorname{Per}$, and the invariant is $\mathcal{H}_g$.

Bases and coordinates. U always denotes a connected open subset of $\mathbb{C}^+$, and D a connected conjugation-invariant open subset of $\mathbb{C}$ meeting $\mathbb{R}$; Ω_U and Ω_D are their circularizations. The planar coordinate is $z = x + \iota y$ with $y > 0$, and the quaternionic point is $q = x + Iy$. *The axial radius is y* ; the letters r and R are reserved for radii of balls and tubes. Five of the sources wrote r for the axial radius and five wrote y ; the second choice is adopted.

The singular sphere. The centre is $p = u + \iota v$ with $v > 0$; the sphere is $S_p = u + v\mathbb{S}$; the offsets are $\xi = x - u$ and $\eta = y - v$; and $\rho = |z - p| = \operatorname{dist}(q, S_p)$. **Warning.** The letters s and t are not used for any of these. Three sources gave the pair (s, t) three mutually exclusive meanings — the offsets in one order, the offsets in the opposite order, and the centre — in formulas that this document must place side by side.

The table. Everything else is collected in Table 1, which is the reference for the rest of the document.

Table 1: Notation fixed for the whole document.

Symbol	Meaning
$\mathcal{S}(U) = \mathcal{O}(U, \mathbb{H}_{\mathbb{C}})$	holomorphic stems $F = A + \iota B$ on U , with $A_x = B_y$, $A_y = -B_x$
$\mathcal{I}(F)(x + Iy) = A + IB$	the induced slice function; $\mathcal{SR}$ and $\mathcal{AM}$ are the spaces of slice-regular and axially monogenic functions
$g = P + IQ$	the axially monogenic target; $P, Q : U \rightarrow \mathbb{H}$
$f = \mathcal{I}(F) = A + IB$	a slice-regular primitive, $\Delta_4 f = g$
$C = B/y$, $V = A - xB/y$	the slope potential and the constant potential; $f(q) = qC + V$ (Part II)

Symbol	Meaning
$\omega_g = \frac{1}{2}(-Pdx + Qdy)$	the first (slope) closed form
$\theta_g = \frac{1}{2}((xP + yQ)dx + (yP - xQ)dy)$	the second (constant) closed form, computed from g alone, with <i>no</i> primitive chosen
$\eta_C = (C + \frac{yQ}{2})dx + \frac{yP}{2}dy$	the primitive- <i>dependent</i> second form of the classical local construction (2.54). Warning: $\eta_C \neq \theta_g$; they differ by $d(xC)$, and one source called η_C what four others called θ_g
$a_\gamma = \oint_\gamma \omega_g, \quad b_\gamma = \oint_\gamma \theta_g$	the slope period and the constant period; $\text{Per}(g)$ is the period map
$\text{Aff}_{\mathbb{H}} = \{q \mapsto qa + b : a, b \in \mathbb{H}\}$	the affine kernel; a, b are <i>quaternions</i>
$S, \quad \Pi = \text{id} - S\text{Per}$	the splitting section and the projection onto the range
$\mathcal{H}_g = -\frac{1}{2}(P + yP_y) - \frac{y}{2}P_x$	the holomorphic invariant; $\mathcal{H}_g = F''$ for every local primitive stem
$\mathcal{O}_{\text{mom}}(U)$	the moment-real class: holomorphic $h : U \rightarrow \mathbb{H}_{\mathbb{C}}$ with $\oint_\gamma h dz \in \mathbb{H}$ and $-\oint_\gamma z h dz \in \mathbb{H}$ for every loop
$G_{p;a,b} = \Delta_4 \mathcal{I}(\frac{za+b}{2\pi i} \log(z-p))$	the logarithmic mode with period pair $\text{Ind}(\gamma, p)(a, b)$
$\text{Ind}(\gamma, p)$	winding number
$\ g\ _{\text{ax}}(z) = (P ^2 + Q ^2)^{1/2}$	the <i>axial pair norm</i>
$\sup_{I \in \mathbb{S}} P + IQ $	the <i>spherical supremum</i> at one base point; $\ g\ _{\text{ax}} \leq \sup_I P + IQ \leq \sqrt{2} \ g\ _{\text{ax}}$
$M^{\text{ax}}(\rho), \quad M^{\text{sph}}(\rho)$	the maxima of the two norms over $ z - p = \rho$. Warning, the worst collision in the sources: one letter M_g denoted the first in one manuscript and the second in three others, and the two obey leading-size laws with <i>different constants</i> (Part VII). They are never conflated here
$\mathcal{R}_p(a, b)^2 = ua + b ^2 + v^2 a ^2 = \ pa + b\ _{\mathbb{H}_{\mathbb{C}}}^2$	the residue norm; five sources gave it five names
$\mathbf{n}(\alpha, \beta) = \max_{I \in \mathbb{S}} \beta + I\alpha = (\alpha ^2 + \beta ^2 + 2 \text{Im}_{\mathbb{H}}(\beta\bar{\alpha}))^{1/2}$	the pair norm on $\mathbb{H}^2$ governing the spherical supremum
$\mathcal{E}_g(\varepsilon, R) = \int_{\varepsilon < \rho < R} g ^2 dV$	the truncated <i>energy</i> . Warning: “energy” means the squared L^2 norm, never a Dirichlet integral
$J_g(\rho), \quad E^{\text{ren}}, \quad T_R(S_p)$	the shell density, the renormalized constant, the tube of radius R about S_p
$E(q) = \bar{q}/(2\pi^2 q ^4)$	the Cauchy–Fueter kernel. The letter E also began “energy” in the sources; here the energy always carries a subscript or the decoration $\mathcal{E}_g, E^{\text{ren}}$
m	the rank of a winding basis of the upper meridian $U = D \cap \{y > 0\}$, equivalently the number of conjugation-exchanged pairs of complementary components of D . These two counts agree; the equivalence is proved in Part V and is assumed nowhere before it
$\text{Res}_c f$	the slice Laurent residue at a <i>real</i> isolated singularity (§2.9)
$\text{Res}_a^F f$	the Cauchy–Fueter flux residue at an <i>isolated point</i> (§2.13)
$\text{Res}_{S_p}^{\text{aff}}(g) = (a, b)$	the <i>affine spherical residue</i> at a nonreal sphere (Part VII). These three residues are different objects and are never identified
$\mathcal{P}_k(q) = -\Delta_4(q^{k+2})/(2(k+1)(k+2))$	the entire axially monogenic polynomials of Part IX, normalized by $\mathcal{P}_k(x) = x^k$ on $\mathbb{R}$. The calligraphic letter distinguishes them from the axial component P

Two further conventions are stated once. Integration of a quaternion-valued object always means integration of its four real components with the displayed order of quaternionic factors retained. And a period pair $(a, b) \in \mathbb{H}^2$ is an *ordered pair of quaternions*, carrying eight real parameters, not a single quaternion and not a complex number.

2 Part I. Two regularity theories and the bridge between them

This part supplies, once, the background that Parts II–XI silently assume, and ends with the local theorem that they globalize. It follows the expository source throughout; the nine extension manuscripts reprove none of it, yet each of them uses the stem formalism, the affine kernel, real-centred Taylor expansion (for smoothness across the real axis), Laurent theory, harmonic Liouville and the Green identity. Everything here is established mathematics; the proofs are included so that the later parts depend on no unproved quaternionic machinery. The standard scalar theorems of one-variable complex analysis — Cauchy, the maximum principle, Laurent, Montel, Hurwitz, the normalized Riemann mapping theorem — are prerequisites, as are the divergence theorem and the existence of smooth mollifiers.

2.1 Why there are two theories

Over $\mathbb{C}$, differentiability, local power series, the Cauchy–Riemann equation and several geometric properties fit into one theory. Over $\mathbb{H}$ the literal translations do not fit together, and the reason is a rigidity theorem.

Proposition 2.1 (Rigidity of a direction-independent quotient). *Let $\Omega \subseteq \mathbb{H}$ be connected and open and let $f \in C^2(\Omega, \mathbb{H})$. If for every $q \in \Omega$ the limit*

$$a(q) = \lim_{h \rightarrow 0} h^{-1}(f(q+h) - f(q))$$

exists independently of the direction of h , then $f(q) = qa + b$ for fixed $a, b \in \mathbb{H}$.

Proof. Put $e_0 = 1$, $e_1 = \mathbf{i}$, $e_2 = \mathbf{j}$, $e_3 = \mathbf{k}$. Increments $h = te_\mu$ give $\partial_\mu f = e_\mu a$ and $a = \partial_0 f$. Equality of the mixed second derivatives with ∂_0 gives $\partial_\mu a = e_\mu \partial_0 a$, and then equality of the derivatives with indices 1, 2 gives $\mathbf{ij} \partial_0 a = \mathbf{ji} \partial_0 a$. Since $\mathbf{ij} - \mathbf{ji} = 2\mathbf{k}$ is invertible, $\partial_0 a = 0$; hence all first derivatives of a vanish, connectedness makes a constant, and all first derivatives of $f(q) - qa$ vanish. $\square$

The quotient with h^{-1} on the right gives $f(q) = aq + b$ instead. Either way the class excludes q^2 , so one does not force the complex difference quotient to do everything. *Slice regularity* keeps power series in one quaternionic variable; *Cauchy–Fueter regularity* keeps a first-order elliptic system and a boundary integral in four real dimensions. The relation between them is a differential transform, not an identification of function spaces, and that transform is the subject of §2.14.

2.2 Slice regularity, representation, and stems

Throughout this subsection $D \subseteq \mathbb{C}$ is connected, open, invariant under complex conjugation and meets $\mathbb{R}$; Ω_D is its circularization. It is an axially symmetric slice domain: it contains a whole conjugacy sphere whenever it contains one of its points, and each section $\Omega_D \cap \mathbb{C}_I$ is connected.

Definition 2.2. A C^1 function $f : \Omega_D \rightarrow \mathbb{H}$ is *slice regular* if for each $I \in \mathbb{S}$ its restriction f_I satisfies

$$\bar{\partial}_I f_I = \frac{1}{2}(\partial_x + I\partial_y)f_I = 0. \quad (2.1)$$

Its slice (Cullen) derivative is $\partial_c f = \frac{1}{2}(\partial_x - I\partial_y)f = \partial_x f$.

Lemma 2.3 (Splitting). *Fix orthogonal $I, J \in \mathbb{S}$. Every quaternion-valued function on $\mathbb{C}_I$ is uniquely $F + GJ$ with F, G taking values in $\mathbb{C}_I$. It satisfies (2.1) if and only if F and G are holomorphic, and $|F + GJ|^2 = |F|^2 + |G|^2$.*

Proof. The real basis $1, I, J, IJ$ gives the decomposition; substituting it into (2.1) the two $\mathbb{C}_I$ components vanish independently and give ordinary Cauchy–Riemann equations. The norm identity is orthogonality of the four basis elements. $\square$

Theorem 2.4 (Identity principle). *If two slice-regular functions on Ω_D agree on a subset of one $\mathbb{C}_I$ -section having an accumulation point in that section, they agree on Ω_D .*

Proof. Apply scalar uniqueness to both splitting components of the difference: it vanishes on the whole I -section, hence on a real interval of D . Its restriction to any other section has those same real zeros, and the splitting lemma with scalar uniqueness finishes. $\square$

The phrase “one section” is essential: an arbitrary accumulation of zeros in $\mathbb{H}$ does not suffice, since a whole two-sphere can be a zero set (Theorem 2.16).

Theorem 2.5 (Representation formula and stems). *For slice-regular f on Ω_D and all $I, K \in \mathbb{S}$,*

$$f(x + Ky) = \frac{1 - KI}{2} f(x + Iy) + \frac{1 + KI}{2} f(x - Iy). \quad (2.2)$$

Equivalently there are unique $\mathbb{H}$ -valued $A(x, y), B(x, y)$ with

$$f(x + Ky) = A(x, y) + KB(x, y), \quad (2.3)$$

A even and B odd in y , satisfying

$$A_x = B_y, \quad A_y = -B_x. \quad (2.4)$$

Proof. On one section put $h_{\pm}(x, y) = f(x \pm Iy)$ and $A = \frac{1}{2}(h_+ + h_-)$, $B = -\frac{I}{2}(h_+ - h_-)$; the parities are immediate and (2.1) gives $(h_{\pm})_y = \pm I(h_{\pm})_x$, whence (2.4). Thus $A + KB$ is holomorphic on each K -section and equals $f(x)$ at $y = 0$, so scalar uniqueness identifies it with f . Expanding $A + KB$ gives (2.2); evaluating at K and $-K$ gives uniqueness. $\square$

The function $F(x + \iota y) = A(x, y) + \iota B(x, y)$ is holomorphic with values in $\mathbb{H}_{\mathbb{C}}$ and satisfies

$$F(\bar{z}) = \kappa(F(z)), \quad (2.5)$$

and is called a *holomorphic stem*; conversely a stem induces a slice-regular function by (2.3), an operation written $f = \mathcal{I}(F)$. Away from $\mathbb{R}$ this is (2.4); at a real point the complex Taylor coefficients of F lie in $\mathbb{H}$ by (2.5), so the induced series is real analytic — which is also what justifies C^1 regularity at real points in the converse construction. A slice function is *intrinsic* (slice preserving) when A, B are real valued; its values on each $\mathbb{C}_I$ lie in that plane, and real-coefficient power series are intrinsic.

Proposition 2.6 (The two derivatives at a nonreal point). *Let $q = x + Iy$ with $y > 0$ and split $v \in \mathbb{H}$ as $v_{\parallel} + v_{\perp}$ with $v_{\parallel} \in \mathbb{C}_I$ and $v_{\perp} \perp \mathbb{C}_I$. Then*

$$df_q(v) = v_{\parallel} \partial_c f(q) + v_{\perp} \partial_{\text{sph}} f(q), \quad \partial_{\text{sph}} f(q) = \frac{B(x, y)}{y} = (2Iy)^{-1}(f(q) - f(\bar{q})). \quad (2.6)$$

At a real point, $df_x(v) = v \partial_c f(x)$.

Proof. Within $\mathbb{C}_I$ the Cauchy–Riemann equation gives $\partial_y f = I \partial_x f$, which is the parallel formula. In a perpendicular direction the real part and the length of the imaginary part have zero first variation while I varies by $v_{\perp}/y$; differentiate (2.3). At a real point use the real-centred power series. $\square$

For $f(q) = q^2$ the two right multipliers are $2q$ and $2x$; their norms differ, so slice regularity does not imply four-dimensional conformality. The spherical derivative B/y is exactly the slope potential C of Part II, a coincidence that is not accidental: the first obstruction of the global theory is the obstruction to integrating C .

2.3 Cauchy formula, Taylor series, Morera

For $q \notin [s]$ set

$$\mathcal{C}(q, s) = -(q^2 - 2\operatorname{Re}(s)q + |s|^2)^{-1}(q - \bar{s}), \quad (2.7)$$

the inverse being the ordinary quaternionic inverse of the displayed value. For fixed s this is slice regular in q off $[s]$: it is induced by the stem obtained by replacing q with the central variable z . When $q, s \in \mathbb{C}_I$ commutativity gives $\mathcal{C}(q, s) = (s - q)^{-1}$, and for $|q| < |s|$,

$$\mathcal{C}(q, s) = \sum_{n \geq 0} q^n s^{-n-1}. \quad (2.8)$$

Indeed the absolutely convergent sum R satisfies $Rs - qR = 1$, and $s^2 - 2\operatorname{Re}(s)s + |s|^2 = 0$ then gives $0 = R(s^2 - 2\operatorname{Re}(s)s + |s|^2) = (q^2 - 2\operatorname{Re}(s)q + |s|^2)R + q - \bar{s}$. This is not a licence to commute q with s .

Theorem 2.7 (Slice Cauchy formula). *Let D be bounded with piecewise C^1 boundary and f slice regular on an axially symmetric slice domain containing $\overline{\Omega_D}$. Give ∂D_I its positive complex orientation (with the usual negative orientation on holes) and set $ds_I = -Ids$. Then for all $q \in \Omega_D$,*

$$f(q) = \frac{1}{2\pi} \int_{\partial D_I} \mathcal{C}(q, s) ds_I f(s), \quad (2.9)$$

independently of I . In particular $\int_{\partial V_I} ds_I f(s) = 0$ for any relatively compact planar region V_I in a section on which f is regular up to the boundary.

Proof. On the I -section use Lemma 2.3 and the ordinary Cauchy formula on both components; the normalization $ds_I = -Ids$ places the complex factor $1/I$ correctly. For arbitrary q , both sides are slice functions of q obeying (2.2) — the integral does because every kernel does and the factors $ds_I f(s)$ sit on the right — so equality on one section gives equality everywhere. Conjugation invariance of D keeps the kernel nonsingular for $s \in \partial D_I$. Cauchy's theorem is the same argument applied to the two splitting components. $\square$

Theorem 2.8 (Taylor theorem at real centres). *Let $c \in \mathbb{R}$ and let f be slice regular on $B(c, R)$. There are unique $a_n \in \mathbb{H}$ with*

$$f(q) = \sum_{n \geq 0} (q - c)^n a_n, \quad a_n = \frac{\partial_c^n f(c)}{n!}, \quad (2.10)$$

converging absolutely and uniformly on smaller closed balls, and for $0 < r < R$,

$$a_n = \frac{1}{2\pi} \int_{|s-c|=r, s \in \mathbb{C}_I} (s - c)^{-n-1} ds_I f(s), \quad |a_n| \leq \frac{M_I(r)}{r^n}, \quad M_I(r) = \max_{|s-c|=r, s \in \mathbb{C}_I} |f(s)|. \quad (2.11)$$

Proof. Take ordinary Taylor expansions of both splitting components on one section; their quaternionic combinations give a_n , and real differentiation at c identifies them independently of the section. Scalar Cauchy gives the integral formula, whose norm estimate is $(2\pi)^{-1}(2\pi r)r^{-n-1}M_I(r)$. For $|q - c| \leq \varrho < r$ the resulting series is dominated by a geometric series, defines a slice-regular function agreeing with f on the chosen section, and hence agrees everywhere by Theorem 2.4. $\square$

Consequently every slice-regular function is real analytic: the stem description proves this off $\mathbb{R}$ and (2.10) proves it at $\mathbb{R}$. This is precisely the fact used in Part V, where the two closed one-forms are extended smoothly across the real axis. Ordinary powers centred at a nonreal quaternion are *not* asserted to give a Taylor theorem on a Euclidean ball.

Theorem 2.9 (Morera). *A continuous $f : \Omega_D \rightarrow \mathbb{H}$ with $\int_{\partial T} ds_I f(s) = 0$ for every $I \in \mathbb{S}$ and every triangle T with closure in the I -section is slice regular.*

Proof. Scalar Morera makes both splitting components of f_I holomorphic; construct A, B from that section as in Theorem 2.5 to get a real-analytic slice-regular extension. On every other section the extension and f are holomorphic and agree on a real interval, hence coincide. $\square$

2.4 Maximum modulus, Liouville, Schwarz

Theorem 2.10 (Maximum modulus). *If $f \in \mathcal{SR}(\Omega_D)$ and $|f|$ has a local maximum at an interior point, then f is constant.*

Proof. Let the maximum be at $p \in \mathbb{C}_I$ with $M = |f(p)|$. If $M = 0$ apply Theorem 2.4. Otherwise put $u = f(p)/M$, $g = f\bar{u}$, so g is regular, $g(p) = M$, $|g| = |f|$. Split $g_I = F + GJ$; near p , $|F| \leq |g| \leq M$ and $F(p) = M$, so scalar maximum modulus gives $F \equiv M$ locally and $|g|^2 = |F|^2 + |G|^2 \leq M^2$ forces $G = 0$. Theorem 2.4 finishes. $\square$

Theorem 2.11 (Liouville and polynomial growth). *A bounded entire slice-regular function is constant. More generally if $|f(q)| \leq C(1 + |q|)^\alpha$ on $\mathbb{H}$ with $\alpha \geq 0$, then f is a right-coefficient polynomial of degree at most $\lfloor \alpha \rfloor$.*

Proof. Expand at 0; (2.11) gives $|a_n| \leq C(1 + r)^\alpha r^{-n}$, and $r \rightarrow \infty$ kills every $n > \alpha$. $\square$

Theorem 2.12 (Higher-order Schwarz lemma). *Let $f : \mathbb{B} \rightarrow \mathbb{B}$ be slice regular on the unit ball $\mathbb{B} = B(0, 1)$, with the first $m \geq 1$ Taylor coefficients vanishing. Then*

$$|f(q)| \leq |q|^m, \quad \frac{|\partial_c^m f(0)|}{m!} \leq 1, \quad (2.12)$$

with equality in the first inequality at a nonzero point, or in the second, exactly when $f(q) = q^m u$ for a fixed $|u| = 1$.

Proof. Write $f(q) = q^m g(q)$ from the power series; for $|q| = r < 1$, $|g(q)| \leq r^{-m}$, and maximum modulus on $B(0, r)$ with $r \uparrow 1$ gives $|g| \leq 1$. Either equality forces $|g|$ to attain 1 inside, so Theorem 2.10 makes g a unit constant. $\square$

These proofs illustrate a principle used repeatedly below: use scalar holomorphic components for linear analytic questions, but preserve the quaternionic norm whenever a sharp scalar bound is wanted.

2.5 The regular product, symmetrization, and the geometry of zeros

Pointwise multiplication does not preserve slice regularity: the constant $\mathbf{i}$ and the identity are regular, but on the $\mathbf{j}$ -section $(\partial_x + \mathbf{j}\partial_y)(\mathbf{i}q) = 2\mathbf{i} \neq 0$. The replacement is multiplication before evaluating the stem.

Definition 2.13. For $f = \mathcal{I}(F)$ and $g = \mathcal{I}(G)$ put $f * g = \mathcal{I}(FG)$. Thus with $f = A + IB$ and $g = C' + ID'$,

$$(f * g)(x + Iy) = AC' - BD' + I(AD' + BC'). \quad (2.13)$$

Stem symmetry is preserved because κ is an algebra automorphism, and holomorphy because multiplication is complex bilinear; so $*$ is associative on $\mathcal{SR}$, and for power series

$$\left(\sum_{n \geq 0} q^n a_n \right) * \left(\sum_{n \geq 0} q^n b_n \right) = \sum_{n \geq 0} q^n \sum_{k=0}^n a_k b_{n-k}. \quad (2.14)$$

Intrinsic functions have central stems: if h is intrinsic then $h * f = f * h$ and this equals the *pointwise* product $h(q)f(q)$ — but need not equal $f(q)h(q)$.

Proposition 2.14 (Evaluation of a regular product). *For $q \in \Omega_D$,*

$$(f * g)(q) = \begin{cases} f(q) g(f(q)^{-1} q f(q)), & f(q) \neq 0, \\ 0, & f(q) = 0. \end{cases} \quad (2.15)$$

Proof. Write $q = x + Iy$, $u = f(q) = A + IB$; for $u \neq 0$ put $I' = u^{-1}Iu$. Then $u(C' + I'D') = uC' + IuD' = AC' - BD' + I(BC' + AD')$, which is (2.13); the same expression vanishes when $u = 0$. $\square$

The argument of the second factor rotates within $[q]$: this is exactly why roots of successive factors need not be roots of their product.

Extend quaternionic conjugation complex-linearly to stems by $(a + \iota b)^c = \bar{a} + \iota \bar{b}$, an antiautomorphism, and set $f^c = \mathcal{I}(F^c)$, $f^s = f * f^c = f^c * f$. The equality of the two products follows from

$$\mathcal{N}_f(z) := F(z)F(z)^c = |A|^2 - |B|^2 + 2\iota \operatorname{Re}(A\bar{B}), \quad (2.16)$$

$$f^s(x + Iy) = |A|^2 - |B|^2 + 2I \operatorname{Re}(A\bar{B}). \quad (2.17)$$

The *scalar norm stem* $\mathcal{N}_f$ is complex-scalar valued and holomorphic, so f^s is intrinsic; in a power series $f^c(q) = \sum q^n \bar{a}_n$ and $f^s(q) = \sum q^n \sum_k a_k \bar{a}_{n-k}$ with real coefficients. On the real axis

$$\mathcal{N}_f(x) = |f(x)|^2 \quad (x \in D \cap \mathbb{R}), \quad (2.18)$$

but for general q , $f^s(q) \neq |f(q)|^2$: for $f(q) = q - \mathbf{i}$ one has $f^s(q) = q^2 + 1$, and at $q = \mathbf{j}$ these are 0 and 2.

Proposition 2.15 (Multiplicativity of the norm stem). $\mathcal{N}_{f*g} = \mathcal{N}_f \mathcal{N}_g$ and $(f * g)^s = f^s g^s$; and $f \neq 0$ implies $\mathcal{N}_f \neq 0$.

Proof. $\mathcal{N}_g$ is central, so $(FG)(FG)^c = FGG^c F^c = \mathcal{N}_g FF^c = \mathcal{N}_g \mathcal{N}_f$. If $\mathcal{N}_f \equiv 0$ then (2.18) makes f vanish on a real interval and Theorem 2.4 gives $f \equiv 0$. $\square$

Theorem 2.16 (Zero geometry and detection by symmetrization). *On a nonreal sphere $S = x + y\mathbb{S}$ a slice function has no zero, exactly one zero, or is identically zero there; and*

$$f^s|_S = 0 \iff f \text{ has a zero on } S, \quad (2.19)$$

while for real x , $f^s(x) = 0$ iff $f(x) = 0$. For regular $f \neq 0$ the zero-bearing conjugacy classes are locally finite in Ω_D ; the zeros consist of isolated real or nonreal points and isolated two-spheres.

Proof. On S write $f = A + IB$. If $B = 0$ the claim is immediate. If $B \neq 0$ the only possible zero is at $I = -AB^{-1}$, which is an imaginary unit exactly when $|A| = |B|$ and $\operatorname{Re}(A\bar{B}) = 0$ — its norm is $|A|/|B|$ and its real part is $-\operatorname{Re}(A\bar{B})/|B|^2$ — and these are exactly the two real conditions for (2.17) to vanish. For local finiteness, $\mathcal{N}_f$ is a nonzero scalar holomorphic function with discrete zeros; each conjugate pair is one nonreal sphere and each real zero one real point, and $q \mapsto \operatorname{Re} q \pm \iota |\operatorname{Im} q|$ carries compacta to compacta. $\square$

For example $q^2 + 1$ vanishes on all of $\mathbb{S}$. This is the structural fact that forces the geometric unit of the later theory to be a conjugacy sphere rather than a point.

Proposition 2.17 (Division by a spherical factor). *Let $p = x + Iy$ be nonreal and $\Delta_p(q) = q^2 - 2xq + x^2 + y^2$. A regular f vanishes identically on $[p]$ if and only if $f = \Delta_p * g = \Delta_p g$ for a regular g on Ω_D .*

Proof. Vanishing on the sphere means $A = B = 0$ there, i.e. the stem vanishes at $z_0 = x + \iota y$ and $\bar{z}_0$; divide the four complex components by $(z - z_0)(z - \bar{z}_0)$, whose coefficients are real, so both apparent singularities are removable and the quotient keeps the stem symmetry (2.5). $\square$

2.6 Factorization, total multiplicity, the argument principle, Rouché

Lemma 2.18 (Regular division on a ball). *Let $f(q) = \sum_{n \geq 0} q^n a_n$ converge on $B(0, R)$ and $p \in B(0, R)$. There is a unique regular g on that ball with*

$$f = (q - p) * g + f(p), \quad g(q) = \sum_{n \geq 0} q^n b_n, \quad b_n = \sum_{k=n+1}^{\infty} p^{k-n-1} a_k. \quad (2.20)$$

For a polynomial of degree $n \geq 1$, g has degree $n - 1$.

Proof. For $|p| < r < R$ Cauchy's estimate gives $|a_k| \leq M(r)r^{-k}$, so $|b_n| \leq M(r)r^{-n-1}/(1-|p|/r)$ and the series converges on $|q| < r$, hence on the ball. Directly, $b_{n-1} - pb_n = a_n$ for $n \geq 1$ and $-pb_0 = a_0 - f(p)$, which are the coefficients of (2.20). If two quotients existed their difference h would satisfy $(q - p) * h = 0$, and evaluation at real $x \neq p$ with Theorem 2.4 gives $h = 0$. $\square$

Theorem 2.19 (Quaternionic fundamental theorem of algebra). *Every nonconstant right-coefficient quaternionic polynomial has a zero in $\mathbb{H}$, and if $P(q) = q^n a_n + \cdots + a_0$ with $a_n \neq 0$ then*

$$P = (q - p_1) * \cdots * (q - p_n) a_n \quad (2.21)$$

for suitable $p_1, \dots, p_n \in \mathbb{H}$.

Proof. The real-coefficient polynomial P^s has degree $2n$ and leading coefficient $|a_n|^2$, so its scalar stem has a root z ; a real root gives $P(z) = 0$ and a nonreal root gives a zero on the corresponding sphere by Theorem 2.16. Divide out one linear regular factor by Lemma 2.18 and repeat; leading coefficients identify the remaining constant. $\square$

This concerns the central-variable, right-coefficient polynomial algebra; it is not a claim about expressions with q interspersed among coefficients.

Example 2.20 (Factors do not simply list roots). $P = (q - \mathbf{i}) * (q - \mathbf{j}) = q^2 - q(\mathbf{i} + \mathbf{j}) + \mathbf{k}$ satisfies $P(\mathbf{i}) = 0$ but $P(\mathbf{j}) = 2\mathbf{k} \neq 0$. Its symmetrization is $(q^2 + 1)^2$, and on $\mathbb{S}$ its coefficient $B = -(\mathbf{i} + \mathbf{j})$ is nonzero, so Theorem 2.16 allows only the one pointwise zero $\mathbf{i}$. By contrast $(q - \mathbf{i}) * (q + \mathbf{i}) = q^2 + 1$ vanishes on an entire sphere with the *same* symmetrization.

For nonzero regular f define the total multiplicity of a zero-bearing class by

$$\mu_f([p]) = \begin{cases} \text{ord}_{x+\iota y} \mathcal{N}_f, & p = x + Iy, \ y > 0, \\ \frac{1}{2} \text{ord}_x \mathcal{N}_f, & p = x \in \mathbb{R}. \end{cases} \quad (2.22)$$

The two nonreal conjugate zeros have equal order, and the real order is even: if $f = (q - x)^m u$ with $u(x) \neq 0$ then $\mathcal{N}_f(z) = (z - x)^{2m} \mathcal{N}_u(z)$ with $\mathcal{N}_u(x) = |u(x)|^2 > 0$. A simple isolated linear factor has total multiplicity one and the spherical factor Δ_p has total multiplicity two; this counts algebraic mass in a class, not the cardinality of the pointwise zero set. (The letter μ is used here because m is reserved for the hole count of Table 1.)

Corollary 2.21 (Degree equals total multiplicity). *For a degree- n polynomial, $\sum_{[p]} \mu_P([p]) = n$.*

Proof. By (2.21) and Proposition 2.15, $P^s(q) = |a_n|^2 \prod_{j=1}^n \Delta_{p_j}(q)$; each nonreal factor contributes one to its class and each real factor is $(q - p_j)^2$, contributing one under the real convention. $\square$

Let $U' \Subset D$ be bounded, conjugation invariant, with finitely many piecewise C^1 boundary curves (not necessarily connected), put $\Gamma_{U'} = \{x + Iy : x + \iota y \in \partial U', \ I \in \mathbb{S}\}$, suppose f has no zero on $\Gamma_{U'}$, and let $\nu_f(U')$ be the sum of (2.22) over zero-bearing classes in $\Omega_{U'}$.

Theorem 2.22 (Argument principle). *Under these assumptions*

$$\nu_f(U') = \frac{1}{4\pi i} \int_{\partial U'} \frac{\mathcal{N}'_f(z)}{\mathcal{N}_f(z)} dz. \quad (2.23)$$

Proof. Theorem 2.16 makes $\mathcal{N}_f$ nonzero on $\partial U'$. Near a zero of order μ write $\mathcal{N}_f = (z - z_0)^\mu v$ with $v(z_0) \neq 0$; the ordinary argument principle counts complex multiplicities, which occur in conjugate pairs except at real zeros, whose orders are even. Halving the complex count is exactly (2.22). $\square$

The integrand is scalar complex valued; a pointwise quaternionic expression such as $f'(q)f(q)^{-1}$ cannot replace it in this proof.

Theorem 2.23 (Rouché on circular boundaries). *If f, g are regular on Ω_D and $|g(q) - f(q)| < |f(q)|$ for all $q \in \Gamma_{U'}$, then $\nu_g(U') = \nu_f(U')$.*

Proof. Put $h_t = f + t(g - f)$ for $t \in [0, 1]$; the strict inequality gives $|h_t| \geq |f| - t|g - f| > 0$ on $\Gamma_{U'}$, so $\mathcal{N}_{h_t}$ has no zero on $\partial U'$. The integral (2.23) depends continuously on t and is integer valued, hence constant. $\square$

The hypothesis is imposed on *all* quaternionic points of each boundary sphere, not only on one complex contour; that is what makes the zero-detection argument valid.

Corollary 2.24 (Polynomial localization). *If $P(q) = q^n + \sum_{k < n} q^k a_k$ and $\sum_{k < n} R^k |a_k| < R^n$, then all zeros lie in $|q| < R$ with total multiplicity n ; in particular every zero satisfies $|q| \leq 1 + \max_{k < n} |a_k|$.*

Proof. Apply Theorem 2.23 to q^n and P on $|q| = R$ together with Corollary 2.21; with $M = \max_{k < n} |a_k|$ and $R > 1 + M$, $\sum_{k < n} R^k |a_k| \leq M(R^n - 1)/(R - 1) < R^n$, and let $R \downarrow 1 + M$. $\square$

2.7 The regular reciprocal, minimum modulus, and open mappings

Let $Z(f^s)$ be the union of zero-bearing classes of a nonzero regular f . On $\Omega_D \setminus Z(f^s)$ put

$$f^{-*} = (f^s)^{-1} f^c, \quad (2.24)$$

the first factor being intrinsic and regular; stem multiplication gives $f * f^{-*} = f^{-*} * f = 1$ there. For instance $(q - p)^{-*} = \Delta_p(q)^{-1}(q - \bar{p})$, whose excluded set is the whole sphere $[p]$ even though $q - p$ has only the one pointwise zero p .

Proposition 2.25 (The conjugation change of variable). *On $\Omega_D \setminus Z(f^s)$ set $T_f(q) = f^c(q)^{-1} q f^c(q)$. This is a real-analytic diffeomorphism of that domain onto itself preserving each conjugacy class, with inverse T_{f^c} , and*

$$f^{-*}(q) = f(T_f(q))^{-1}, \quad (f^{-*} * g)(q) = f(T_f(q))^{-1} g(T_f(q)). \quad (2.25)$$

Proof. Nonvanishing of $f^s(q)$ gives nonvanishing of $f^c(q)$ by Theorem 2.16. Let $c = f^c(q)$, $r = T_f(q)$, $n = f^s(q)$; Proposition 2.14 applied to $f^c * f$ gives $n = cf(r)$. Since n is intrinsic it commutes with q , so $T_{f^c}(r) = f(r)^{-1} r f(r) = n^{-1} c (c^{-1} q c) c^{-1} n = q$. Also $n^{-1} c = (cf(r))^{-1} c = f(r)^{-1}$, which is the first identity; for the second, the conjugating value is $u = f^{-*}(q) = n^{-1} c$ and $u^{-1} q u = c^{-1} q c = T_f(q)$. $\square$

So the reciprocal is a pointwise inverse only *after* a sphere-preserving change of variable.

Theorem 2.26 (Minimum modulus). *If $f \in \mathcal{SR}(\Omega_D)$ and $|f|$ has a local minimum at p , then $f(p) = 0$ or f is constant.*

Proof. Assume $f(p) \neq 0$. First, $[p]$ contains no zero: on a nonreal sphere $|A + IB|^2 = |A|^2 + |B|^2 - 2\operatorname{Re}(A\bar{B}I)$ is an affine real function of $I \in \mathbb{S}$, and a local minimum of an affine function on a sphere is global, so the value at p is no larger than anywhere on $[p]$. Next, $D' = D \setminus Z(\mathcal{N}_f)$ is connected (perturb a polygonal path around a locally finite set), conjugation invariant and still meets $\mathbb{R}$, so f^{-*} is regular on $\Omega_{D'}$. By Proposition 2.25, $|f^{-*}| = 1/|f \circ T_f|$ has a local maximum, so Theorem 2.10 makes f^{-*} a nonzero constant c ; then $f(q)c = 1$, and Theorem 2.4 propagates this. $\square$

Define the *degenerate set* $\Sigma_f = \bigcup\{x + y\mathbb{S} : y > 0, f \text{ constant on } x + y\mathbb{S}\}$, the union of spheres on which $B(x, y) = 0$.

Theorem 2.27 (Open mapping with spherical qualifications). *Let $f \in \mathcal{SR}(\Omega_D)$ be nonconstant.*

- (i) *At every $p \notin \Sigma_f$ the image of every sufficiently small Euclidean ball centred at p contains a neighbourhood of $f(p)$; consequently f restricted to $\Omega_D \setminus \overline{\Sigma_f}$ is open.*
- (ii) *If $V \subseteq \Omega_D$ is open and axially symmetric then $f(V)$ is open; in particular $f(\Omega_D)$ is open.*

Proof. For (i), $h = f - f(p)$ has an isolated zero at p (a real zero is isolated; a nonreal zero is the unique zero on a nondegenerate sphere, and zero-bearing classes are locally finite). Take a closed ball whose boundary carries no zero of h and put $m_0 = \min_{\partial B(p, r)} |h| > 0$. If $|w - f(p)| < m_0/2$ then $|f - w| > m_0/2$ on the boundary while $|f(p) - w| < m_0/2$, so $|f - w|$ attains an interior minimum, which by Theorem 2.26 is zero. For (ii), choose a relatively compact axially symmetric neighbourhood of the whole class $[p]$ inside V whose boundary avoids the zeros of $f - f(p)$, and repeat. $\square$

Example 2.28 (Why the unqualified open mapping theorem fails). The entire regular $f(q) = q^2$ is not open at $p = \mathbf{i}$. If $q^2 = -1 + t\mathbf{j}$ with $t \neq 0$ and $q = x + v$, then $\operatorname{Im}(q^2) = 2xv = t\mathbf{j}$, so $x \neq 0$ and v is parallel to $\mathbf{j}$; in particular $|q - \mathbf{i}| \geq 1$. Hence $f(B(\mathbf{i}, 1/2))$ contains -1 but no point $-1 + t\mathbf{j}$, $t \neq 0$. The missing behaviour sits on a sphere that f collapses to a point. The closure in (i) is also needed: for q^2 the degenerate spheres form the nonzero imaginary hyperplane, whose closure contains 0, and removing only the nonreal ones leaves 0 while deleting all preimages of negative reals.

2.8 Schwarz–Pick and an intrinsic Riemann mapping theorem

For $a \in \mathbb{B}$ define the regular Möbius map

$$\mathcal{M}_a(q) = (1 - q\bar{a})^{-*} * (q - a). \quad (2.26)$$

Its denominator has no pointwise zero in $\mathbb{B}$ because $|q\bar{a}| < 1$, so Theorem 2.16 makes the regular reciprocal well defined on the whole ball; the two factors commute under $*$ because their coefficients lie in one commutative plane.

Lemma 2.29 (Ball and boundary estimates). *$\mathcal{M}_a$ maps $\mathbb{B}$ into $\mathbb{B}$ with the single zero a , and for $|a| < r < 1$,*

$$\min_{|q|=r} |\mathcal{M}_a(q)| = \frac{r - |a|}{1 - r|a|}. \quad (2.27)$$

Proof. With $h(q) = 1 - q\bar{a}$, (2.25) gives $\mathcal{M}_a(q) = (1 - \varpi\bar{a})^{-1}(\varpi - a)$ with $\varpi = T_h(q)$, and the elementary identity

$$|1 - \varpi\bar{a}|^2 - |\varpi - a|^2 = (1 - |\varpi|^2)(1 - |a|^2) \quad (2.28)$$

bounds the norm by one. T_h is a bijection and $h^c(a) = 1 - a^2$ commutes with a , so $T_h(a) = a$. For $|\varpi| = r$ and $t = \operatorname{Re}(\varpi\bar{a})$, $|\varpi - a|^2/|1 - \varpi\bar{a}|^2 = (r^2 + |a|^2 - 2t)/(1 + r^2|a|^2 - 2t)$ is decreasing in t (numerator minus denominator is $-(1 - r^2)(1 - |a|^2)$), so the minimum is at $t = r|a|$. $\square$

Theorem 2.30 (Schwarz lemma with an arbitrary zero). *If $F : \mathbb{B} \rightarrow \mathbb{B}$ is regular with $F(a) = 0$, then $|F(q)| \leq |\mathcal{M}_a(q)|$ on $\mathbb{B}$, with equality at some $q \neq a$ exactly when $F = \mathcal{M}_a u$ for a constant $|u| = 1$.*

Proof. Regular division gives $F = (q - a) * G$; put $H = (1 - q\bar{a}) * G$, regular on $\mathbb{B}$, so $F = \mathcal{M}_a * H$ and $H = \mathcal{M}_a^{-*} * F$ off the zero-bearing sphere. For $r > |a|$, quotient evaluation and (2.27) give $|H| \leq (1 - r|a|)/(r - |a|)$ on $|q| = r$; maximum modulus and $r \uparrow 1$ give $|H| \leq 1$, and Proposition 2.14 finishes. Equality forces $|H| = 1$ at an interior point. $\square$

Theorem 2.31 (Regular Schwarz–Pick inequality). *Let $f : \mathbb{B} \rightarrow \mathbb{B}$ be regular, $a \in \mathbb{B}$, $b = f(a)$, and $\Phi_{f,b} = (1 - f\bar{b})^{-*} * (f - b)$. Then $|\Phi_{f,b}(q)| \leq |\mathcal{M}_a(q)|$ on $\mathbb{B}$, with equality at some $q \neq a$ exactly when $\Phi_{f,b} = \mathcal{M}_a u$, $|u| = 1$. If a is real,*

$$\frac{|\partial_c f(a)|}{1 - |f(a)|^2} \leq \frac{1}{1 - a^2}. \quad (2.29)$$

Proof. $h = 1 - f\bar{b}$ is regular and zero free since $|f(q)||b| < 1$, so its reciprocal is regular; quotient evaluation and (2.28) give $|\Phi_{f,b}| < 1$. Evaluating $h * \Phi_{f,b} = f - b$ at a gives $(1 - |b|^2)\Phi_{f,b}(a) = 0$, because $h(a) = 1 - |b|^2$ is a nonzero real scalar and does not rotate the argument; now apply Theorem 2.30. For real a , restrict $h * \Phi_{f,b} = f - b$ to $\mathbb{R}$ and differentiate: $\Phi_{f,b}(a) = 0$ gives $\partial_c \Phi_{f,b}(a) = \partial_c f(a)/(1 - |b|^2)$, while $\mathcal{M}_a(x) = (x - a)/(1 - ax)$ has $\mathcal{M}'_a(a) = 1/(1 - a^2)$. $\square$

At nonreal base points the real differential has the two pieces of (2.6), and the simple differential inequality (2.29) is deliberately *not* asserted there.

Theorem 2.32 (Intrinsic Riemann mapping theorem). *Let $D \subsetneq \mathbb{C}$ be simply connected, open, conjugation invariant, and let $c \in D \cap \mathbb{R}$. There is a unique intrinsic slice-regular bijection $\varphi : \Omega_D \rightarrow \mathbb{B}$ with $\varphi(c) = 0$ and $\partial_c \varphi(c) > 0$ real; its inverse is intrinsic and slice regular.*

Proof. The normalized scalar Riemann mapping theorem gives a unique biholomorphism $\phi : D \rightarrow \{|z| < 1\}$ with $\phi(c) = 0$, $\phi'(c) > 0$; since $z \mapsto \bar{\phi}(\bar{z})$ has the same normalization, uniqueness gives $\phi(\bar{z}) = \overline{\phi(z)}$, and likewise for $\psi = \phi^{-1}$. Both therefore have intrinsic slice extensions, which on every $\mathbb{C}_I$ are the scalar maps under $x + \iota y \leftrightarrow x + Iy$, so they are mutually inverse section by section. For uniqueness, an intrinsic bijection with this normalization restricts to a biholomorphism on each section — a nonreal point cannot map to a real value, since its whole sphere would then have the same image — so scalar uniqueness on one section and (2.2) finish. $\square$

This asserts a slice-analytic equivalence, not conformal uniformization in $\mathbb{R}^4$: for an intrinsic function induced by $\phi = u + \iota v$ the two stretch factors in (2.6) are $|\phi'(z)|$ and $|v(x, y)/y|$, and for q^2 at $1 + \mathbf{i}$ they are $2\sqrt{2}$ and 2. Note also that ordinary composition of regular functions need not be regular: $q \mapsto q\mathbf{i}$ and $q \mapsto q^2$ are regular but their composite is $-|q|^2$ on $\mathbb{C}_j$. Composition works when the inner function is intrinsic.

2.9 Laurent expansions, residues, and essential singularities

The statements here concern isolated singularities at *real* points, which is what makes ordinary radial annuli and ordinary integer powers the right objects. Nonreal singularities require an entire conjugacy sphere and are the subject of Part VII, whose residue is a different object.

Theorem 2.33 (Laurent expansion on a spherical annulus). *Let f be slice regular on $A_{r,R} = \{r < |q| < R\}$, $0 \leq r < R \leq \infty$. Then $f(q) = \sum_{n \in \mathbb{Z}} q^n a_n$ with unique $a_n \in \mathbb{H}$, converging absolutely and uniformly on compact subannuli, and for $r < \varrho < R$,*

$$a_n = \frac{1}{2\pi} \int_{|s|=\varrho, s \in \mathbb{C}_I} s^{-n-1} ds_I f(s), \quad |a_n| \leq M_I(\varrho) \varrho^{-n}, \quad (2.30)$$

independently of ϱ and I .

Proof. Apply the scalar Laurent theorem to the four complex coordinates of the stem F ; the symmetry (2.5) and uniqueness of Laurent coefficients make each coefficient κ -fixed, hence in $\mathbb{H}$. Evaluating the stem series gives the expansion, and the splitting lemma gives (2.30). Coefficient estimates on a smaller radius control the negative tail and on a larger radius the positive tail, both geometrically. $\square$

Translation by a real centre c replaces q^n by $(q - c)^n$ verbatim.

Corollary 2.34 (Classification and Riemann removability at a real point). *For f regular on $0 < |q| < R$ the singularity at 0 is removable if all negative coefficients vanish, a pole if finitely many but at least one are nonzero, and essential otherwise. A bounded singularity is removable, and for a pole of order $\mathfrak{m}$,*

$$|f(q)| = |q|^{-\mathfrak{m}}(|a_{-\mathfrak{m}}| + o(1)) \quad (q \rightarrow 0). \quad (2.31)$$

Proof. Under a local bound M , (2.30) gives $|a_{-\mathfrak{m}}| \leq M|q|^{\mathfrak{m}} \rightarrow 0$. For a pole, $q^{\mathfrak{m}}f(q)$ extends regularly with nonzero value $a_{-\mathfrak{m}}$, and multiplicativity of the norm gives (2.31). $\square$

Theorem 2.35 (Residue theorem for isolated real singularities). *Let f be regular near $\overline{\Omega_{U'}} \setminus \{c_1, \dots, c_N\}$ with real interior c_j and U' having piecewise smooth positively oriented boundary, and let $\text{Res}_{c_j} f$ be the coefficient of $(q - c_j)^{-1}$. Then*

$$\frac{1}{2\pi} \int_{\partial U'} ds_I f(s) = \sum_{j=1}^N \text{Res}_{c_j} f. \quad (2.32)$$

Proof. Remove small disks around the c_j in the I -section and apply Cauchy's theorem to both splitting components; on each small circle the Laurent series converges uniformly, all powers integrate to zero except $(s - c_j)^{-1}$ whose normalized integral is 1, and the right coefficients stay on the right. $\square$

Theorem 2.36 (Casorati–Weierstrass at a real essential singularity). *If f has an essential singularity at $0 \in \mathbb{R}$ then $f(\{0 < |q| < r\})$ is dense in $\mathbb{H}$ for every $0 < r < R$.*

Proof. If a ball $B(a, \varepsilon)$ were omitted, $g = f - a$ would be zero free with $|g| \geq \varepsilon$, so $|g^{-*}(q)| = |g(T_g(q))|^{-1} \leq \varepsilon^{-1}$ because T_g preserves each radius. By removability $h = g^{-*}$ extends regularly and is not identically zero. If $h(0) \neq 0$ then h^{-*} is regular near 0 and agrees with g , so the singularity is removable; if $h(0) = 0$ write $h = q^{\mathfrak{m}}u$ with $u(0) \neq 0$, whence $g = h^{-*} = q^{-\mathfrak{m}}u^{-*}$ has a pole. Both contradict essentiality. $\square$

2.10 Normal families and Hurwitz

Theorem 2.37 (Weierstrass convergence and Montel). *A locally uniform limit of slice-regular functions on Ω_D is slice regular, and every family bounded uniformly on compacta is normal.*

Proof. Split each restriction to $\mathbb{C}_I$ into two complex components: scalar Weierstrass convergence makes the limit components holomorphic, and their slice extension equals the limit by (2.2). For normality, scalar Montel on both components with successive extraction gives a subsequence converging on compacta of that section; to propagate to a compact $K \subset \Omega_D$, use the compact two-point projection $K_I = \{\text{Re } q \pm I|\text{Im } q| : q \in K\}$, and note that the coefficients in (2.2) have norm at most one, so the uniform difference of two members on K is at most twice their uniform difference on K_I . $\square$

Theorem 2.38 (Hurwitz). *If $f_n \in \mathcal{SR}(\Omega_D)$ have no zeros and converge locally uniformly to f , then $f \equiv 0$ or f is zero free.*

Proof. The representation formula recovers A_n, B_n from values on a fixed section, so (2.16) gives $\mathcal{N}_{f_n} \rightarrow \mathcal{N}_f$ locally uniformly on D , and each $\mathcal{N}_{f_n}$ is zero free by Theorem 2.16. Scalar Hurwitz then gives the dichotomy, and (2.18) with Theorem 2.4 handles the degenerate case. $\square$

Viewing f_I as a holomorphic map into $\mathbb{C}^2$ would not suffice: the maps $z \mapsto (z, 1/n)$ are nonvanishing but converge to $(z, 0)$, which vanishes. Zero-freeness on *all* spheres, encoded by $\mathcal{N}_f$, is the extra information.

2.11 Cauchy–Fueter regularity and the four-dimensional Cauchy formula

Now change the notion of regularity. Let $\Omega \subset \mathbb{H} \cong \mathbb{R}^4$ be an arbitrary connected open set; no axial symmetry is assumed. A C^1 function $f : \Omega \rightarrow \mathbb{H}$ is *left Cauchy–Fueter regular* (left monogenic) when $\mathcal{D}f = 0$; for a factor on the other side one uses the right-acting operator $g\mathcal{D} = \partial_{x_0}g + (\partial_{x_1}g)\mathbf{i} + (\partial_{x_2}g)\mathbf{j} + (\partial_{x_3}g)\mathbf{k}$. Anticommutation gives $\overline{\mathcal{D}}\mathcal{D} = \mathcal{D}\overline{\mathcal{D}} = \Delta_4$ as identities of constant-coefficient operators. Note at once that $\mathcal{D}q = 1 - 1 - 1 - 1 = -2$: the identity function, the basic example of slice analysis, is not Fueter regular.

Lemma 2.39 (Quaternionic Green identity). *Let Ω be bounded with C^1 boundary, $n = n_0 + \mathbf{i}n_1 + \mathbf{j}n_2 + \mathbf{k}n_3$ the outward unit normal, and f, g of class C^1 near $\overline{\Omega}$. Then*

$$\int_{\partial\Omega} g(y)n(y)f(y)dS_y = \int_{\Omega} ((g\mathcal{D})(y)f(y) + g(y)(\mathcal{D}f)(y))dV_y. \quad (2.33)$$

Proof. Apply the divergence theorem componentwise to the four-vector with quaternionic components $ge_\mu f$: its divergence is $\sum_\mu (\partial_{x_\mu}g)e_\mu f + \sum_\mu ge_\mu \partial_{x_\mu}f = (g\mathcal{D})f + g(\mathcal{D}f)$, which fixes the order of the three factors on the boundary. $\square$

In particular a Fueter-regular function satisfies the Cauchy theorem

$$\int_{\partial\Omega} n(y)f(y)dS_y = 0, \quad (2.34)$$

the integration cycle being three-dimensional. The fundamental kernel is

$$E(q) = \frac{\bar{q}}{2\pi^2|q|^4}, \quad q \neq 0, \quad (2.35)$$

left and right Fueter regular away from 0: indeed $\mathcal{D}\bar{q} = 4$ and $\sum_\mu e_\mu \bar{q} \partial_{x_\mu}|q|^{-4} = -4q\bar{q}|q|^{-6} = -4|q|^{-4}$ cancels it, and the right calculation is identical with the order reversed. On $|q| = r$ the outward normal is q/r , so

$$E(q)n(q) = n(q)E(q) = \frac{1}{2\pi^2 r^3}, \quad \int_{|q|=r} E(q)n(q)dS_q = 1, \quad (2.36)$$

since the unit three-sphere has area $2\pi^2$.

Theorem 2.40 (Cauchy–Pompeiu and Cauchy–Fueter). *Under the boundary hypotheses of Lemma 2.39, for $f \in C^1$ near $\overline{\Omega}$ and $x \in \Omega$,*

$$f(x) = \int_{\partial\Omega} E(y-x)n(y)f(y)dS_y - \int_{\Omega} E(y-x)(\mathcal{D}f)(y)dV_y, \quad (2.37)$$

and consequently, if $\mathcal{D}f = 0$,

$$f(x) = \int_{\partial\Omega} E(y-x)n(y)f(y)dS_y. \quad (2.38)$$

Proof. Apply (2.33) with $g(y) = E(y-x)$ on $\Omega \setminus \overline{B(x, \varepsilon)}$, where $g\mathcal{D} = 0$; the inner normal is the negative of the small ball's outward normal. The inner boundary term tends to $f(x)$ by (2.36) and continuity, and the volume integral converges because $|E(y-x)| = O(|y-x|^{-3})$ is locally integrable in four dimensions. $\square$

Equation (2.38) represents functions already known to be regular; it is not a claim that arbitrary boundary values admit a regular extension.

2.12 Fueter analogues of the basic analytic theorems

Proposition 2.41. *Every C^1 Fueter-regular function is real analytic, and each of its four real components is harmonic.*

Proof. Use (2.38) on a closed ball inside the domain; the kernel is smooth in x uniformly for y on the boundary, so all derivatives pass under the integral. For analyticity near the centre a , expand

$$|y - x|^{-4} = |y - a|^{-4} \left(1 + \frac{-2\langle y - a, x - a \rangle + |x - a|^2}{|y - a|^2} \right)^{-2},$$

whose binomial series converges absolutely and uniformly on the integration sphere for $|x - a|$ small; after multiplication by $\overline{y - x}$ it is a convergent power series in the four coordinates of $x - a$. Finally $\Delta_4 f = \overline{\mathcal{D}}(\mathcal{D}f) = 0$. $\square$

Theorem 2.42 (Cauchy derivative estimates). *If f is Fueter regular near $\overline{B(a, R)}$ and $M_R = \max_{|q-a|=R} |f(q)|$, then for every multi-index α*

$$|\partial^\alpha f(a)| \leq C_\alpha \frac{M_R}{R^{|\alpha|}}, \quad C_\alpha = 2\pi^2 \max_{|\omega|=1} |\partial^\alpha E(\omega)|, \quad (2.39)$$

with $C_0 = 1$ and $C_\alpha = 5$ admissible for each first partial derivative.

Proof. Differentiate (2.38); E is homogeneous of degree -3 , so $\partial^\alpha E$ is homogeneous of degree $-3 - |\alpha|$, and the area factor is $2\pi^2 R^3$. Explicitly $\partial_{x_\mu} E(q) = (2\pi^2)^{-1} (\bar{e}_\mu |q|^{-4} - 4x_\mu \bar{q} |q|^{-6})$, of norm at most $5/(2\pi^2 |q|^4)$. No sharpness is asserted for these constants. $\square$

Corollary 2.43 (Liouville and polynomial growth). *A bounded entire Fueter-regular function is constant; if $|f(q)| \leq C(1 + |q|)^\beta$ with $\beta \geq 0$ then f is a polynomial in x_0, x_1, x_2, x_3 of total degree at most $\lfloor \beta \rfloor$ satisfying $\mathcal{D}f = 0$.*

Proof. Let $R \rightarrow \infty$ in (2.39): every derivative of order $> \beta$ vanishes at every centre, and the real Taylor expansion terminates. $\square$

The conclusion differs from Theorem 2.11: here a polynomial means a polynomial in the four real coordinates, not necessarily a right-coefficient polynomial in q . The axially monogenic entire functions of polynomial growth are classified in Part IX by the polynomials $\mathcal{P}_k$.

Theorem 2.44 (Mean value and maximum modulus). *If f is Fueter regular near $\overline{B(a, R)}$ then*

$$f(a) = \frac{1}{2\pi^2 R^3} \int_{|y-a|=R} f(y) dS_y, \quad (2.40)$$

and if $|f|$ has a local maximum in a connected domain then f is constant.

Proof. (2.40) is (2.38) with (2.36). If $|f| \leq M = |f(a)|$ near a with $M > 0$, the real-linear functional $\ell(w) = \operatorname{Re}(w \overline{f(a)})/M$ satisfies $\ell(w) \leq |w|$, and on every small sphere the average of $\ell(f)$ is M while $\ell(f) \leq |f| \leq M$; continuity forces equality pointwise, and equality in real Cauchy–Schwarz with $|f| \leq M$ gives $f \equiv f(a)$. Real analyticity propagates. $\square$

Theorem 2.45 (Fueter–Morera). *A continuous $f : \Omega \rightarrow \mathbb{H}$ is Fueter regular if and only if $\int_{\partial B} n(y) f(y) dS_y = 0$ for every ball with closure in Ω .*

Proof. Necessity is (2.34). For sufficiency mollify: by Fubini the flux of f_ε through a ball is the average of fluxes of f through translated balls, hence zero, so the divergence theorem gives $\int_B \mathcal{D}f_\varepsilon dV = 0$ for every such ball and shrinking balls gives $\mathcal{D}f_\varepsilon = 0$. Apply (2.38) to f_ε on a fixed smaller ball and pass to the limit using uniform boundary convergence; the limit integral is regular inside because $\mathcal{D}_x E(y - x) = 0$ there. $\square$

2.13 Removable singularities, the flux residue, and two counterexamples

Theorem 2.46 (Sharp growth threshold for removability at a point). *Suppose f is Fueter regular on $B(a, R) \setminus \{a\}$ and*

$$\sup_{|q-a|=r} |f(q)| = o(r^{-3}) \quad (r \downarrow 0). \quad (2.41)$$

Then f extends uniquely to a Fueter-regular function on $B(a, R)$; in particular a bounded isolated singularity is removable. The exponent 3 cannot be replaced by a statement allowing all $O(r^{-3})$ growth.

Proof. Apply (2.38) on the punctured ball with outer radius $R_0 < R$ and inner radius ε at a fixed $x \neq a$. For small ε the kernel $E(y - x)$ is bounded on the inner sphere, so that contribution is at most a constant times $\varepsilon^3 \sup_{|y-a|=\varepsilon} |f(y)| \rightarrow 0$. Hence $f(x)$ is the outer boundary integral alone, which is regular throughout $B(a, R_0)$. Sharpness: $E(q - a)c$ with $c \neq 0$ is a nonremovable example of exact order $|q - a|^{-3}$. $\square$

The exponent 3 here belongs to an *isolated point* in $\mathbb{R}^4$. Part VII proves a threshold with exponent 1 at a singular two-sphere; the two exponents describe singular sets of real codimension four and two and are never to be confused.

Definition 2.47 (Flux residue). For a Fueter-regular function with an isolated singularity at a put

$$\text{Res}_a^F f = \int_{|q-a|=r} n(q) f(q) dS_q, \quad (2.42)$$

for any sufficiently small $r > 0$, the normal pointing away from a . The normalization keeps the $2\pi^2$ in the kernel, not in the residue.

The value is independent of r by the Green identity on an annulus.

Theorem 2.48 (Fueter residue theorem). *Let Ω be a bounded C^1 domain and f Fueter regular near $\bar{\Omega}$ except at finitely many interior points $a_1, \dots, a_N$, where it is regular in punctured neighbourhoods. Then*

$$\int_{\partial\Omega} n(q) f(q) dS_q = \sum_{j=1}^N \text{Res}_{a_j}^F f, \quad (2.43)$$

and $\text{Res}_a^F (E(q - a)c) = c$.

Proof. Remove disjoint small balls and apply (2.34); the inner normals are the negatives of those in (2.42). The last assertion follows from $nE = 1/(2\pi^2 r^3)$ on the sphere. $\square$

The next two examples are the reason every quaternionic statement in this document has the shape it has. Both are needed later and neither is reproved anywhere else.

Example 2.49 (A zero flux residue without removability). $\partial_{x_0} E$ is Fueter regular away from 0, is nonzero, is homogeneous of degree -4 , and is singular at 0. Since E is odd, $\partial_{x_0} E$ is even, whereas n is odd on a centred sphere; hence its flux, and therefore its residue (2.42), vanishes. So

$$\text{Res}_0^F (\partial_{x_0} E) = 0 \quad \text{while the singularity at 0 is not removable.}$$

A single flux residue does not encode the whole principal part of a singularity. This is the isolated-point ancestor of the fact, proved in Part VIII, that the total spherical flux $8\pi v(ua + b)$ misses the slope residue a : it is exactly why the spherical residue of Part VII must be a *pair* (a, b) , defined by two period integrals rather than by one flux, and why a genuinely second conserved flux with a nonconstant regular weight is needed to recover a .

Example 2.50 (Vanishing on a plane: the identity and open mapping theorems fail). The function

$$L(q) = x_1 - \mathbf{i}x_0 \quad (2.44)$$

is Fueter regular, since $\mathcal{D}L = -\mathbf{i} + \mathbf{i} = 0$. It vanishes on the entire plane $\{x_0 = x_1 = 0\}$ without being identically zero, and its image lies inside the single slice $\mathbb{C}_{\mathbf{i}}$ and hence has empty interior in $\mathbb{H}$. Therefore *neither* the accumulation-point identity theorem (contrast Theorem 2.4) *nor* the nonconstant open mapping theorem (contrast Theorem 2.27) holds for arbitrary Cauchy–Fueter-regular functions. The appropriate elementary uniqueness statement is vanishing on a nonempty open set, which does force global vanishing by Proposition 2.41.

2.14 The bridge: axial identities, the Vekua system, and Fueter’s mapping theorem

The two theories are connected by a second-order operator. Everything in this subsection is used by every later part; it is stated once here.

Away from the real axis write $q = x + Iy$ with $y = |\operatorname{Im} q| > 0$ and consider an *axial* function $f(q) = A(x, y) + IB(x, y)$ with $\mathbb{H}$ -valued A, B .

Lemma 2.51 (Axial differential identities and the Vekua system). *For smooth axial $A + IB$ with $y > 0$,*

$$\mathcal{D}(A + IB) = \left(A_x - B_y - \frac{2B}{y}\right) + I(B_x + A_y), \quad (2.45)$$

$$\Delta_4(A + IB) = A_{xx} + A_{yy} + \frac{2A_y}{y} + I\left(B_{xx} + B_{yy} + \frac{2B_y}{y} - \frac{2B}{y^2}\right). \quad (2.46)$$

Consequently $g = P + IQ$ is *axially monogenic* if and only if the Vekua system

$$P_x - Q_y = \frac{2Q}{y}, \quad Q_x + P_y = 0 \quad (2.47)$$

holds on the base. For a slice-regular $f = \mathcal{I}(F)$ with $F = A + \iota B$,

$$\mathcal{D}f = -\frac{2B}{y}, \quad \Delta_4 f = \frac{2A_y}{y} + I\left(\frac{2B_y}{y} - \frac{2B}{y^2}\right), \quad (2.48)$$

that is, $g = \Delta_4 \mathcal{I}(F) = P + IQ$ with $P = 2A_y/y$ and $Q = 2B_y/y - 2B/y^2$; equivalently, in complexified form,

$$Q - \iota P = \frac{2}{y}F' - \frac{2}{y^2}\operatorname{Im}_{\mathbb{C}} F, \quad \|g\|_{\text{ax}} \leq \frac{2}{y}\|F'\|_{\mathbb{H}_{\mathbb{C}}} + \frac{2}{y^2}\|F\|_{\mathbb{H}_{\mathbb{C}}}. \quad (2.49)$$

Proof. Put $\mathbf{v} = \operatorname{Im} q = \mathbf{i}x_1 + \mathbf{j}x_2 + \mathbf{k}x_3$, so $I = \mathbf{v}/y$, and for $\ell = 1, 2, 3$

$$\partial_{x_\ell} y = \frac{x_\ell}{y}, \quad \partial_{x_\ell} I = \frac{e_\ell}{y} - \frac{\mathbf{v}x_\ell}{y^3}.$$

Hence $\sum_{\ell=1}^3 e_\ell \partial_{x_\ell} I = -2/y$, the angular Laplacian contributes $\Delta_{\mathbb{R}^3} I = -2I/y^2$, and the mixed radial–angular term $\sum_{\ell} (\partial_{x_\ell} I)(\partial_{x_\ell} y)$ vanishes. The product rule together with the ordinary radial Laplacian in three dimensions gives (2.45)–(2.46), with all quaternionic factors on their displayed sides. An expression $C_1 + IC_2$ vanishing for every $I \in \mathbb{S}$ has $C_1 = C_2 = 0$, by comparing I and $-I$; applied to (2.45) this gives (2.47). For a slice-regular f the stem equations (2.4) give $A_x = B_y$, $A_y = -B_x$ and $A_{xx} + A_{yy} = B_{xx} + B_{yy} = 0$, which turn (2.45)–(2.46) into (2.48). Finally $F' = A_x + \iota B_x = B_y - \iota A_y$, so $\frac{2}{y}F' - \frac{2}{y^2}\operatorname{Im}_{\mathbb{C}} F = \frac{2B_y}{y} - \frac{2B}{y^2} - \iota \frac{2A_y}{y} = Q - \iota P$, and the estimate follows since $\|Q - \iota P\|_{\mathbb{H}_{\mathbb{C}}}^2 = |P|^2 + |Q|^2 = \|g\|_{\text{ax}}^2$. $\square$

All ten source manuscripts contain this computation with identical signs and coefficients; it is stated here once and used without further comment in Parts II–IX. Identity (2.49) explains, incidentally, why off the real axis the order of a singularity increases by *one* under a map defined by a four-dimensional operator of order two.

Theorem 2.52 (Fueter mapping, biharmonicity, and the affine kernel). *If $f \in \mathcal{SR}(\Omega_D)$ then*

$$\mathcal{D}(\Delta_4 f) = 0, \quad \Delta_4^2 f = 0. \quad (2.50)$$

On a connected slice domain the kernel of $f \mapsto \Delta_4 f$ is exactly

$$\text{Aff}_{\mathbb{H}} = \{q \mapsto qa + b : a, b \in \mathbb{H}\},$$

with a and b quaternions. Moreover a function that is both slice regular and left Fueter regular is constant.

Proof. Away from $\mathbb{R}$ write $\Delta_4 f = P + IQ$ with $P = 2A_y/y$ and $Q = 2B_y/y - 2B/y^2$ as in (2.48). The stem equations give

$$P_x = \frac{2B_{yy}}{y} = Q_y + \frac{2Q}{y}, \quad Q_x + P_y = \frac{2(B_{yx} + A_{yy})}{y} - \frac{2(B_x + A_y)}{y^2} = 0,$$

which is exactly the Vekua system (2.47); by (2.45) this says $\mathcal{D}(\Delta_4 f) = 0$, and applying $\bar{\mathcal{D}}$ gives $\Delta_4^2 f = 0$. Slice-regular functions are smooth across $\mathbb{R}$ by Theorem 2.8, so continuity extends both identities across it.

If $\Delta_4 f = 0$ then comparing the values at I and $-I$ gives $P = Q = 0$, i.e. $A_y = 0$ and $yB_y = B$. Locally in the upper half-plane $B = yb(x)$, and the stem equations then give $b'(x) = 0$ and $A_x = b$, so $f(q) = qa + b_0$ on a nonempty open set for fixed $a, b_0 \in \mathbb{H}$; Theorem 2.4 propagates this through Ω_D . Conversely affine functions have zero Laplacian. Finally $\mathcal{D}f = 0$ in (2.48) forces $B = 0$, the stem equations force A to be locally constant, and Theorem 2.4 makes it globally constant. $\square$

Remark 2.53 (Why the coefficients may not be complexified). The kernel would be twice as large, and the entire obstruction theory of Parts II–IX would disappear, if a, b were allowed to lie in $\mathbb{H}_{\mathbb{C}}$. They may not. A direct computation on the product-domain stem $\iota(za + b)$, whose components are $A = -ya$ and $B = xa + b$, gives

$$\Delta_4 \mathcal{I}(\iota(za + b)) = -\frac{2a}{y} - I \frac{2(xa + b)}{y^2} \neq 0 \quad \text{unless } a = b = 0. \quad (2.51)$$

The special case $a = 0, b = 1$ is the stem $F = \iota$, giving $f(q) = I$ and $\Delta_4 f = -2I/y^2$. Note that such an f exists only on a product domain: on a slice domain meeting $\mathbb{R}$ it is not even continuous. Part III identifies the right-hand side of (2.51) as exactly the kernel of the holomorphic invariant $\mathcal{H}_g$, and Part X uses it as a separating example.

The two Cauchy kernels are linked by the same operator. Writing $\mathbf{v} = \text{Im } q$ and $x = x_0$, direct computation gives

$$\Delta_4 1 = \Delta_4 q = 0, \quad \Delta_4 q^2 = -4, \quad \Delta_4 q^3 = -4(3x + \mathbf{v}), \quad \Delta_4(q^{-1}) = -\frac{4\bar{q}}{|q|^4} = -8\pi^2 E(q) \quad (q \neq 0). \quad (2.52)$$

For the last identity write $q^{-1} = \bar{q}|q|^{-2}$; in four dimensions $\Delta_4 |q|^{-2} = 0$ away from 0, and the cross term of the product rule is $2 \sum_{\mu} \bar{e}_{\mu} \partial_{x_{\mu}} |q|^{-2} = -4\bar{q}|q|^{-4}$. Thus the slice kernel of homogeneity -1 becomes, under Δ_4 , the Fueter kernel of homogeneity -3 . The third identity in (2.52) is not decoration: Part VIII uses $-\frac{1}{4}\Delta_4((q - u)^3) = 3(\text{Re } q - u) + \text{Im } q$ as the unique weight making the second spherical flux conserved, and Part IX identifies it with the degree-one entire axially monogenic polynomial $3\mathcal{P}_1(q - u)$. The normalization $\Delta_4 q^2 = -4$ recorded here is what fixes every numerical constant in those parts.

2.15 The local inverse by two integrations, and the seam

A local inverse of Δ_4 should only be expected within the axial class. In the following theorem the planar set lies strictly above the real axis, so its circularization is a *product domain* rather than a slice domain meeting $\mathbb{R}$; the stem definition is used there, and the global identity and maximum principles proved above are not asserted for arbitrary product domains.

Theorem 2.54 (Constructive axial inverse). *Let $U \subset \mathbb{C}^+$ be nonempty, open and simply connected, and let*

$$g(x + Iy) = P(x, y) + IQ(x, y)$$

be a smooth axially monogenic function on Ω_U . Then there is a slice-regular axial f on the same set with $\Delta_4 f = g$, unique up to addition of $qa + b$ with $a, b \in \mathbb{H}$.

Proof. By Lemma 2.51 the target satisfies the Vekua system (2.47). The quaternion-valued one-form

$$\omega_g = -\frac{P}{2}dx + \frac{Q}{2}dy \quad (2.53)$$

is closed, by the second Vekua equation. Integrating its four real components on the simply connected U gives a potential C with $C_x = -P/2$ and $C_y = Q/2$; set $B = yC$. Next consider

$$\eta_C = \left(C + \frac{yQ}{2}\right)dx + \frac{yP}{2}dy. \quad (2.54)$$

It is closed because, using the first Vekua equation,

$$\partial_y \left(C + \frac{yQ}{2}\right) = \frac{Q}{2} + \frac{Q}{2} + \frac{yQ_y}{2} = \frac{yP_x}{2} = \partial_x \left(\frac{yP}{2}\right).$$

Choose a primitive A of η_C . Then

$$A_x = C + \frac{yQ}{2} = B_y, \quad A_y = \frac{yP}{2} = -B_x,$$

so $F = A + \iota B$ is a holomorphic stem on U , and (2.48) gives

$$\Delta_4 \mathcal{I}(F) = \frac{2A_y}{y} + I \left(\frac{2B_y}{y} - \frac{2B}{y^2} \right) = P + IQ = g.$$

Changing C by a constant quaternion a changes B by ya and changes A by $xa + b$, which is precisely the addition of $qa + b$; conversely the kernel computation in Theorem 2.52, carried out locally on U , shows that the difference of any two primitives has this form, and connectedness makes the constants global. $\square$

Remark 2.55 (The seam). This construction explains both the affine ambiguity and the topological hypothesis, and it is exactly where the rest of the document begins. On a base that is *not* simply connected, the closed forms (2.53) and (2.54) can have nonzero periods, and the theorem claims no global primitive without period conditions; nor does it claim a primitive for a general nonaxial Cauchy–Fueter-regular function. Two features of the construction are what the extension replaces.

First, the two integrations here are *sequential*: the second form η_C of (2.54) depends on the first primitive C , so on a multiply connected base it is not even defined until the first integration has succeeded. Part II removes this by exhibiting a primitive-*independent* closed form θ_g , computed from P and Q alone, related to η_C by $\eta_C - d(xC) = \theta_g$; the two obstructions then become simultaneous rather than sequential, and the resulting criterion needs no simple connectivity, no finite topology and no boundary regularity. The reader is warned that one of the ten source manuscripts used the letter η for θ_g and four used it for η_C ; in this document η_C means (2.54) and nothing else.

Second, the normalization of the primitive. Fixing $C(z_*) = V(z_*) = 0$ at a base point $z_* = x_* + \iota y_*$ selects a unique primitive, and Part II records the two equivalent descriptions of that choice: it is the unique primitive vanishing on the *entire* sphere $x_* + y_*\mathbb{S}$, equivalently $F(z_*) = 0$ in $\mathbb{H}_{\mathbb{C}}$ — *eight* real conditions, not the four real conditions $f(x_* + I_* y_*) = 0$ at a single quaternionic point.

2.16 What survives, what changes, and three cautions

The proofs above identify two different mechanisms. Slice analysis retains one-complex-variable holomorphy in each section and coordinates the sections through a stem and the representation formula. Fueter analysis retains a first-order elliptic operator and its fundamental solution, but no longer admits all ordinary powers of q .

Classical feature	Slice-regular counterpart	Fueter-regular counterpart
Cauchy formula	planar contour, regular resolvent kernel (§2.3)	three-dimensional boundary, Dirac kernel (§2.11)
Taylor theory	powers $(q - c)^n a_n$ at real centres (§2.3)	real-coordinate analyticity and derivative estimates (§2.12)
Harmonicity	biharmonic: $\Delta_4^2 f = 0$ (§2.14)	harmonic: $\Delta_4 f = 0$ (§2.12)
Zeros	isolated points or isolated whole spheres; the scalar norm stem controls multiplicity (§§2.5–2.6)	a nonzero function can vanish on a plane (Example 2.50)
Maximum, Liouville	both hold (§2.4)	both hold (§2.12)
Open mapping	holds off the closure of the degenerate spheres and for symmetric open sets (§2.7)	fails in general (Example 2.50)
Singularities	Laurent series and residues at real isolated singularities (§2.9)	flux residues and a sharp $o(r^{-3})$ removability criterion (§2.13)
Conformal mapping	intrinsic slice-wise Riemann mapping, not arbitrary four-dimensional conformality (§2.8)	no corresponding claim follows from the Fueter equation alone

Three cautions recur, and they govern every statement in Parts II–XI. *First*, a right coefficient cannot be moved to the left without changing the problem; this is why the period pair (a_γ, b_γ) acts by $f \mapsto f + qa_\gamma + b_\gamma$ with the coefficients on the right, and why the monodromy of Part II is a translation in a fixed eight-real-dimensional space rather than a multiplicative holonomy. *Second*, the geometric unit for many slice theorems is the whole conjugacy sphere and not an individual point; this is why the singular set of Part VII is a two-sphere of real codimension two, with critical exponent 1, and not an isolated point with critical exponent 3. *Third*, a quaternionic theorem must specify which regularity it uses — the source g is axially *monogenic* and the primitive f is slice *regular*, and by Theorem 2.52 nothing nonconstant is both.

3 Two closed forms and the complete global criterion

3.1 The setting, and what is recalled from Part I

Throughout this part U is a connected open subset of the upper half-plane $\mathbb{C}^+ = \{z = x + \iota y : y > 0\}$, and

$$\Omega_U = \{q = x + Iy : x + \iota y \in U, I \in \mathbb{S}\}$$

is its circularization. A holomorphic $F = A + \iota B : U \rightarrow \mathbb{H}_{\mathbb{C}}$ induces the slice-regular function $\mathcal{I}(F)(x + Iy) = A(x, y) + IB(x, y)$; all quaternionic coefficients multiply on the right, and ι is the central imaginary unit of $\mathbb{H}_{\mathbb{C}}$, never an element of $\mathbb{S}$. We write $\mathcal{SR}(\Omega_U)$ for the induced slice-regular functions and $\mathcal{AM}(\Omega_U)$ for the right $\mathbb{H}$ -vector space of smooth axially monogenic fields $g(x + Iy) = P(x, y) + IQ(x, y)$, that is, those annihilated by $\mathcal{D}$. Domains meeting the real axis are treated separately in Part V; nothing in the present part requires U to be simply connected, finitely connected, bounded, or to have regular boundary.

Two facts established in Part I are used constantly. The first is the Vekua system: for $y > 0$, $g = P + IQ$ is axially monogenic exactly when

$$P_x - Q_y = \frac{2Q}{y}, \quad Q_x + P_y = 0. \quad (3.1)$$

The second is the Fueter formula: for $f = \mathcal{I}(F) = A + IB$ with F holomorphic,

$$\Delta_4 f = \frac{2A_y}{y} + I \left(\frac{2B_y}{y} - \frac{2B}{y^2} \right), \quad (3.2)$$

and $\Delta_4 f$ is again axially monogenic. In particular $\Delta_4 q^2 = -4$: the operator $\mathcal{D} = \partial_{x_0} + i\partial_{x_1} + j\partial_{x_2} + k\partial_{x_3}$ carries no factor $\frac{1}{2}$, and neither does $\Delta_4 = \overline{\mathcal{D}}\mathcal{D}$.

3.2 The change of potentials

The local inverse of Part I integrates twice. It first produces $C = B/y$ from a closed form built out of g , and then produces A from a second form that already contains the function C just constructed. That is the source of the apparent asymmetry between the two integrations, and it is what the following change of unknowns removes.

Definition 3.1 (The two potentials). For $f = \mathcal{I}(F) \in \mathcal{SR}(\Omega_U)$ with $F = A + \iota B$, set

$$C = \frac{B}{y} \quad (\text{the slope potential}), \quad V = A - \frac{xB}{y} = A - xC \quad (\text{the constant potential}). \quad (3.3)$$

Both are $\mathbb{H}$ -valued. Then $F(z) = zC(z) + V(z)$ and

$$f(x + Iy) = (x + Iy)C(x, y) + V(x, y) = qC + V. \quad (3.4)$$

The passage from (A, B) to (C, V) is a real-linear isomorphism at each point of U , because $y > 0$; nothing is lost. Its point is that (3.4) exhibits f in the shape of the affine kernel with variable coefficients, so that the two coefficient functions are precisely the two objects whose global existence can fail.

3.3 The two closed one-forms

Definition 3.2 (The slope form and the constant form). For $g = P + IQ \in \mathcal{AM}(\Omega_U)$ put

$$\omega_g = \frac{1}{2}(-Pdx + Qdy), \quad (3.5)$$

$$\theta_g = \frac{1}{2}((xP + yQ)dx + (yP - xQ)dy). \quad (3.6)$$

These are $\mathbb{H}$ -valued one-forms on U , integrated componentwise over real curves; since the coefficients multiply real differentials, no ordering ambiguity arises. Both are defined directly from the target g : no primitive, no branch, and no derivative of g occurs in them.

Lemma 3.3 (Closedness and the potential identities). *Let $g = P + IQ \in \mathcal{AM}(\Omega_U)$.*

- (i) ω_g and θ_g are closed on U .

(ii) If $f = \mathcal{I}(F) \in \mathcal{SR}(\Omega_U)$ satisfies $\Delta_4 f = g$, then the potentials (3.3) satisfy

$$dC = \omega_g, \quad dV = \theta_g, \quad f(q) = qC + V. \quad (3.7)$$

(iii) Conversely, if $C, V : U \rightarrow \mathbb{H}$ are any C^1 functions with $dC = \omega_g$ and $dV = \theta_g$, then $A = xC + V$, $B = yC$ define a holomorphic stem $F = A + \iota B$ on U , and $\Delta_4 \mathcal{I}(F) = g$.

Proof. (i) The coefficient of $dx \wedge dy$ in $d\omega_g$ is $\frac{1}{2}(Q_x + P_y)$, which vanishes by the second Vekua equation in (3.1). For the second form, twice the corresponding coefficient is

$$\partial_y(xP + yQ) - \partial_x(yP - xQ) = x(P_y + Q_x) + 2Q + y(Q_y - P_x) = -\{y(P_x - Q_y) - 2Q - x(Q_x + P_y)\},$$

which vanishes by both equations of (3.1). All computations are componentwise in the real vector space $\mathbb{H}$.

(ii) From $B = yC$ and the Fueter formula (3.2), $Q = 2B_y/y - 2B/y^2$ gives $C_y = B_y/y - B/y^2 = Q/2$, while the Cauchy–Riemann equation $B_x = -A_y$ together with $P = 2A_y/y$ gives $C_x = B_x/y = -P/2$. Hence $dC = \omega_g$. For the second potential,

$$V_x = A_x - C - xC_x = B_y - C + \frac{xP}{2} = \frac{xP + yQ}{2}, \quad V_y = A_y - xC_y = \frac{yP - xQ}{2},$$

using $A_x = B_y$ and $B_y = C + yC_y = C + yQ/2$. Hence $dV = \theta_g$.

(iii) With $A = xC + V$ and $B = yC$,

$$A_x = C + xC_x + V_x = C + \frac{yQ}{2} = B_y, \quad A_y = xC_y + V_y = \frac{yP}{2} = -B_x.$$

So $F = A + \iota B$ is holomorphic, and substituting into (3.2) returns $P + IQ$. $\square$

3.4 Elimination of the first primitive

The second form of the classical local construction (Part I) is the primitive-*dependent* one-form

$$\eta_C = \left(C + \frac{yQ}{2}\right)dx + \frac{yP}{2}dy, \quad (3.8)$$

whose primitive is A and whose very definition presupposes that a first primitive C of ω_g has already been chosen. The two second forms are different one-forms, and they are related by an exact term.

Proposition 3.4 (The elimination identity). *Wherever a primitive C of ω_g exists,*

$$\boxed{\eta_C - d(xC) = \theta_g} \quad (3.9)$$

In particular θ_g is the primitive-independent representative of the cohomology class of η_C on any subdomain where ω_g is exact.

Proof. Using $dC = \omega_g$,

$$d(xC) = Cdx + x\omega_g = \left(C - \frac{xP}{2}\right)dx + \frac{xQ}{2}dy.$$

Subtracting this from (3.8) gives $\frac{1}{2}(xP + yQ)dx + \frac{1}{2}(yP - xQ)dy = \theta_g$. $\square$

Remark 3.5 (The two obstructions are simultaneous, not sequential). Identity (3.9) is the pivot of the whole theory, and it is worth saying exactly what it buys and what it does not. The form η_C is defined only after ω_g has been integrated, so on a domain where ω_g is not exact it is not even globally defined; the construction then stalls at the first step and the second obstruction cannot be formulated at all. By contrast θ_g in (3.6) is written down from P and Q alone and is closed on all of U by Lemma 3.3(i), whatever the periods of ω_g may be. Consequently the two obstructions of Theorem 3.7 are simultaneous conditions on g , not a condition followed by a conditional second condition, and no choice of first primitive can remove a nonzero period of θ_g : by (3.9) a different choice of C changes η_C by an exact form and leaves θ_g untouched. This is also the reason that *two* periods, rather than one, are the correct invariant: they are the two coefficient functions of (3.4), and Example (i) of Part X separates them.

3.5 The global period criterion

Definition 3.6 (The period pair). For a closed piecewise C^1 curve $\gamma \subset U$ and $g \in \mathcal{AM}(\Omega_U)$ put

$$a_\gamma(g) = \oint_\gamma \omega_g, \quad b_\gamma(g) = \oint_\gamma \theta_g, \quad \text{Per}_g(\gamma) = (a_\gamma(g), b_\gamma(g)) \in \mathbb{H}^2. \quad (3.10)$$

We call a_γ the *slope period* and b_γ the *constant period*. By Lemma 3.3(i) both depend only on the integral homology class of γ .

Theorem 3.7 (The complete same-domain criterion). *Let $U \subset \mathbb{C}^+$ be any connected open set and let $g \in \mathcal{AM}(\Omega_U)$. The following are equivalent.*

- (i) *There is a single-valued $f \in \mathcal{SR}(\Omega_U)$ with $\Delta_4 f = g$.*
- (ii) *Both ω_g and θ_g are exact on U .*
- (iii) *$a_\gamma(g) = b_\gamma(g) = 0$ for every closed piecewise C^1 curve $\gamma \subset U$.*

When these hold, fix $z_ \in U$ and $c_0, e_0 \in \mathbb{H}$; every solution is obtained exactly once from*

$$C(z) = c_0 + \int_{z_*}^z \omega_g, \quad V(z) = e_0 + \int_{z_*}^z \theta_g, \quad f(x + Iy) = (x + Iy)C(z) + V(z), \quad (3.11)$$

the integrals being taken along any path in U . The solution is unique modulo $\text{Aff}_{\mathbb{H}}$.

Proof. (i) $\Leftrightarrow$ (ii) is Lemma 3.3(ii)–(iii): a primitive supplies primitives of the two forms, and primitives of the two forms supply a holomorphic stem on U , hence, after reflecting by $F(\bar{z}) = \kappa F(z)$, a symmetric stem on $U \cup \bar{U}$ and a single-valued $f \in \mathcal{SR}(\Omega_U)$ with $\Delta_4 f = g$.

(ii) $\Rightarrow$ (iii) is immediate. For (iii) $\Rightarrow$ (ii), note that a connected open subset of the plane is polygonally path connected; define the two path integrals in (3.11). Two paths with the same endpoints differ by a closed curve, so (iii) makes both values independent of the path, for each of the four real components separately. On a small disk the usual differentiation of a path integral shows that the resulting functions are primitives of the corresponding forms.

For uniqueness, two pairs of primitives differ by constants $a, b \in \mathbb{H}$, and (3.4) turns that difference into $qa + b$; conversely every such affine addition is annihilated by Δ_4 (Corollary 3.9). $\square$

Remark 3.8 (No topological and no boundary hypothesis). Theorem 3.7 assumes nothing about U beyond connectedness and openness. In particular it does not assume finite connectivity, a rectifiable or accessible boundary, boundedness, or simple connectivity of the complementary components; it is not restricted to a rectangle or a disk, and it is an equivalence rather than a sufficient condition. The price is that condition (iii) quantifies over all loops. Converting it into a *finite* computation requires a topological hypothesis, and exactly that hypothesis — a finite winding basis — is isolated in Part IV; the cokernel computation lives there, not here.

Corollary 3.9 (The affine kernel). *On connected U ,*

$$\ker(\Delta_4|_{\mathcal{SR}(\Omega_U)}) = \text{Aff}_{\mathbb{H}} = \{q \mapsto qa + b : a, b \in \mathbb{H}\}. \quad (3.12)$$

The coefficients are quaternions, not elements of $\mathbb{H}_{\mathbb{C}}$.

Proof. If $\Delta_4 f = 0$ then $g = 0$, so $dC = \omega_0 = 0$ and $dV = \theta_0 = 0$ by Lemma 3.3(ii); connectedness makes $C \equiv a$ and $V \equiv b$ constant quaternions, and (3.4) gives $f(q) = qa + b$. The converse is (3.2). $\square$

The restriction to *real* quaternionic coefficients in (3.12) is not a matter of taste: enlarging $\text{Aff}_{\mathbb{H}}$ to the complex-affine stems would double the kernel and erase the obstruction. The relevant counterexample is the stem $F = \iota$, which induces $f(q) = I$ and has $\Delta_4 f = -2I/y^2 \neq 0$; it reappears in Part III as Theorem 4.6, where the whole eight-real-dimensional family $\Delta_4 \mathcal{I}(\iota(za + b))$ is identified as the kernel of the holomorphic invariant.

Corollary 3.10 (Affine monodromy). *Without any vanishing hypothesis, a local Fueter primitive of $g \in \mathcal{AM}(\Omega_U)$ continues analytically along every path in U . Continuation around a based loop γ changes it by*

$$F \mapsto F + z a_{\gamma}(g) + b_{\gamma}(g), \quad f \mapsto f + q a_{\gamma}(g) + b_{\gamma}(g). \quad (3.13)$$

Consequently the full monodromy of the local inverse is the additive homomorphism

$$\text{Per}_g : H_1(U; \mathbb{Z}) \longrightarrow \mathbb{H} \oplus \mathbb{H}, \quad [\gamma] \mapsto (a_{\gamma}, b_{\gamma}), \quad (3.14)$$

and there are no further local compatibility obstructions.

Proof. Pull the two closed forms back to the universal cover of U , where they are exact; equivalently, integrate them along paths from a fixed base point while retaining the homotopy class. Lemma 3.3(iii) makes the resulting $F = zC + V$ holomorphic on each chart, and it agrees with the prescribed primitive after a correct choice of the two initial constants. Appending γ to a path adds a_{γ} to C and b_{γ} to V , which is (3.13) by (3.4). Concatenation of loops adds periods and reversal negates them, and closed forms integrate homologically, so (3.14) is a well-defined homomorphism. The last sentence is the content of Theorem 3.7: the two periods are the only obstructions, and the construction supplies an actual continued primitive along every path. $\square$

Remark 3.11 (What is, and is not, noncommutative here). The monodromy (3.13) is a *translation* inside the fixed eight-real-dimensional space $\text{Aff}_{\mathbb{H}}$. It is not a quaternionic fractional-linear action, not a path-ordered exponential, and not a multiplicative holonomy; noncommutativity of $\mathbb{H}$ plays no role in the composition law, because the coefficients remain on the right and the group operation is addition. In particular the obstruction group is abelian, and every commutator loop in $\pi_1(U)$ acts trivially. What is genuinely quaternionic is the *size* of the obstruction — two quaternionic coordinates rather than one complex residue — and the geometry of the conjugacy spheres, which re-enters essentially in the singularity norms and the energy law of Parts VII and VIII.

Theorem 3.7 and Corollary 3.10 may be assembled into an exact sequence of right quaternionic vector spaces,

$$0 \longrightarrow \mathbb{H}^2 \xrightarrow{(a,b) \mapsto qa+b} \mathcal{SR}(\Omega_U) \xrightarrow{\Delta_4} \mathcal{AM}(\Omega_U) \xrightarrow{\text{Per}} \text{Hom}(H_1(U; \mathbb{Z}), \mathbb{H}^2), \quad (3.15)$$

exact at $\mathcal{SR}(\Omega_U)$ and at $\mathcal{AM}(\Omega_U)$. We do *not* assert here that the last arrow is surjective for an arbitrary, possibly infinitely connected, U ; surjectivity is proved in Part IV under an explicit finite-topology hypothesis, and the modes that realize the periods are constructed there.

3.6 Normalization: eight real conditions, not four

Corollary 3.12 (The normalized primitive). *Fix $z_* = x_* + iy_* \in U$ and suppose $g \in \mathcal{AM}(\Omega_U)$ has a global primitive. The following three descriptions single out one and the same primitive f .*

- (a) $C(z_*) = V(z_*) = 0$, i.e. the choice $c_0 = e_0 = 0$ in (3.11).
- (b) $F(z_*) = 0$ in $\mathbb{H}_{\mathbb{C}}$, which is eight real conditions on the stem.
- (c) f vanishes on the entire base sphere $x_* + y_*\mathbb{S} \subset \Omega_U$.

It is the unique primitive with any one of these properties.

Proof. (a) $\Leftrightarrow$ (b): the map $\mathbb{H}^2 \rightarrow \mathbb{H}_{\mathbb{C}}$, $(a, b) \mapsto z_*a + b$, is a real isomorphism because $y_* > 0$ — it is the change of potentials (3.3) evaluated at z_* — so $F(z_*) = z_*C(z_*) + V(z_*)$ vanishes precisely when both potentials do. (a) $\Rightarrow$ (c): $f(x_* + Iy_*) = (x_* + Iy_*)C(z_*) + V(z_*) = 0$ for every $I \in \mathbb{S}$. (c) $\Rightarrow$ (a): if $A(z_*) + IB(z_*) = 0$ for all $I \in \mathbb{S}$, subtract the values at I and $-I$ to get $2IB(z_*) = 0$, hence $B(z_*) = 0$ and then $A(z_*) = 0$. Uniqueness: two primitives differ by $qa + b$ by Corollary 3.9, and an affine function vanishing at all of $x_* + y_*\mathbb{S}$ has $a = b = 0$ by the same subtraction. $\square$

Remark 3.13 (A normalization that is easy to get wrong). Descriptions (b) and (c) look like conditions of different sizes and are often stated as if they were: (b) is manifestly eight real conditions, (c) looks like the vanishing of a function on a two-sphere. They coincide, and both are strictly stronger than the four-real-dimensional condition $f(q_*) = 0$ at a *single* quaternionic point $q_* = x_* + I_*y_*$. That weaker condition does not determine f : the elements of $\text{Aff}_{\mathbb{H}}$ vanishing at one fixed q_* are exactly $q \mapsto qa - q_*a$ with $a \in \mathbb{H}$ arbitrary (note $q_*a \in \mathbb{H}$), a four-real-dimensional space, so a pointwise normalization leaves a four-parameter ambiguity and one must pass to the whole conjugacy sphere $x_* + y_*\mathbb{S}$ to pin the primitive down. Installing the pointwise normalization in place of (b) or (c) produces a normalized inverse operator that is not well defined. The quantitative consequences of the normalization — continuity of the resulting inverse on the zero-period subspace and its domain-dependent constants — belong to Part VI.

3.7 Change of the real origin

Proposition 3.14 (Translation covariance of the period pair). *Let $c \in \mathbb{R}$ and let $x' = x - c$, $y' = y$ be the translated planar coordinates, so that $q' = q - c$. For the same field, $\omega'_g = \omega_g$ and*

$$\theta'_g = \theta_g + c\omega_g, \quad \text{hence} \quad a'_\gamma = a_\gamma, \quad b'_\gamma = b_\gamma + ca_\gamma. \quad (3.16)$$

Consequently, for data attached to a sphere with centre $p = u + iv$, whose centre becomes $u' = u - c$, the pair

$$(a, ua + b) \quad \text{is invariant, since } u'a + b' = (u - c)a + (b + ca) = ua + b.$$

Proof. ω_g contains no x . Substituting $x = x' + c$ in (3.6),

$$\theta_g = \frac{1}{2}((x'P + yQ)dx' + (yP - x'Q)dy) + \frac{c}{2}(Pdx' - Qdy) = \theta'_g - c\omega_g,$$

which is (3.16). Integrate over γ . $\square$

Thus a and b are affine-monodromy coordinates, not two independently translation-invariant scalars; it is the combination $ua + b$, and not b itself, that is intrinsic to the sphere. This is exactly why the residue norm $\mathcal{R}_p(a, b)^2 = |ua + b|^2 + v^2|a|^2$ of Part VIII is translation-invariant, and why the energy coefficient it controls does not depend on a choice of real origin.

4 The holomorphic invariant, its exact image and kernel

The period criterion of Part II uses only first-order real data and no derivatives of g . For meromorphic and rational examples, and for every residue computation in Parts VII and IX, a second description is more efficient: a single holomorphic $\mathbb{H}_{\mathbb{C}}$ -valued function attached to g , whose ordinary complex residues are the two quaternionic periods and whose higher Laurent coefficients are the higher multipoles.

4.1 Definition, and why five formulas define one object

Definition 4.1 (The holomorphic invariant). For $g = P + IQ \in \mathcal{AM}(\Omega_U)$ put

$$\mathcal{H}_g(z) = -\frac{1}{2}(P(x, y) + yP_y(x, y)) - \frac{\iota y}{2}P_x(x, y). \quad (4.1)$$

Lemma 4.2 (One object, two formulas). On $\mathcal{AM}(\Omega_U)$,

$$\mathcal{H}_g = \frac{1}{2}(-P + yQ_x) - \frac{\iota y}{2}P_x. \quad (4.2)$$

Proof. The second Vekua equation $Q_x + P_y = 0$ gives $yQ_x = -yP_y$. $\square$

This is worth stating explicitly, because the two displays (4.1) and (4.2) occur in the literature as if they were definitions of two different invariants; they are the same function of g , and every identity below may be verified from either. Note also that $\mathcal{H}_g$ depends on Q only through a derivative that the Vekua system re-expresses in P : the invariant is determined by the single axial component P together with axial monogenicity.

4.2 Holomorphy, the second derivative, and the complex periods

Theorem 4.3 (The target-derived second derivative). Let $g \in \mathcal{AM}(\Omega_U)$. Then $\mathcal{H}_g$ is a single-valued holomorphic $\mathbb{H}_{\mathbb{C}}$ -valued function on U ; every local holomorphic stem F of a Fueter primitive of g satisfies

$$F'' = \mathcal{H}_g, \quad (4.3)$$

and for every closed piecewise C^1 curve $\gamma \subset U$,

$$a_\gamma(g) = \oint_\gamma \mathcal{H}_g(z)dz, \quad b_\gamma(g) = -\oint_\gamma z \mathcal{H}_g(z)dz. \quad (4.4)$$

In particular both of these $\mathbb{H}_{\mathbb{C}}$ -valued integrals lie in the real subspace $\mathbb{H} \subset \mathbb{H}_{\mathbb{C}}$.

Proof. First proof of holomorphy (structural). By Theorem 3.7 a primitive exists on every disk $\subset U$. With $F = A + \iota B$, $A = xC + V$, $B = yC$ and the derivative identities in the proof of Lemma 3.3,

$$F' = A_x + \iota B_x = C + \mathcal{M}_g, \quad \mathcal{M}_g := \frac{y}{2}Q - \frac{\iota y}{2}P. \quad (4.5)$$

Differentiating once more in x and using $C_x = -P/2$ and $Q_x = -P_y$,

$$F'' = -\frac{P}{2} + \frac{y}{2}Q_x - \frac{\iota y}{2}P_x = -\frac{1}{2}(P + yP_y) - \frac{\iota y}{2}P_x = \mathcal{H}_g.$$

The right-hand side is defined directly from g on all of U , and local primitives differ by $za + b$, whose second derivative vanishes; so the local identifications agree on overlaps and $\mathcal{H}_g$ is globally holomorphic and independent of every choice.

Second proof of holomorphy (direct, without any primitive). Write $\mathcal{H}_g = J_0 + \iota J_1$ with $J_0 = \frac{1}{2}(-P + yQ_x)$ and $J_1 = -\frac{1}{2}yP_x$, and verify the Cauchy–Riemann equations from (3.1) alone. First,

$$(J_0)_x - (J_1)_y = \frac{1}{2}(-P_x + yQ_{xx}) + \frac{1}{2}(P_x + yP_{xy}) = \frac{y}{2}(Q_{xx} + P_{xy}) = 0,$$

by differentiating $Q_x + P_y = 0$ in x . Second,

$$(J_0)_y + (J_1)_x = \frac{1}{2}(-P_y + Q_x + yQ_{xy}) - \frac{y}{2}P_{xx} = Q_x + \frac{y}{2}(Q_{xy} - P_{xx}) = 0,$$

using $-P_y = Q_x$ in the first bracket and, for the second, the x -derivative of $P_x - Q_y = 2Q/y$, namely $P_{xx} - Q_{xy} = 2Q_x/y$. This proof uses no primitive at all, so the global holomorphy of $\mathcal{H}_g$ is logically independent of the period theorem; it is the reason $\mathcal{H}_g$ may be used to *prove* statements about primitives rather than only to record them.

The complex period formulas. As identities of $\mathbb{H}_{\mathbb{C}}$ -valued one-forms on U ,

$$\mathcal{H}_g dz = \omega_g + d\mathcal{M}_g, \quad z\mathcal{H}_g dz = d(z\mathcal{M}_g) - \theta_g. \quad (4.6)$$

The first is (4.5) differentiated, since $dF' = F''dz = \mathcal{H}_g dz$ and $dC = \omega_g$. For the second, $F = zC + V$ gives $zF' - F = z\mathcal{M}_g - V$, and differentiating, $z\mathcal{H}_g dz = d(z\mathcal{M}_g - F) = d(z\mathcal{M}_g) - \theta_g$. The functions $\mathcal{M}_g$ and $z\mathcal{M}_g$ are globally single-valued on U , being written directly in terms of P, Q ; hence their differentials integrate to zero around any closed curve, and (4.6) yields (4.4). The final reality assertion is immediate from the original definition (3.10), in which the integrands are $\mathbb{H}$ -valued. $\square$

Remark 4.4 (The mechanism, and what it costs). The pair (4.6) is the exact mechanism converting the two real period integrals into complex contour integrals: $\mathcal{H}_g dz$ and ω_g differ by an *exact* $\mathbb{H}_{\mathbb{C}}$ -valued form, as do $-z\mathcal{H}_g dz$ and θ_g , and the correction $\mathcal{M}_g$ is built from g with no integration. The cost of the holomorphic description is two derivatives: ω_g, θ_g are algebraic in (P, Q) , whereas $\mathcal{H}_g$ involves P_x and P_y . For a numerically sampled field the real forms are therefore the stable objects, and $\mathcal{H}_g$ is the efficient one when a closed formula is available.

4.3 The exact image: the moment-real class

Theorem 4.3 says that the two contour integrals of $\mathcal{H}_g$ land in $\mathbb{H}$ although $\mathcal{H}_g$ itself is $\mathbb{H}_{\mathbb{C}}$ -valued. This is a genuine restriction, and it turns out to be the only one.

Definition 4.5 (The moment-real class). Let

$$\mathcal{O}_{\text{mom}}(U) = \left\{ h : U \rightarrow \mathbb{H}_{\mathbb{C}} \text{ holomorphic} : \oint_{\gamma} h dz \in \mathbb{H} \text{ and } -\oint_{\gamma} z h dz \in \mathbb{H} \text{ for every closed } \gamma \subset U \right\}. \quad (4.7)$$

The name records that the two lowest complex *moments* of h along every cycle are real-quaternionic. It does *not* say that h is $\mathbb{H}$ -valued, nor that h is κ -symmetric; on a simply connected U the condition is vacuous and $\mathcal{O}_{\text{mom}}(U)$ is the full space $\mathcal{O}(U; \mathbb{H}_{\mathbb{C}})$.

Theorem 4.6 (Exact image and exact kernel of the invariant). *The map $g \mapsto \mathcal{H}_g$ sends $\mathcal{AM}(\Omega_U)$ onto $\mathcal{O}_{\text{mom}}(U)$, and its kernel is exactly the eight-real-parameter family*

$$\Delta_4 \mathcal{I}(\iota(za + b))(x + Iy) = -\frac{2a}{y} - I \frac{2(xa + b)}{y^2}, \quad a, b \in \mathbb{H}. \quad (4.8)$$

Thus $\ker(g \mapsto \mathcal{H}_g)$ has real dimension exactly eight on a product domain, and each of its members is itself a global Fueter image, with global stem $\iota(za + b)$.

Proof. Necessity of the moment condition is (4.4).

Surjectivity. Let $h \in \mathcal{O}_{\text{mom}}(U)$. Integrate twice on the universal cover $\tilde{U}$ to obtain a holomorphic F with $F'' = h$. Continuation around a lifted loop γ leaves F'' unchanged, so it changes F by $za + b$ with $a, b \in \mathbb{H}_{\mathbb{C}}$ a priori. Integrating $dF' = h dz$ and $d(zF' - F) = z h dz$ identifies these coefficients as

$$a = \oint_{\gamma} h dz, \quad b = - \oint_{\gamma} z h dz,$$

which lie in $\mathbb{H}$ by hypothesis. By Corollary 3.9 the local fields $\Delta_4 \mathcal{I}(F)$ are therefore unchanged by every continuation; they glue to a single-valued $g \in \mathcal{AM}(\Omega_U)$, and $\mathcal{H}_g = F'' = h$ locally, hence globally.

Kernel. If $\mathcal{H}_g = 0$ then every local primitive stem has $F'' = 0$, so $F = zc + d$ with $c, d \in \mathbb{H}_{\mathbb{C}}$. Write $c = a_0 + \iota a$ and $d = b_0 + \iota b$ with $a_0, a, b_0, b \in \mathbb{H}$. The term $za_0 + b_0$ is killed by the Fueter map (Corollary 3.9); for the rest, $\iota(za + b)$ has components $A = -ya$, $B = xa + b$, and (3.2) gives

$$P = \frac{2A_y}{y} = -\frac{2a}{y}, \quad Q = \frac{2B_y}{y} - \frac{2B}{y^2} = -\frac{2(xa + b)}{y^2},$$

which is (4.8). Conversely each such field satisfies (3.1) and has $P_x = 0$ and $P + yP_y = 0$, so $\mathcal{H}_g = 0$. The parameters are determined by the field — a from P , then b from Q — so local choices agree on overlaps and are global on connected U ; this also shows the family is exactly eight-real-dimensional. $\square$

Remark 4.7 (The invariant is not injective, and where the kernel is prohibited). Theorem 4.6 identifies $\mathcal{AM}(\Omega_U)$ with a constrained holomorphic function *plus eight real parameters*; $\mathcal{H}_g = 0$ does not imply $g = 0$. Equivalently, solving (4.1) directly gives $P = c/y$, $Q = (cx + d)/y^2$ with $c = -2a$, $d = -2b$ constant on connected U . Three points deserve emphasis.

- (a) There is no conflict with Theorem 3.7: the fields (4.8) have both periods zero, being global Fueter images. The invariant loses exactly the part of g that carries no obstruction.
- (b) The special case $a = 0$, $b = 1$ is the stem $F = \iota$, whose induced function is $f(q) = I$ with $\Delta_4 f = -2I/y^2 \neq 0$. This is the standing counterexample to enlarging $\text{Aff}_{\mathbb{H}}$ to complex-affine stems.
- (c) On a slice domain meeting $\mathbb{R}$ the kernel is *vacuous*: every member of (4.8) blows up like y^{-1} or y^{-2} as $y \rightarrow 0$, so smooth continuation through a real interval forces $a = b = 0$. On such domains $g \mapsto \mathcal{H}_g$ is injective. The reflected statement is taken up in Part V.

Remark 4.8 (A constructive test for proposed holomorphic encodings). Theorem 4.6 is the precise sense in which axially monogenic fields are “complexified”. It is also a usable test: an arbitrary holomorphic $\mathbb{H}_{\mathbb{C}}$ -valued function is *not* the invariant of an axially monogenic field. Arbitrary complex residues need not satisfy (4.7), and Theorem 4.9 below computes exactly which ones do at an isolated nonreal point.

4.4 Residues, multipoles, and the reality constraint

Fix $p = u + \iota v$ with $v > 0$, let $0 < R < v$, and let g be axially monogenic on the circularization of the punctured disk $0 < |z - p| < R$ — that is, on a punctured tubular neighbourhood of the sphere $S_p = u + v\mathbb{S}$. Recall from Part VII that the *affine spherical residue* of g at S_p is the pair

$$\text{Res}_{S_p}^{\text{aff}}(g) = (a, b) = \left(\oint_{|z-p|=s} \omega_g, \oint_{|z-p|=s} \theta_g \right), \quad 0 < s < R,$$

independent of s by closedness, and that Part VII’s hypothesis-free normal form writes g uniquely as a regular field, plus the logarithmic mode $G_{p;a,b}$ of Part IV, plus $\Delta_4 \mathcal{I}(\sum_{k \geq 1} (z - p)^{-k} c_k)$ with $c_k \in \mathbb{H}_{\mathbb{C}}$. We now recover *all* of these data from $\mathcal{H}_g$ by ordinary contour integration.

Theorem 4.9 (Contour extraction of every multipole, and the reality constraint). *Let $\mathcal{H}_g(z) = \sum_{n \in \mathbb{Z}} h_n(z-p)^n$, $h_n \in \mathbb{H}_{\mathbb{C}}$, be the Laurent expansion on $0 < |z-p| < R$. Then, with the circle positively oriented,*

$$a = 2\pi\iota h_{-1}, \quad b = -2\pi\iota(ph_{-1} + h_{-2}), \quad (4.9)$$

$$c_k = \frac{h_{-k-2}}{k(k+1)} = \frac{1}{2\pi\iota k(k+1)} \oint_{|z-p|=s} (z-p)^{k+1} \mathcal{H}_g(z) dz, \quad k \geq 1. \quad (4.10)$$

Equivalently, a and $-b$ are $2\pi\iota$ times the ordinary complex residues at p of $\mathcal{H}_g$ and of $z\mathcal{H}_g$, computed componentwise in $\mathbb{H}_{\mathbb{C}}$; both results lie in $\mathbb{H}$ although the individual coefficients h_n do not. Moreover the two lowest negative coefficients are not independent:

$$\boxed{h_{-1} = \frac{\iota}{v} \operatorname{Re}_{\mathbb{C}}(h_{-2})} \quad (4.11)$$

Explicitly, writing $h_{-2} = c + \iota d$ with $c, d \in \mathbb{H}$,

$$a = -\frac{2\pi}{v} c, \quad b = 2\pi d + \frac{2\pi u}{v} c. \quad (4.12)$$

Conversely, a holomorphic h on the punctured disk is the invariant of some axially monogenic field there if and only if its Laurent coefficients satisfy (4.11); the remaining coefficients are unrestricted in $\mathbb{H}_{\mathbb{C}}$.

Proof. Formulas (4.9) are the componentwise complex residue theorem applied to (4.4), the punctured disk having a single homology generator. Only the Laurent expansion is used, so they hold also when $\mathcal{H}_g$ has an essential singularity at p .

For (4.10), apply $g \mapsto \mathcal{H}_g$ to the normal form of Part VII. The regular summand has a holomorphic invariant. The mode $G_{p;a,b}$ has the rational invariant $\frac{1}{2\pi\iota} \left(\frac{a}{z-p} - \frac{pa+b}{(z-p)^2} \right)$ of Part IV, which contributes nothing below index -2 . Finally, by (4.3) applied to the stem $(z-p)^{-k} c_k$,

$$\mathcal{H}_g[\Delta_4 \mathcal{I}((z-p)^{-k} c_k)] = \frac{d^2}{dz^2} ((z-p)^{-k} c_k) = k(k+1)(z-p)^{-k-2} c_k,$$

so the coefficient of $(z-p)^{-k-2}$ in $\mathcal{H}_g$ is $k(k+1)c_k$ for $k \geq 1$; extracting it by the Cauchy coefficient formula gives the displayed contour integral.

For (4.11): by (4.9) and $a \in \mathbb{H}$, the first formula forces $h_{-1} = \iota\alpha$ with $\alpha = -a/(2\pi) \in \mathbb{H}$. Write $h_{-2} = c + \iota d$ with $c, d \in \mathbb{H}$. Since $p = u + \iota v$,

$$ph_{-1} + h_{-2} = \iota u\alpha - v\alpha + c + \iota d = (c - v\alpha) + \iota(d + u\alpha).$$

The requirement $b = -2\pi\iota(ph_{-1} + h_{-2}) \in \mathbb{H}$ kills the $\mathbb{H}$ -component of $ph_{-1} + h_{-2}$, that is, $c = v\alpha$. Hence $\alpha = c/v$, so $h_{-1} = \iota c/v = (\iota/v) \operatorname{Re}_{\mathbb{C}}(h_{-2})$, and $a = -2\pi\alpha = -(2\pi/v)c$, while $b = -2\pi\iota \cdot \iota(d + u\alpha) = 2\pi d + (2\pi u/v)c$. This is (4.12).

Conversely, on a punctured disk every loop is homologous to a multiple of one circle, so the two conditions defining $\mathcal{O}_{\text{mom}}$ reduce to $a \in \mathbb{H}$ and $b \in \mathbb{H}$ for that circle, which by the computation just made is exactly (4.11). Theorem 4.6 then produces an axially monogenic g with $\mathcal{H}_g = h$. $\square$

Corollary 4.10 (The sharp vanishing criterion at a punctured tube). *For g axially monogenic on a punctured tube around S_p , the following are equivalent:*

- (i) g has a single-valued slice-regular Fueter primitive on the punctured tube;
- (ii) $(a, b) = (0, 0)$;

- (iii) $h_{-2} = 0$;
- (iv) $h_{-1} = h_{-2} = 0$.

In particular $\mathcal{H}_g$ can never have an isolated pole of order exactly one at a nonreal p , and a single coefficient $h_{-2} \in \mathbb{H}_{\mathbb{C}}$ carries all eight real obstruction coordinates.

Proof. (i) $\Leftrightarrow$ (ii) is Theorem 3.7 on the punctured disk, whose first homology is generated by one circle. (ii) $\Leftrightarrow$ (iv) is (4.9), since $2\pi\iota$ is invertible in $\mathbb{H}_{\mathbb{C}}$. (iii) $\Rightarrow$ (iv) is (4.11), and (iv) $\Rightarrow$ (iii) is trivial. For the last two assertions: a simple pole would mean $h_{-2} = 0$ and $h_{-1} \neq 0$, which (4.11) forbids; and (4.12) is a real-linear bijection $h_{-2} \mapsto (a, b)$ from $\mathbb{H}_{\mathbb{C}} \cong \mathbb{R}^8$ onto $\mathbb{H}^2 \cong \mathbb{R}^8$, with inverse $c = -va/(2\pi)$, $d = (b + ua)/(2\pi)$ — note that it is $ua + b$, in accordance with Proposition 3.14, that appears. $\square$

Remark 4.11 (A true but redundant form of the criterion). Criterion (iv) — the conjunction $h_{-1} = h_{-2} = 0$, with the two polar coefficients presented as independent data — is the form in which the vanishing criterion is most often stated, and it is correct. The reality constraint (4.11) shows it is redundant: $h_{-2} = 0$ alone already forces $h_{-1} = 0$ and hence both periods to vanish. The sharp form is (iii). The same constraint shows that the double-pole coefficient h_{-2} is free in $\mathbb{H}_{\mathbb{C}}$ while the simple-pole coefficient is not free at all, so a classification that treats (h_{-1}, h_{-2}) as sixteen free real parameters overcounts the local obstruction by a factor of two.

Remark 4.12 (What the constraint does *not* make redundant). Constraint (4.11) is specific to an *isolated nonreal parameter point*, where the local first homology is cyclic and the two periods are attached to the same circle. It does not make the two one-forms of Part II redundant. On a general U an arbitrary loop still carries two independent quaternionic periods — Part IV realizes every pair $(a, b) \in \mathbb{H}^2$ at a single hole — and testing the ordinary residue h_{-1} alone still fails to detect the constant obstruction, since $h_{-1} = 0$ with $h_{-2} = \iota d \neq 0$ gives $a = 0$ and $b = 2\pi d \neq 0$. What (4.11) says is only that the eight real coordinates of the pair (a, b) are repackaged, at one isolated sphere, into the eight real coordinates of one element of $\mathbb{H}_{\mathbb{C}}$.

4.5 Faithful detection of singularities

Corollary 4.13 (Removability is a statement about $\mathcal{H}_g$, with no growth hypothesis). *Let g be axially monogenic on a punctured tube around S_p . Then g extends to an axially monogenic field across S_p if and only if $\mathcal{H}_g$ extends holomorphically across p . No growth, integrability, or meromorphy hypothesis is needed in either direction.*

Proof. If g extends, (4.1) defines $\mathcal{H}_g$ on the full disk and Theorem 4.3 makes it holomorphic there. Conversely, suppose $\mathcal{H}_g$ extends holomorphically to the disk $|z - p| < R$. Choose a holomorphic double primitive F_0 of the extended function on that (simply connected) disk and put $g_0 = \Delta_4 \mathcal{I}(F_0)$, which is axially monogenic on the whole circularized disk, the disk staying in $\{y > 0\}$. Then $g - g_0$ has vanishing invariant on the punctured disk, so by Theorem 4.6 it equals $\Delta_4 \mathcal{I}(\iota(za + b))$ for some $a, b \in \mathbb{H}$; that field is smooth across the centre. Hence $g = g_0 + \Delta_4 \mathcal{I}(\iota(za + b))$ extends. $\square$

Remark 4.14 (Scope). Corollary 4.13 is a clean equivalence, but it transfers the problem rather than solving it: deciding whether $\mathcal{H}_g$ extends still requires information about g . What the classification of Part VII adds is the dictionary between hypotheses on g — local integrability, a pointwise growth bound, finite energy — and the pole order of $\mathcal{H}_g$. Two boundary cases should be kept in view here. First, $\mathcal{H}_g$ may have an essential singularity at p : (4.9) and hence Corollary 4.10 remain valid, so a field with an essential singularity can have both periods zero and a single-valued primitive on the punctured tube while being wildly nonremovable. Second, the statements of this part are proved in the *axial* class and at a *nonreal* p ; the constants in (4.12) contain $1/v$ and degenerate as $v \rightarrow 0$, and no claim is made here about real points, about nonaxial Cauchy–Fueter fields, or about uniformity as the sphere collapses onto the real axis.

5 Logarithmic modes and the split exact sequence

Part II produced a criterion: a single-valued slice-regular Fueter primitive exists on Ω_U exactly when the two closed one-forms ω_g and θ_g have vanishing periods on U . A criterion by itself does not say that every formally admissible period vector is attained, nor does it produce a normal form for the failure. This part supplies both. We exhibit, for each pair $(a, b) \in \mathbb{H}^2$ and each puncture p , a single explicit globally single-valued axially monogenic field whose period pair is (a, b) ; we assemble these fields into a right inverse of the period map; and we conclude that the cokernel of the Fueter map on a base with m independent holes has real dimension exactly $8m$, with the direct-sum decomposition realized by concrete functions rather than by an abstract quotient.

Throughout, $U \subset \mathbb{C}^+$ is a connected open set, $\Omega_U = \{x + Iy : x + \iota y \in U, I \in \mathbb{S}\}$ is its circularization, $\mathcal{SR}(\Omega_U)$ is the space of slice-regular functions induced by holomorphic stems on U , and $\mathcal{AM}(\Omega_U)$ is the space of axially monogenic (axially Fueter-regular) fields $g = P + IQ$ on Ω_U . All quaternionic coefficients multiply on the right, and ι is the central complex unit of $\mathbb{H}_{\mathbb{C}}$, never an element of $\mathbb{S}$.

5.1 The two-parameter logarithmic mode

Fix $p = u + \iota v \in \mathbb{C} \setminus U$ and $a, b \in \mathbb{H}$. On any simply connected patch of $\mathbb{C} \setminus \{p\}$ choose a branch of $\log(z - p)$ and set

$$F_{p;a,b}(z) = \frac{za + b}{2\pi\iota} \log(z - p), \quad a, b \in \mathbb{H}. \quad (5.1)$$

This is a holomorphic $\mathbb{H}_{\mathbb{C}}$ -valued stem on the patch. Two branches differ by $2\pi\iota n$ with $n \in \mathbb{Z}$, so the two stems differ by $n(za + b)$, which is a complex-affine stem with *quaternionic* coefficients; by the affine-kernel corollary of Part II (Corollary 3.9) it is annihilated by $\Delta_4 \mathcal{I}(\cdot)$. Hence the local fields $\Delta_4 \mathcal{I}(F_{p;a,b})$ agree on overlaps and define one single-valued field.

Proposition 5.1 (Branch-free formula, potentials, and period normalization). *Let $p = u + \iota v$, let $\xi = x - u$ and $\eta = y - v$ be the offsets from the centre, and let $\rho = |z - p| = (\xi^2 + \eta^2)^{1/2}$. Write $X = xa + b$. Then the local fields $\Delta_4 \mathcal{I}(F_{p;a,b})$ patch to a single smooth axially monogenic field*

$$G_{p;a,b} = \Delta_4 \mathcal{I}\left(\frac{za + b}{2\pi\iota} \log(z - p)\right) \in \mathcal{AM}(\Omega_U) \quad \text{for every } U \subset \mathbb{C}^+ \setminus \{p\},$$

whose axial components are given by the branch-free expressions

$$P_{p;a,b} = \frac{1}{\pi y} \left[\frac{X\xi + ya\eta}{\rho^2} + a \log \rho \right], \quad (5.2)$$

$$Q_{p;a,b} = \frac{ya\xi - X\eta}{\pi y \rho^2} + \frac{X \log \rho}{\pi y^2}. \quad (5.3)$$

Its two potentials, computed from any branch $\log(z - p) = \log \rho + \iota\vartheta$, are

$$C = \frac{1}{2\pi} \left(a\vartheta - \frac{X \log \rho}{y} \right), \quad V = \frac{1}{2\pi} \left(b\vartheta + \left(ya + \frac{xX}{y} \right) \log \rho \right). \quad (5.4)$$

For every closed piecewise- C^1 curve $\gamma \subset U$,

$$\left(\oint_{\gamma} \omega_{G_{p;a,b}}, \oint_{\gamma} \theta_{G_{p;a,b}} \right) = \text{Ind}(\gamma, p) (a, b). \quad (5.5)$$

The map $(a, b) \mapsto G_{p;a,b}$ is right $\mathbb{H}$ -linear. If $\text{Im}_{\mathbb{C}} p < 0$, a branch of $\log(z - p)$ exists on all of $\mathbb{C}^+$, so $G_{p;a,b}$ is defined on all of $\Omega_{\mathbb{C}^+}$ and both of its periods vanish there.

Proof. Write $\log(z - p) = \log \rho + \iota \vartheta$ on a branch. Separating (5.1) into $F = A + \iota B$ and using $1/\iota = -\iota$,

$$A = \frac{X\vartheta + ya \log \rho}{2\pi}, \quad B = \frac{ya\vartheta - X \log \rho}{2\pi},$$

where the coefficient $za + b$ has been split as $(xa + b) + \iota ya = X + \iota ya$. The change of potentials $C = B/y$, $V = A - xB/y$ of Part II, (3.3), gives (5.4) at once; in V the multivalued terms combine as $X\vartheta - xa\vartheta = b\vartheta$.

For the components, use $P = -2C_x$ and $Q = 2C_y$, which is the content of $dC = \omega_g$ in (3.7). With $\vartheta_x = -\eta/\rho^2$, $\vartheta_y = \xi/\rho^2$, $(\log \rho)_x = \xi/\rho^2$, $(\log \rho)_y = \eta/\rho^2$ and $X_x = a$,

$$C_x = \frac{1}{2\pi} \left[-\frac{a\eta}{\rho^2} - \frac{a \log \rho}{y} - \frac{X\xi}{y\rho^2} \right], \quad C_y = \frac{1}{2\pi} \left[\frac{a\xi}{\rho^2} - \frac{X\eta}{y\rho^2} + \frac{X \log \rho}{y^2} \right],$$

which are exactly (5.2)–(5.3). Every term involves only the real logarithm of a positive number: the argument ϑ has cancelled, so (5.2)–(5.3) are unambiguous functions on U , and they are the promised single-valued representative of the branchwise construction.

The periods are read off from (5.4). Continuing along γ returns $\log \rho$ to its initial value and increases ϑ by $2\pi \operatorname{Ind}(\gamma, p)$. Hence C gains $\operatorname{Ind}(\gamma, p) a$ and V gains $\operatorname{Ind}(\gamma, p) b$, and since $\oint_\gamma \omega_g$ and $\oint_\gamma \theta_g$ are precisely these increments, (5.5) follows. (Equivalently, continuation of (5.1) changes the stem by $\operatorname{Ind}(\gamma, p)(za + b)$, and the affine-monodromy corollary of Part II, Corollary 3.10, identifies the period pair.) Right linearity is visible in (5.1). The final assertion holds because a branch of the logarithm defined on the whole simply connected half-plane makes $F_{p;a,b}$ a genuine single-valued stem there, so $G_{p;a,b}$ is itself a global Fueter image. $\square$

Remark 5.2 (One formula, nine sources). Formulas (5.2)–(5.3) are the single branch-free expression that the source manuscripts wrote in nine different variable conventions. Each of them either splits the mode into two one-parameter kernels carrying respectively the slope and the constant period, or writes the two-parameter family directly; the two-parameter form is adopted here because it makes (5.5) a single line and removes an inversion hazard present in the split notations, where the letter A carries the slope charge in one convention and the constant charge in another. The specializations are recovered by $G_{p;a,b} = G_{p;1,0} a + G_{p;0,1} b$. The potentials (5.4) give a second, independent check of *both* period normalizations, since C gains a and V gains b on one circuit; they are also the form in which the mode should be evaluated numerically near the real axis, where (5.3) suffers cancellation.

Corollary 5.3 (Holomorphic encoding of a mode). *With $\mathcal{H}_g$ the holomorphic invariant of Part III,*

$$\mathcal{H}_{G_{p;a,b}}(z) = \left(\frac{za + b}{2\pi\iota} \log(z - p) \right)' = \frac{1}{2\pi\iota} \left(\frac{a}{z - p} - \frac{pa + b}{(z - p)^2} \right). \quad (5.6)$$

Proof. Theorem 4.3 of Part III proves $\mathcal{H}_g = F''$ for every local Fueter primitive stem F of g , and (5.1) is such a stem for $G_{p;a,b}$ on every patch; the right-hand side of (5.6) is single-valued, so the identity is global. Differentiating (5.1) twice,

$$\left(\frac{za + b}{2\pi\iota} \log(z - p) \right)' = \frac{a \log(z - p)}{2\pi\iota} + \frac{za + b}{2\pi\iota(z - p)},$$

and one more differentiation gives $\frac{2a}{2\pi\iota(z - p)} - \frac{za + b}{2\pi\iota(z - p)^2}$, which equals the stated expression because $za + b = (z - p)a + (pa + b)$. $\square$

Formula (5.6) is the point of contact between this part and Part III: the Laurent coefficients of $\mathcal{H}_{G_{p;a,b}}$ at p are $h_{-1} = a/(2\pi\iota)$ and $h_{-2} = -(pa + b)/(2\pi\iota)$, and one checks directly that they satisfy the reality constraint $h_{-1} = (\iota/v) \operatorname{Re}_{\mathbb{C}}(h_{-2})$ of Theorem 4.9. In particular the eight real parameters of a mode are already carried by the single $\mathbb{H}_{\mathbb{C}}$ coefficient h_{-2} .

Remark 5.4 (The mode functions themselves are not claimed to be new). For $p = \iota$ the reflected logarithm of Part V has derivative $\pi^{-1}(1+z^2)^{-1}$, so a local primitive is $\pi^{-1} \arctan z$ up to a real constant, and the functions with local primitives $(2\pi)^{-1} \arctan z$ and $(2\pi)^{-1} z \arctan z$ are the spherical Cauchy kernels N^+ and N^- of the inverse-Fueter literature. In that normalization $G_{\iota;0,1}^{\text{sym}} = 2N^+$ and $G_{\iota;1,0}^{\text{sym}} = 2N^-$. We do not claim the invention of these kernels, and we record that the source manuscripts of this merge disagree about which published article is the primary locus for them: some cite a 2011 paper on sphere-integrated Cauchy kernels, others a 2014 example on inverse Fueter kernels. That attribution is not settled here, and the identification with $N^\pm$ is asserted only with the normalization just spelled out; see the literature paragraph of Part XI. What is proposed here is their normalization by *two* global periods, the resulting completeness statement (5.8), and the classification and energy laws of the later parts.

5.2 Finite winding bases

The splitting theorem needs a topological hypothesis. We state it as a hypothesis about cycles and winding numbers, not about the regularity of the complement, so that no boundary smoothness is imposed on an arbitrary closed set.

Definition 5.5 (Finite winding basis). A connected open $U \subset \mathbb{C}^+$ has a *finite winding basis* of size m if there are points $p_1, \dots, p_m \in \mathbb{C}^+ \setminus U$ and piecewise- C^1 cycles $\gamma_1, \dots, \gamma_m \subset U$ whose homology classes form a $\mathbb{Z}$ -basis of $H_1(U; \mathbb{Z})$ and satisfy

$$\text{Ind}(\gamma_j, p_k) = \delta_{jk}, \quad 1 \leq j, k \leq m. \quad (5.7)$$

The case $m = 0$ means $H_1(U; \mathbb{Z}) = 0$.

Every upper domain obtained by deleting m points from a simply connected upper domain has such a basis, as does one obtained by deleting m disjoint closed Jordan regions, with one point chosen inside each. The hypothesis is used only through (5.7); nothing below requires the complementary set to have rectifiable or even locally connected boundary.

5.3 The split range theorem

Theorem 5.6 (Split exact sequence and the $8m$ cokernel). *Let $U \subset \mathbb{C}^+$ be connected with a finite winding basis $(\gamma_j, p_j)_{j=1}^m$, and define the period map*

$$\text{Per}(g) = (a_{\gamma_j}(g), b_{\gamma_j}(g))_{j=1}^m \in \mathbb{H}^{2m}, \quad a_\gamma(g) = \oint_\gamma \omega_g, \quad b_\gamma(g) = \oint_\gamma \theta_g.$$

Then the sequence of right quaternionic vector spaces

$$0 \longrightarrow \text{Aff}_{\mathbb{H}} \longrightarrow \mathcal{SR}(\Omega_U) \xrightarrow{\Delta_4 \mathcal{I}(\cdot)} \mathcal{AM}(\Omega_U) \xrightarrow{\text{Per}} \mathbb{H}^{2m} \longrightarrow 0 \quad (5.8)$$

is exact, where the first map is $(a, b) \mapsto (q \mapsto qa + b)$. The final surjection has the explicit right $\mathbb{H}$ -linear section

$$S((a_j, b_j)_{j=1}^m) = \sum_{j=1}^m G_{p_j; a_j, b_j}, \quad \text{Per} \circ S = \text{id}_{\mathbb{H}^{2m}}. \quad (5.9)$$

Consequently

$$\mathcal{AM}(\Omega_U) = \Delta_4 \mathcal{I}(\mathcal{SR}(\Omega_U)) \oplus S(\mathbb{H}^{2m}), \quad \dim_{\mathbb{R}} \frac{\mathcal{AM}(\Omega_U)}{\Delta_4 \mathcal{I}(\mathcal{SR}(\Omega_U))} = 8m, \quad (5.10)$$

every $g \in \mathcal{AM}(\Omega_U)$ has the unique decomposition

$$g = \Delta_4 \mathcal{I}(F) + \sum_{j=1}^m G_{p_j; a_{\gamma_j}(g), b_{\gamma_j}(g)}, \quad (5.11)$$

with F unique modulo a stem $za + b$, and $\Pi = \text{id} - S \circ \text{Per}$ is a right-linear projection of $\mathcal{AM}(\Omega_U)$ onto the space of targets possessing a single-valued global Fueter primitive.

Proof. Exactness at $\mathcal{SR}(\Omega_U)$ is Corollary 3.9 of Part II: a slice-regular f induced by a stem F satisfies $\Delta_4 f = 0$ on a connected base exactly when $dC = dV = 0$, i.e. when $F = za + b$ with $a, b \in \mathbb{H}$; the coefficients are quaternionic and not merely elements of $\mathbb{H}_{\mathbb{C}}$, which is what makes the sequence's left-hand term eight-real-dimensional rather than sixteen.

Exactness at $\mathcal{AM}(\Omega_U)$ is the global period criterion Theorem 3.7 of Part II, combined with the basis hypothesis. The period functionals a_γ and b_γ depend only on the homology class of γ , because ω_g and θ_g are closed; so $\text{Per}(g) = 0$ forces both periods to vanish on every cycle of U , and the criterion produces a single-valued primitive. Conversely the periods of a Fueter image vanish, since $\omega_g = dC$ and $\theta_g = dV$ globally in that case.

Surjectivity of Per and the section identity follow from Proposition 5.1 together with (5.7): the k th period pair of $S((a_j, b_j)_j)$ is $\sum_j \text{Ind}(\gamma_k, p_j)(a_j, b_j) = (a_k, b_k)$. Because γ_k lies in U and $p_j \notin U$, each mode in the sum is regular on Ω_U , so the sum is a legitimate element of $\mathcal{AM}(\Omega_U)$.

For the direct sum, write $g = (g - S\text{Per}g) + S\text{Per}g$; the first summand lies in $\ker \text{Per} = \Delta_4 \mathcal{I}(\mathcal{SR})$ by the previous paragraph. If a vector $\mathbf{w} \in \mathbb{H}^{2m}$ has $S\mathbf{w} \in \ker \text{Per}$ then $\mathbf{w} = \text{Per}S\mathbf{w} = 0$, so the sum is direct. Applying Per to any decomposition of the shape (5.11) recovers the mode coefficients, which proves their uniqueness; the remaining ambiguity in F is the affine kernel. Finally $\mathbb{H}^{2m}$ has real dimension $8m$, and $\Pi^2 = \Pi$ because $\text{Per}S = \text{id}$. $\square$

Remark 5.7 (What is canonical and what is a choice). The quotient $\mathcal{AM}(\Omega_U)/\Delta_4 \mathcal{I}(\mathcal{SR}(\Omega_U))$ and the period coordinates on it are canonical: they depend only on U and on the chosen homology basis, not on the auxiliary points. The complement $S(\mathbb{H}^{2m})$ is not canonical. Moving p_j to another point p'_j of the same complementary component changes the representative $G_{p_j; a, b}$ to $G_{p'_j; a, b}$, and the difference has zero periods on every cycle of U — the two points have the same winding numbers against $\gamma_1, \dots, \gamma_m$ — hence is itself a global Fueter image. So the splitting moves but the decomposition class does not. Continuity of Π and S for compact-open convergence, and the corresponding closed range statement, are proved in Part VI; nothing in Theorem 5.6 asserts boundedness in any weighted or Hilbert norm.

Example 5.8 (Neither obstruction can be dropped). Let $U = \mathbb{C}^+ \setminus \{p\}$ and let γ be a small positively oriented circle about p . By (5.5),

$$\text{Per}(G_{p;0,1}) = (0, 1), \quad \text{Per}(G_{p;1,0}) = (1, 0).$$

Thus $G_{p;0,1}$ has an exact first form, so the slope potential C exists globally on U and the first of the two classical integrations succeeds; nevertheless no global slice-regular Fueter primitive exists, because the second period is 1. The field $G_{p;1,0}$ fails already at the first integration. Neither test implies the other, and neither may be omitted. Right multiplication by $1, i, j, k$ turns each of the two families into four real directions, giving the eight independent real directions counted by (5.10) at a single hole. This also shows that the second obstruction is not a residue of a badly chosen first primitive: by the elimination identity $\eta_C - d(xC) = \theta_g$ of Part II (Proposition 3.4), θ_g is computed from the target alone, and it is nonexact here for every choice of C .

6 Reflection: domains meeting the real axis

Everything so far took place over a base $U \subset \mathbb{C}^+$, so the circularized domain Ω_U avoided the real axis of $\mathbb{H}$. One might suspect that the obstruction of Theorem 5.6 is an artifact of that avoidance — that a domain which contains real points, and whose slice functions are therefore rigid there, would have no room for a period. It does not. The obstruction survives verbatim, with a count that is *not* the first Betti number of the base and is completely insensitive to holes that conjugation fixes.

6.1 Symmetric bases, parity, and invariance of the two forms

Let $D \subset \mathbb{C}$ be open, connected, invariant under $\tau(z) = \bar{z}$, and meeting $\mathbb{R}$. Its circularization

$$\Omega_D = \{x + Iy : x + iy \in D, I \in \mathbb{S}\} \subset \mathbb{H}$$

contains the real interval $D \cap \mathbb{R}$. A holomorphic stem on D is required to satisfy the reflection-reality condition $F(\bar{z}) = \kappa(F(z))$, where κ is the conjugation of $\mathbb{H}_{\mathbb{C}}$ fixing $\mathbb{H}$; equivalently A is even and B is odd in y . This is the usual notion of a slice-regular function on an axially symmetric slice domain.

For $g \in \mathcal{AM}(\Omega_D)$ the axial components are recovered from a single slice by

$$P(x, y) = \frac{g(x + Iy) + g(x - Iy)}{2}, \quad Q(x, y) = -\frac{I}{2}(g(x + Iy) - g(x - Iy)) \quad (6.1)$$

for any fixed $I \in \mathbb{S}$, which exhibits P as even and Q as odd in y and smooth across $y = 0$.

Lemma 6.1 (The two forms extend and are τ -invariant). *For $g \in \mathcal{AM}(\Omega_D)$ the one-forms $\omega_g = \frac{1}{2}(-P dx + Q dy)$ and $\theta_g = \frac{1}{2}((xP + yQ)dx + (yP - xQ)dy)$ are smooth on all of D , closed on all of D , and satisfy*

$$\tau^* \omega_g = \omega_g, \quad \tau^* \theta_g = \theta_g. \quad (6.2)$$

Proof. Neither form contains a division by y , so smoothness across the axis follows from (6.1). Closedness is Lemma 3.3(i) of Part II, valid where the Vekua system holds, i.e. for $y \neq 0$; the $dx \wedge dy$ coefficients $(Q_x + P_y)/2$ and $\frac{1}{2}\{y(P_x - Q_y) - 2Q - x(Q_x + P_y)\}$ are continuous on D and vanish off the axis, hence vanish on it by continuity. For (6.2): under τ one has $dx \mapsto dx$ and $dy \mapsto -dy$, while P is even and Q is odd. The dx coefficients $-P/2$ and $(xP + yQ)/2$ are therefore even, and the dy coefficients $Q/2$ and $(yP - xQ)/2$ are odd; each form is thus preserved. $\square$

Lemma 6.2 (The upper meridian is connected). *If D is connected, open and τ -invariant, then $U = D \cap \mathbb{C}^+$ is connected.*

Proof. Let $z_0, z_1 \in U$ and join them by a path in D . Fold the path upward by replacing $x + iy$ with $x + i|y|$; the result is a continuous path in $D \cap \{y \geq 0\}$, because D is τ -invariant. Its image is compact and therefore has positive distance δ from the closed complement of D . Translating the whole folded path by $i\delta/2$ keeps it in D and places it strictly above $\mathbb{R}$. Two short vertical segments, from z_0 and z_1 up to the translated endpoints, stay in $D \cap \{y > 0\}$ for δ small enough. The concatenation is a path in U . $\square$

6.2 The criterion across the real axis

Theorem 6.3 (Real-axis period criterion). *Let D be connected, open, τ -invariant and meeting $\mathbb{R}$, and let $g \in \mathcal{AM}(\Omega_D)$. There exists $f \in \mathcal{SR}(\Omega_D)$ with $\Delta_4 f = g$ if and only if ω_g and θ_g have vanishing periods on D , equivalently if and only if they have vanishing periods on the upper meridian $U = D \cap \mathbb{C}^+$. Such an f is unique modulo $q \mapsto qa + b$, $a, b \in \mathbb{H}$. No finite-topology hypothesis and no boundary regularity hypothesis is used.*

Proof. Necessity is the restriction of the criterion of Part II to Ω_U , together with the observation that for a global primitive the potentials $C = B/y$ and $V = A - xB/y$ extend smoothly and evenly across $y = 0$: locally $B(x, y)/y = \int_0^1 B_y(x, ty) dt$ because B is odd, and the identities (3.7) then hold on all of D by continuity. This also gives uniqueness, since a difference of two primitives has both potentials locally constant, hence constant on the connected D .

For sufficiency assume only the weaker hypothesis, that both periods vanish on U ; the construction will produce single-valued potentials on all of D , so that the periods on D vanish

as well and the two conditions are indeed equivalent. Lemma 6.2 makes U connected, and Part II supplies potentials C, V on U with $dC = \omega_g$, $dV = \theta_g$.

The extension across the axis is the symmetrization argument. Fix a real point $x_* \in D \cap \mathbb{R}$ and a small disk $\Delta \subset D$ centred there and invariant under τ . By Lemma 6.1 both forms are smooth and closed on Δ , which is simply connected, so each has a primitive there; call them C_0, V_0 , normalized to agree with C, V on the (connected) upper half-disk. Replacing C_0 by $\frac{1}{2}(C_0 + C_0 \circ \tau)$ and V_0 likewise does not change their differentials, by (6.2), and makes both even in y ; the normalization is unaffected because the averaged function differs from the original by a constant that vanishes on the real diameter. Hence C, V extend to Δ as even functions, and any two such local extensions agree on overlaps because they agree above the axis and are even. Finally extend C, V to the remaining lower part of D by $C(\tau z) = C(z)$, $V(\tau z) = V(z)$; this is consistent with the extensions already made near the axis and, by (6.2), still satisfies $dC = \omega_g$ and $dV = \theta_g$. The potentials are now single-valued on all of D , so both forms are exact there and every period on D vanishes.

Set $A = xC + V$ and $B = yC$ on the extended domain. Then A is even, B is odd, and the Cauchy–Riemann system $A_x = B_y$, $A_y = -B_x$ holds off the axis by Lemma 3.3(iii); both sides are continuous, so it holds on the axis too. Therefore $F = A + \iota B$ is a holomorphic stem on D satisfying $F(\bar{z}) = \kappa(F(z))$, its induced slice function f is slice-regular on Ω_D , and $\Delta_4 f = g$ off the axis, hence everywhere by continuity. $\square$

6.3 Which holes actually obstruct

We now fix a finite-type class in which the count can be made. The complement of D in the Riemann sphere has finitely many components; each bounded component (a *hole*) is a point or a closed Jordan region, and the unbounded component contains ∞ and may reduce to $\{\infty\}$. No assertion about wild boundaries is needed, and the same conclusions hold for a finite winding basis in the sense of Definition 5.5. Conjugation permutes the holes: it either fixes a hole or exchanges it with a distinct one. Write s for the number of fixed holes and m for the number of exchanged pairs, so that the first Betti number of D is $s + 2m$.

Lemma 6.4 (Reflection on planar homology). *The classes of positively oriented loops around the holes freely generate $H_1(D; \mathbb{Z})$, and for a hole K with class e_K ,*

$$\tau_* e_K = -e_{\tau K}.$$

Consequently a τ -invariant closed form has period zero around a conjugation-fixed hole, and around a conjugate pair its counterclockwise periods are negatives of one another. In particular

$$\dim_{\mathbb{R}} H_{\text{dR}}^1(D; \mathbb{R})^+ = m, \tag{6.3}$$

where the superscript denotes the $+1$ eigenspace of τ^ .*

Proof. Cutting D along disjoint arcs joining each hole to the unbounded complementary component leaves a simply connected region; restoring the arcs one at a time produces exactly one independent loop per hole, and these classes are detected by the integer winding numbers about one selected point in each hole. Conjugation reverses planar orientation and permutes the selected points, which gives $\tau_* e_K = -e_{\tau K}$. For an invariant closed form α and a cycle γ , $\int_{\tau\gamma} \alpha = \int_{\gamma} \tau^* \alpha = \int_{\gamma} \alpha$; applied to e_K with $\tau K = K$ this gives $\int_{e_K} \alpha = -\int_{e_K} \alpha$, hence 0, there being no 2-torsion in $(\mathbb{H}, +)$ or $(\mathbb{R}, +)$. Applied to an exchanged pair it gives the opposite-period relation. Those are the only relations imposed on the free basis, so one real coordinate survives per exchanged pair, which is (6.3). $\square$

The two families of source manuscripts state the resulting count in two different ways: some count m as the rank of a winding basis of the upper meridian U , others as the number of

conjugation-exchanged pairs of holes of D . Neither family proves that the two indices agree. The missing link is the following elementary observation, which we isolate because without it the reflection theorem and the product-domain theorem read as two different theorems about two different integers.

Lemma 6.5 (Invariant holes meet the real axis). *Let $K \subset \mathbb{C}$ be nonempty, connected and τ -invariant. Then $K \cap \mathbb{R} \neq \emptyset$. Consequently every conjugation-fixed hole of D meets $\mathbb{R}$, while the two members of an exchanged pair are disjoint from $\mathbb{R}$.*

Proof. Suppose $K \cap \mathbb{R} = \emptyset$. Then $K = (K \cap \mathbb{C}^+) \sqcup (K \cap \{y < 0\})$ is a partition of K into two relatively open subsets. Both are nonempty: if K were contained in one open half-plane, $\tau K = K$ would be contained in the other, forcing $K = \emptyset$. This contradicts connectedness. For the last claim, if K is one member of an exchanged pair and $x \in K \cap \mathbb{R}$, then $x = \tau(x) \in \tau K$, so $K \cap \tau K \neq \emptyset$; but distinct complementary components are disjoint. $\square$

Proposition 6.6 (The two hole counts agree). *Let D be as above, with s fixed holes $K_1, \dots, K_s$ and m exchanged pairs, and choose the upper member $L_j \subset \mathbb{C}^+$ of the j th pair together with a point $p_j \in L_j$. Then $U = D \cap \mathbb{C}^+$ is connected and has a finite winding basis of size m , realized by $p_1, \dots, p_m$ and small positively oriented loops $\gamma_1, \dots, \gamma_m$ in U encircling $L_1, \dots, L_m$. In particular the rank of a winding basis of the upper meridian equals the number of conjugation-exchanged pairs of holes of D .*

Proof. Connectedness is Lemma 6.2. Compute the complement of U in the Riemann sphere:

$$\widehat{\mathbb{C}} \setminus U = (\widehat{\mathbb{C}} \setminus D) \cup (\{y \leq 0\} \cup \{\infty\}).$$

The set $\{y \leq 0\} \cup \{\infty\}$ is connected and contains ∞ , so it lies in one component together with the unbounded complementary component of D . By Lemma 6.5 each fixed hole K_i meets $\mathbb{R} \subset \{y \leq 0\}$ and is connected, so it lies in that same component. Again by Lemma 6.5 each L_j is a connected compact subset of the open half-plane $\mathbb{C}^+$, disjoint from $\{y \leq 0\}$, from the other holes, and from the unbounded component; hence each L_j is a component of $\widehat{\mathbb{C}} \setminus U$ on its own, as is the lower member τL_j — which, however, lies in $\{y < 0\}$ and so has already been absorbed into the big component. Therefore $\widehat{\mathbb{C}} \setminus U$ has exactly $m + 1$ components, and $H_1(U; \mathbb{Z})$ is free of rank m , generated by loops separating $L_1, \dots, L_m$ from ∞ . Small loops γ_j around L_j inside U represent those generators and satisfy $\text{Ind}(\gamma_j, p_k) = \delta_{jk}$. $\square$

6.4 Reflected modes and the range theorem across the axis

The modes of Part IV are defined on $\mathbb{C}^+ \setminus \{p\}$ and need not extend across $\mathbb{R}$. Their reflected differences do.

Proposition 6.7 (Reflected modes). *For $p = u + \iota v$ with $v > 0$ and $a, b \in \mathbb{H}$, put*

$$G_{p;a,b}^{\text{sym}} = G_{p;a,b} - G_{\bar{p};a,b} \quad \text{on } y > 0, \quad F_{p;a,b}^{\text{sym}}(z) = \frac{za + b}{2\pi\iota} \log \frac{z - p}{z - \bar{p}}. \quad (6.4)$$

Then $G_{p;a,b}^{\text{sym}}$ extends to a smooth axially monogenic field on $\mathbb{H} \setminus S_p$, where $S_p = u + v\mathbb{S}$; it is locally $\Delta_4 \mathcal{I}(F_{p;a,b}^{\text{sym}})$. Its period pair on an upper counterclockwise loop γ is $\text{Ind}(\gamma, p)(a, b)$, and on a counterclockwise loop about $\bar{p}$ it is $(-a, -b)$. Its restriction to the real axis is

$$G_{p;a,b}^{\text{sym}}(x) = -\frac{4v}{\pi d} a + \frac{4v(x - u)}{\pi d^2} (xa + b), \quad d = (x - u)^2 + v^2. \quad (6.5)$$

Proof. Away from $\mathbb{R}$ the field is the difference of two modes, hence axially monogenic; by the last clause of Proposition 5.1 the subtracted term $G_{\bar{p};a,b}$ is regular on all of $\Omega_{\mathbb{C}^+}$ and has zero

periods there, so the period statement on upper loops follows from (5.5). Near a real point $x_* \in \mathbb{R}$ the quotient $R(z) = (z - p)/(z - \bar{p})$ is holomorphic and nonzero, and $|R| = 1$ on $\mathbb{R}$; choose the local branch L of $\log R$ with purely imaginary value at x_* . Uniqueness of the branch then gives $L(\bar{z}) = -\overline{L(z)}$, so $L/(2\pi i)$ is real on the real diameter, and (6.4) is a reflection-real holomorphic stem on that disk. Its Fueter image is smooth across $\mathbb{R}$. Two admissible branches differ by an integer multiple of $za + b$, whose image is zero, so the local images agree with each other and, above the axis, with $G_{p;a,b} - G_{\bar{p};a,b}$. For the period around $\bar{p}$, the winding numbers of numerator and denominator of R are exchanged.

For (6.5), use the real-trace identity (10.3) of Part IX, $(\Delta_4 \mathcal{I}F)(x) = -2F''(x)$, together with Corollary 5.3 applied to both p and $\bar{p}$:

$$(F_{p;a,b}^{\text{sym}})''(x) = \frac{1}{2\pi i} \left[\frac{a}{x-p} - \frac{pa+b}{(x-p)^2} - \frac{a}{x-\bar{p}} + \frac{\bar{p}a+b}{(x-\bar{p})^2} \right].$$

Put $\xi = x - u$, $w = \xi - i\nu$, $w' = \xi + i\nu$, so $ww' = d$; and split the quaternionic data as $\alpha = a$, $\beta = ua + b$, whence $pa + b = \beta + i\nu\alpha$ and $\bar{p}a + b = \beta - i\nu\alpha$. All the complex factors below are central in $\mathbb{H}_{\mathbb{C}}$, so the right-hand coefficients may be kept on the right throughout. From $w' - w = 2i\nu$ and $w' + w = 2\xi$,

$$\frac{1}{w} - \frac{1}{w'} = \frac{2i\nu}{d}, \quad \frac{1}{w^2} - \frac{1}{w'^2} = \frac{4i\nu\xi}{d^2}, \quad \frac{1}{w^2} + \frac{1}{w'^2} = \frac{2(\xi^2 - \nu^2)}{d^2}.$$

Therefore the bracket equals

$$\frac{2i\nu}{d}\alpha - \frac{4i\nu\xi}{d^2}\beta - i\nu \frac{2(\xi^2 - \nu^2)}{d^2}\alpha = \frac{2i\nu}{d^2}[(d - \xi^2 + \nu^2)\alpha - 2\xi\beta] = \frac{4i\nu}{d^2}[v^2\alpha - \xi\beta],$$

using $d - \xi^2 + \nu^2 = 2v^2$. Dividing by $2\pi i$ and multiplying by -2 ,

$$G_{p;a,b}^{\text{sym}}(x) = -\frac{4v}{\pi d^2}(v^2a - \xi(ua + b)),$$

which is (6.5): indeed $-\frac{4v}{\pi d}a + \frac{4v\xi}{\pi d^2}(xa + b) = \frac{4v}{\pi d^2}[(\xi^2 - d)a + \xi(ua + b)] = \frac{4v}{\pi d^2}[-v^2a + \xi(ua + b)]$, since $xa + b = \xi a + (ua + b)$. The same formula follows, without the trace identity, by expanding the single-valued imaginary part of (6.4), $-\frac{1}{4\pi} \log \frac{(x-u)^2 + (y-v)^2}{(x-u)^2 + (y+v)^2} = \frac{vy}{\pi d} + O(y^3)$, in (5.2). $\square$

Theorem 6.8 (Reflection-sensitive range theorem). *Let D be as in Proposition 6.6, with s conjugation fixed holes and m exchanged pairs, and let γ_j, p_j be the upper detecting loops and points. Then*

$$0 \longrightarrow \text{Aff}_{\mathbb{H}} \longrightarrow \mathcal{SR}(\Omega_D) \xrightarrow{\Delta_4 \mathcal{I}(\cdot)} \mathcal{AM}(\Omega_D) \xrightarrow{\text{Per}} \mathbb{H}^{2m} \longrightarrow 0 \quad (6.6)$$

is exact, with $\text{Per}(g) = (a_{\gamma_j}(g), b_{\gamma_j}(g))_{j=1}^m$ and with the explicit right inverse $S((a_j, b_j)_j) = \sum_j G_{p_j; a_j, b_j}^{\text{sym}}$. Hence every $g \in \mathcal{AM}(\Omega_D)$ is uniquely

$$g = \Delta_4 \mathcal{I}(F) + \sum_{j=1}^m G_{p_j; a_{\gamma_j}(g), b_{\gamma_j}(g)}^{\text{sym}}, \quad \dim_{\mathbb{R}} \text{coker } \Delta_4 \mathcal{I}(\cdot) = 8m, \quad (6.7)$$

and the cokernel is canonically $H_{\text{dR}}^1(D; \mathbb{R})^+ \otimes_{\mathbb{R}} \mathbb{H}^2$ via $g \mapsto ([\omega_g], [\theta_g])$. The count $8m$ is independent of s : it is not $8(s + 2m)$, and it is not eight times the first Betti number of D .

Proof. Exactness at $\mathcal{SR}(\Omega_D)$ is the affine kernel again, now on the connected base D . Exactness at $\mathcal{AM}(\Omega_D)$: if the m selected upper period pairs vanish, then by Lemma 6.4 all periods of the invariant forms ω_g, θ_g vanish on every generator of $H_1(D; \mathbb{Z})$ — automatically around each fixed

hole, and by the opposite-period relation around each lower hole — so Theorem 6.3 produces a global primitive. Conversely the periods of a Fueter image vanish.

For surjectivity, $G_{p_j; a, b}^{\text{sym}}$ is regular on Ω_D because its singular sphere S_{p_j} lies over the missing hole L_j , and by Proposition 6.7 its selected period pair is (a, b) at the j th pair and $(0, 0)$ at all the others. Thus $\text{Per} S = \text{id}$ and the argument of Theorem 5.6 applies verbatim, giving (6.7).

The cohomological description is a restatement of what has just been proved, not an appeal to an unproved quaternionic sheaf theory: by Lemma 6.1 the classes $[\omega_g], [\theta_g]$ lie in the τ -invariant part of $H_{\text{dR}}^1(D; \mathbb{H})$, and (6.3) gives $\dim_{\mathbb{R}}(H_{\text{dR}}^1(D; \mathbb{R})^+ \otimes \mathbb{H}^2) = 8m$. Explicit representatives are furnished by S . $\square$

Corollary 6.9 (Exact topological criterion for surjectivity). *In this finite-type class, $\Delta_4 \mathcal{I}(\cdot) : \mathcal{SR}(\Omega_D) \rightarrow \mathcal{AM}(\Omega_D)$ is surjective if and only if every complementary component of D is fixed by complex conjugation, i.e. iff $m = 0$. Simple connectivity of the components of D is sufficient but not necessary.*

Proof. Immediate from (6.7); for necessity, a nonzero reflected mode placed in a missing pair is an explicit non-image. $\square$

6.5 Worked cases

Example 6.10 (The four-dimensional shell: a hole that does not obstruct). Let $D = \{z : R_1 < |z| < R_2\}$ with $0 < R_1 < R_2$, so $\Omega_D = \{q \in \mathbb{H} : R_1 < |q| < R_2\}$. Its one bounded hole $\{|z| \leq R_1\}$ is conjugation-fixed, so $s = 1$, $m = 0$, and every axially monogenic field on the shell has a global slice-regular Fueter primitive — although $b_1(D) = 1$. The upper-meridian picture agrees: U is parametrized by $(\log|z|, \arg z)$ with $\arg z \in (0, \pi)$ and is simply connected, so it has a winding basis of size 0, as Proposition 6.6 predicts. A blanket rule “ $8 \dim_{\mathbb{R}} H^1(D; \mathbb{R})$ ” would give 8 here, and would be wrong.

Corollary 6.11 (Deleted spheres and deleted real points). *Let $p_1, \dots, p_m \in \mathbb{C}^+$ be distinct and let $c_1, \dots, c_\ell \in \mathbb{R}$ be distinct. On*

$$\Omega = \mathbb{H} \setminus \left(\bigcup_{j=1}^m S_{p_j} \cup \{c_1, \dots, c_\ell\} \right)$$

the Fueter image in $\mathcal{AM}(\Omega)$ has real codimension $8m$, independent of ℓ . In particular, for $m = 0$ every axially monogenic field on $\mathbb{H}$ minus finitely many real points has a global slice-regular Fueter primitive there.

Proof. The symmetric planar base is $D = \mathbb{C} \setminus (\{p_j, \bar{p}_j\}_j \cup \{c_k\}_k)$. Its holes are the ℓ real punctures, each conjugation-fixed, and the m exchanged pairs $\{p_j, \bar{p}_j\}$. Apply Theorem 6.8; equivalently, $U = \mathbb{C}^+ \setminus \{p_1, \dots, p_m\}$ by Proposition 6.6, with small circles as a winding basis of size m . A real puncture lies on the boundary of the upper parameter domain, not inside it, and so creates no affine-monodromy cycle. $\square$

Remark 6.12 (What Corollary 6.11 does *not* say). It does not assert that real-point singularities are removable, and it does not assert that a field on $\mathbb{H}$ minus a real point has no principal part there. It asserts only that such a field lies in the image of the Fueter map on that same punctured domain. The distinction is exhibited by the identity, valid for real a ,

$$\Delta_4((q - a)^{-1}) = -\frac{4\overline{q - a}}{|q - a|^4}, \quad (6.8)$$

whose right-hand side is, up to the constant $-8\pi^2$, the Cauchy–Fueter kernel at a : it is a nonremovable singular axially monogenic field on $\mathbb{H} \setminus \{a\}$ that nevertheless has the global slice-regular primitive $(q - a)^{-1}$. Global invertibility on a punctured domain and removability through the puncture are different statements, and only the first is being claimed.

Example 6.13 (One sphere removed: an eight-dimensional cokernel). Let $D = \mathbb{C} \setminus \{\iota, -\iota\}$, so $\Omega_D = \mathbb{H} \setminus \mathbb{S}$. There is exactly one exchanged pair, hence $m = 1$ and $\dim_{\mathbb{R}} \text{coker} = 8$ by Theorem 6.8. The two normalized generators are

$$g_b = G_{\iota;0,1}^{\text{sym}}, \quad g_a = G_{\iota;1,0}^{\text{sym}},$$

with upper period pairs $(0, 1)$ and $(1, 0)$ respectively. Both are smooth axially monogenic on all of $\mathbb{H} \setminus \mathbb{S}$, including the real axis, yet neither is $\Delta_4 f$ for any single-valued $f \in \mathcal{SR}(\Omega_D)$. Specializing (6.5) at $u = 0$, $v = 1$, $d = 1 + x^2$,

$$g_b(x) = \frac{4x}{\pi(1+x^2)^2}, \quad g_a(x) = -\frac{4}{\pi(1+x^2)^2}. \quad (6.9)$$

The obstruction is therefore not caused by a concealed singularity on $\mathbb{R}$.

6.6 The obstructed field on $\mathbb{H} \setminus \mathbb{S}$ is an averaged Cauchy–Fueter kernel

It is worth insisting that g_b is a concrete, branch-free function and not a multivalued object disguised by notation. Its first potential is globally single-valued. Writing $q = x + Iy$ and $d_{\mp}^2 = x^2 + (y \mp 1)^2$,

$$C_b(x, y) = -\frac{1}{2\pi y} \log \frac{d_-}{d_+} = -\frac{1}{4\pi y} \log \frac{x^2 + (y-1)^2}{x^2 + (y+1)^2}, \quad g_b = -2(C_b)_x + 2I(C_b)_y. \quad (6.10)$$

The apparent singularity at $y = 0$ is removable, with $C_b(x, 0) = 1/(\pi(1+x^2))$; expanding the logarithm to first order in y gives this at once, and differentiating recovers (6.9). Thus even the *first* potential can be globally single-valued while the second is not — the concrete form of Example 5.8 on a domain meeting $\mathbb{R}$.

Proposition 6.14 (Sphere-average representation). *Let $d\sigma$ be surface measure on $\mathbb{S}$, with $\sigma(\mathbb{S}) = 4\pi$, and let $E(q) = \bar{q}/(2\pi^2|q|^4)$ be the Cauchy–Fueter kernel. Then, for $q \notin \mathbb{S}$,*

$$C_b(q) = \frac{1}{4\pi^2} \int_{\mathbb{S}} \frac{d\sigma(J)}{|q-J|^2}, \quad g_b(q) = 2 \int_{\mathbb{S}} E(q-J) d\sigma(J) = -2\bar{\mathcal{D}} C_b(q). \quad (6.11)$$

Proof. For $q = x + Iy$ integrate over the angle $t = \langle I, J \rangle$:

$$\int_{\mathbb{S}} \frac{d\sigma(J)}{|q-J|^2} = 2\pi \int_{-1}^1 \frac{dt}{x^2 + y^2 + 1 - 2yt} = \frac{\pi}{y} \log \frac{x^2 + (y+1)^2}{x^2 + (y-1)^2},$$

which after division by $4\pi^2$ is exactly (6.10). For the second formula, $-2\bar{\mathcal{D}}(4\pi^2|q-J|^2)^{-1} = 2E(q-J)$, and differentiation under the integral sign is legitimate on every compact set disjoint from $\mathbb{S}$. The relation $g_b = -2\bar{\mathcal{D}}C_b$ is (6.10) rewritten in quaternionic form. $\square$

So the obstructed target is a Cauchy–Fueter kernel averaged over the singular sphere; smoothness and monogenicity off $\mathbb{S}$ are transparent from (6.11), since the kernel is monogenic away from its pole. This is the special case, computed by hand, of the general layer representation, Theorem 9.3 of Part VIII.

Proposition 6.15 (An independent complex-analytic verification). *There is no reflection-real holomorphic stem F on $D = \mathbb{C} \setminus \{\iota, -\iota\}$ with $\Delta_4 \mathcal{I}(F) = g_b$.*

Proof. The real-trace identity (10.3) and (6.9) would force $F''(x) = -2x(\pi(1+x^2)^2)^{-1}$ for all real $x \in D$. Applying the identity theorem to the four complex components of the holomorphic function F'' on the connected set D extends this to

$$F''(z) = -\frac{2z}{\pi(1+z^2)^2} = \left(\frac{1}{\pi(1+z^2)} \right)', \quad \text{hence} \quad F'(z) = \frac{1}{\pi(1+z^2)} + A$$

for some constant $A \in \mathbb{H}_\mathbb{C}$, the difference having vanishing derivative on a connected set. But F' is the derivative of a single-valued function, so $\oint_\gamma F'(z) dz = 0$ for every closed $\gamma \subset D$, whereas around a small positively oriented circle about ι the displayed right-hand side integrates to $2\pi\iota \cdot \frac{1}{2\pi\iota} = 1$. Contradiction. $\square$

This argument uses no period theorem, no reflection lemma and no classification: it is an independent proof that same-domain surjectivity fails on $\mathbb{H} \setminus \mathbb{S}$.

Remark 6.16 (The editorial position taken by this document). Corollary 6.9 gives a necessary and sufficient topological criterion, and Example 6.13 together with Proposition 6.15 exhibits explicit axially monogenic fields on $\mathbb{H} \setminus \mathbb{S}$ that are not Fueter images on that same domain. It follows that an *unqualified* assertion of same-domain surjectivity is false as stated. The sufficient condition “every component of the symmetric planar set is simply connected” is correct as hypothesized, and Corollary 6.9 shows it is not necessary. We deliberately do not adjudicate whether any particular published theorem is incorrect: that would require auditing its own domain, locality and branch hypotheses, which we have not done. The source manuscripts of this merge are split on exactly this point — one of them names a published sentence as requiring a topological qualification, while four others explicitly decline to assert that any published theorem is wrong — and the position adopted here is the second. See Part XI.

7 Quantitative theory: stability, closed range, and period-corrected approximation

The criterion of Part II (Theorem 3.7) is an exact dichotomy: a single-valued slice-regular Fueter primitive exists on the circularization Ω_U if and only if both period functionals annihilate g . A dichotomy alone is fragile. It does not say whether a computed period of size 10^{-6} is a genuine obstruction or an artifact of quadrature, and it does not say whether a target with a nonzero period might nevertheless be approximated, arbitrarily well and uniformly, by Fueter images of globally single-valued primitives. This part answers both questions. The period functionals are continuous in the crudest reasonable topology, the range is closed, and a nonzero period is a quantitative *error floor* on every detecting loop. The part closes with the approximation theorem that this floor forces: Runge [11] approximation is possible for every axially monogenic target only after a fixed correction space of real dimension exactly $8m$ is adjoined.

Throughout, $U \subset \mathbb{C}^+$ is a connected open set, $g = P + IQ \in \mathcal{AM}(\Omega_U)$, and ω_g, θ_g are the two closed one-forms of Part II (Definition 3.2, Lemma 3.3),

$$\omega_g = \frac{1}{2}(-P dx + Q dy), \quad \theta_g = \frac{1}{2}((xP + yQ) dx + (yP - xQ) dy). \quad (7.1)$$

Recall from Part II that θ_g is computed from the target alone, and is related to the primitive-dependent second form η_C of the classical two-integration construction by the elimination identity (3.9), $\eta_C - d(xC) = \theta_g$; nothing below uses η_C .

7.1 The two norms on an axial field

The sources measure an axial field in two inequivalent ways, and the difference is the single most dangerous point in this theory. We name both and never conflate them.

Definition 7.1 (Axial pair norm and spherical supremum). For $g = P + IQ$ axial on Ω_U put, at a base point $z = x + \iota y \in U$,

$$\|g\|_{\text{ax}}(z) = (|P(x, y)|^2 + |Q(x, y)|^2)^{1/2}, \quad |g|_{\text{sph}}(z) = \sup_{I \in \mathbb{S}} |P(x, y) + IQ(x, y)|. \quad (7.2)$$

For a compact $K \subset U$ we abbreviate $\sup_K \|g\|_{\text{ax}} = \sup_{z \in K} \|g\|_{\text{ax}}(z)$. Near an isolated singular sphere (Part VII) the two associated radial maxima are

$$M_g^{\text{ax}}(\rho) = \max_{|z-p|=\rho} \|g\|_{\text{ax}}(z), \quad M_g^{\text{sph}}(\rho) = \max_{|z-p|=\rho} \max_{I \in \mathbb{S}} |P + IQ|. \quad (7.3)$$

Lemma 7.2 (Norms over the sphere of imaginary units). *Let $P, Q \in \mathbb{H}$ and let $d\sigma$ be ordinary surface measure on $\mathbb{S}$, of total mass 4π . Then*

$$\frac{1}{4\pi} \int_{\mathbb{S}} |P + IQ|^2 d\sigma(I) = |P|^2 + |Q|^2, \quad (7.4)$$

$$\max_{I \in \mathbb{S}} |P + IQ|^2 = |P|^2 + |Q|^2 + 2 |\text{Im}_{\mathbb{H}}(P\bar{Q})|. \quad (7.5)$$

Consequently

$$\|g\|_{\text{ax}}(z) \leq |g|_{\text{sph}}(z) \leq \sqrt{2} \|g\|_{\text{ax}}(z), \quad (7.6)$$

and both inequalities are attained. Moreover, for $X, Y \in \mathbb{H}$ and real α, β ,

$$\text{Im}_{\mathbb{H}}((\alpha X - \beta Y)(\overline{\beta X + \alpha Y})) = (\alpha^2 + \beta^2) \text{Im}_{\mathbb{H}}(X\bar{Y}). \quad (7.7)$$

Finally, for all $x, y \in \mathbb{R}$ and $P, Q \in \mathbb{H}$,

$$|xP + yQ|^2 + |yP - xQ|^2 = (x^2 + y^2)(|P|^2 + |Q|^2) = |z|^2 \|g\|_{\text{ax}}(z)^2. \quad (7.8)$$

Proof. Expanding the quaternionic square, $|P + IQ|^2 = |P|^2 + |Q|^2 - 2 \text{Re}_{\mathbb{H}}(P\bar{Q}I)$, and the last term is twice the Euclidean inner product of the imaginary vector $\text{Im}_{\mathbb{H}}(P\bar{Q})$ with $I \in \mathbb{S}$. Its mean over $\mathbb{S}$ vanishes and its maximum is twice the length of that vector, which gives (7.4) and (7.5). The left inequality in (7.6) follows from (7.4) (a supremum dominates a mean), the right from (7.5) and $2|\text{Im}_{\mathbb{H}}(P\bar{Q})| \leq 2|P||Q| \leq |P|^2 + |Q|^2$. Equality on the left holds exactly when $\text{Im}_{\mathbb{H}}(P\bar{Q}) = 0$, on the right exactly when $|P| = |Q|$ and $P\bar{Q}$ is purely imaginary. For (7.7), expand the product: the terms $X\bar{X}$ and $Y\bar{Y}$ are real and drop out of $\text{Im}_{\mathbb{H}}$, while $\text{Im}_{\mathbb{H}}(Y\bar{X}) = -\text{Im}_{\mathbb{H}}(X\bar{Y})$; collecting gives the stated factor. Identity (7.8) is the same computation with the real orthogonal matrix $\begin{pmatrix} x & y \\ y & -x \end{pmatrix}$, whose two rows are orthogonal of common length $|z|$, so the cross terms $2xy \text{Re}_{\mathbb{H}}(P\bar{Q})$ cancel. $\square$

Remark 7.3 (The equivalence of norms is not an equivalence of constants). Inequality (7.6) makes $\|\cdot\|_{\text{ax}}$ and $|\cdot|_{\text{sph}}$ equivalent, so they define the same topology, the same notion of closed range, and the same class of continuous functionals. They do *not* give the same asymptotic constants. The two exact leading-size laws at a critical singular sphere (Theorem 8.21 below) have genuinely different limits in the two norms, differing by the term $2v|\text{Im}_{\mathbb{H}}(b\bar{a})|$ under the square root. Every estimate in this part is stated in the axial pair norm, which is the sharper of the two on the period side; every asymptotic law in Part VII is stated twice, once in each norm.

7.2 Period bounds and the distance to the range

Proposition 7.4 (Quantitative stability of the obstruction). *Let $\gamma \subset U$ be a closed piecewise C^1 curve of length L_γ , and put $R_\gamma = \max_{z \in \gamma} |z| > 0$. Write $(a_\gamma, b_\gamma) = (\int_\gamma \omega_g, \int_\gamma \theta_g)$ for the period pair of Definition 3.6. Then*

$$|a_\gamma| \leq \frac{L_\gamma}{2} \sup_\gamma \|g\|_{\text{ax}}, \quad |b_\gamma| \leq \frac{L_\gamma R_\gamma}{2} \sup_\gamma \|g\|_{\text{ax}}. \quad (7.9)$$

Consequently, for every globally single-valued $f \in \mathcal{SR}(\Omega_U)$,

$$\sup_\gamma \|g - \Delta_4 f\|_{\text{ax}} \geq \max \left\{ \frac{2|a_\gamma|}{L_\gamma}, \frac{2|b_\gamma|}{L_\gamma R_\gamma} \right\}. \quad (7.10)$$

Proof. Parametrize γ by $(x(t), y(t))$ with unit speed. The value of $2\omega_g$ on the tangent is $-P\dot{x} + Q\dot{y}$, a right $\mathbb{H}$ -linear combination of P, Q with real coefficient vector $(-\dot{x}, \dot{y})$ of Euclidean length 1; Cauchy–Schwarz in $\mathbb{H} \oplus \mathbb{H} \cong \mathbb{R}^8$ gives $|\omega_g(\dot{\gamma})| \leq \frac{1}{2}\|g\|_{\text{ax}}$. For θ_g the coefficient vector of (P, Q) in $2\theta_g(\dot{\gamma})$ is $(x\dot{x} + y\dot{y}, y\dot{x} - x\dot{y})$, whose Euclidean length is $|z|$ by (7.8); hence $|\theta_g(\dot{\gamma})| \leq \frac{1}{2}|z|\|g\|_{\text{ax}}$. Integrating over γ proves (7.9). By Theorem 3.7 every $\Delta_4 f$ with $f \in \mathcal{SR}(\Omega_U)$ has vanishing periods, so $g - \Delta_4 f$ has the periods of g ; applying (7.9) to $g - \Delta_4 f$ and solving for $\sup_\gamma \|g - \Delta_4 f\|_{\text{ax}}$ gives (7.10). $\square$

Inequality (7.10) is the quantitative content of the whole theory. A target with a nonzero period around γ cannot be uniformly approximated on the circularized loop by Fueter images of global primitives: there is a positive error floor, computable from the two period integrals and the elementary geometry of γ alone, below which no such approximation can go. This is strictly stronger than the statement that no exact global solution exists.

Remark 7.5 (Two weaker forms in circulation). Manuscript 10 states the lower bound in the combined form $2(|a_\gamma|^2 + |b_\gamma|^2)^{1/2}(L_\gamma\sqrt{1 + R_\gamma^2})^{-1}$. That is implied by, and in general strictly weaker than, the max-form (7.10), since $\max\{s, t\} \geq (s^2 + t^2)^{1/2}/\sqrt{2}$ and the denominators differ. The bounds (7.9) are also frequently written with the spherical supremum in place of $\|\cdot\|_{\text{ax}}$; by (7.6) that version is weaker by a factor at most $\sqrt{2}$. The pair-norm form is adopted here because it is the sharper one and because it is what the singular asymptotics of Part VII are calibrated against.

Corollary 7.6 (Closed range, with no topological hypothesis). *On every connected open $U \subset \mathbb{C}^+$ — including one whose complement has infinitely many components — the range $\Delta_4 \mathcal{SR}(\Omega_U)$ is closed in $\mathcal{AM}(\Omega_U)$ for locally uniform convergence, and a fortiori for the C^∞ compact-open topology.*

Proof. By Theorem 3.7, $\Delta_4 \mathcal{SR}(\Omega_U) = \bigcap_\gamma \ker(g \mapsto (a_\gamma, b_\gamma))$, the intersection taken over all closed piecewise C^1 curves in U . Each such functional is right $\mathbb{H}$ -linear and, by (7.9), continuous for uniform convergence on the compact set γ . An intersection of kernels of continuous functionals is closed. No finiteness of the winding basis is used, and no boundary regularity of U is used. $\square$

Corollary 7.7 (Continuity of the splitting, in the finite case). *Suppose U carries a finite winding basis of size m in the sense of Definition 5.5, with omitted points $p_1, \dots, p_m$ and dual loops $\gamma_1, \dots, \gamma_m$, and let S and $\Pi = \text{id} - S \text{Per}$ be the section (5.9) and the projection of the split range Theorem 5.6. Then Per , S and Π are continuous for the compact-open topology and for the C^∞ compact-open topology, $\Pi^2 = \Pi$, Π has range $\Delta_4 \mathcal{SR}(\Omega_U)$ and kernel $S(\mathbb{H}^{2m})$, and for every compact $K \subset U$*

$$\sup_K \|S \text{Per} g\|_{\text{ax}} \leq \sum_{j=1}^m \left(|a_{\gamma_j}(g)| \sup_K \|G_{p_j;1,0}\|_{\text{ax}} + |b_{\gamma_j}(g)| \sup_K \|G_{p_j;0,1}\|_{\text{ax}} \right). \quad (7.11)$$

Proof. Each of the $2m$ period coordinates is continuous by (7.9). S sends a coefficient vector to a finite right $\mathbb{H}$ -combination of the fixed smooth modes $G_{p_j;1,0}, G_{p_j;0,1}$ of Proposition 5.1, so it is continuous from $\mathbb{H}^{2m}$ with any norm; (7.11) is the triangle inequality in $\mathbb{H} \oplus \mathbb{H}$ together with $|uw| = |u||w|$ for quaternions. The algebraic identities follow from $\text{Per}S = \text{id}$. $\square$

7.3 The normalized inverse as a continuous operator

Closedness of the range says nothing about how large a primitive must be. The next theorem packages the inverse as a named operator with a stated domain, a stated normalization, and an explicit estimate — together with an explicit statement of what that estimate does *not* give.

Theorem 7.8 (The normalized inverse and its pathwise estimate). *Fix $z_* = x_* + \iota y_* \in U$. On the zero-period subspace $\mathcal{AM}_0(\Omega_U) = \ker \text{Per} \subset \mathcal{AM}(\Omega_U)$ there is a unique right $\mathbb{H}$ -linear inverse*

$$\mathcal{N} : \mathcal{AM}_0(\Omega_U) \longrightarrow \mathcal{SR}(\Omega_U), \quad \Delta_4 \mathcal{N}g = g,$$

normalized by $C(z_) = V(z_*) = 0$ for the two potentials of Definition 3.1; equivalently, $\mathcal{N}g$ is the unique primitive vanishing on the entire base sphere $x_* + y_*\mathbb{S}$, equivalently the unique primitive whose stem satisfies $F(z_*) = 0$ in $\mathbb{H}_{\mathbb{C}}$ — eight real conditions, not the four real conditions of vanishing at a single quaternionic point (Corollary 3.12 and Remark 3.13).*

Let $z \in U$ be joined to z_ by a path $\sigma \subset U$ of length at most L , all of whose points satisfy $|w| \leq R_*$, and suppose $\|g\|_{\text{ax}} \leq M$ on σ . Then*

$$|C(z)| \leq \frac{LM}{2}, \quad |V(z)| \leq \frac{LR_*M}{2}, \quad |(\mathcal{N}g)(q)| \leq \frac{(|q| + R_*)LM}{2} \quad (q = x + \iota y), \quad (7.12)$$

and the stem derivative obeys $\|F'(z)\|_{\mathbb{H}_{\mathbb{C}}} \leq \frac{1}{2}(L + y)M$. For every compact $K \subset U$ there exist a compact $K_ \subset U$ containing $K \cup \{z_*\}$ and a finite L_K for which the path hypothesis holds simultaneously at every $z \in K$; with $R_* = \max_{K_*} |w|$ one obtains the operator estimate*

$$\sup_{z \in K, I \in \mathbb{S}} |(\mathcal{N}g)(x + \iota y)| \leq L_K R_* \sup_{K_*} \|g\|_{\text{ax}}. \quad (7.13)$$

Finally $\mathcal{N}$ is continuous for the C^∞ compact-open topologies.

Proof. Existence and the formula $\mathcal{N}g(q) = qC(x, y) + V(x, y)$ are the Theorem 3.7, with C and V obtained by path integration of the exact forms ω_g and θ_g from z_* . For uniqueness of the normalization: an element $q \mapsto qa + b$ of $\text{Aff}_{\mathbb{H}}$ vanishing on the whole sphere $x_* + y_*\mathbb{S}$ has, on subtracting its values at I and $-I$, $2Iy_*a = 0$, and $y_* > 0$ forces $a = 0$ and then $b = 0$; equivalently the real-linear map $(a, b) \mapsto z_*a + b$ from $\mathbb{H}^2$ to $\mathbb{H}_{\mathbb{C}}$ is an isomorphism precisely because $y_* > 0$, which is the reconciliation of the "vanishes on a sphere" and " $F(z_*) = 0$ " descriptions of the same eight conditions.

For the estimates, integrate the pointwise bounds established in the proof of Proposition 7.4 along σ : $|\omega_g(\dot{\sigma})| \leq \frac{1}{2}\|g\|_{\text{ax}}$ and $|\theta_g(\dot{\sigma})| \leq \frac{1}{2}|w|\|g\|_{\text{ax}} \leq \frac{1}{2}R_*\|g\|_{\text{ax}}$. This gives the first two inequalities in (7.12), and $\mathcal{N}g = qC + V$ with $|q| = |z|$ gives the third. The stem bound follows from the identity (4.5) of Part III, $F' = C + \frac{y}{2}(Q - \iota P)$, together with $\|Q - \iota P\|_{\mathbb{H}_{\mathbb{C}}} = \|g\|_{\text{ax}}$.

For the uniform path family, cover K by finitely many open disks with compact closures in U , join z_* to each centre by a fixed polygonal path in U , and append a final segment inside the relevant disk. The union of these paths with the closed disks is a compact $K_* \subset U$, and their lengths have a common bound L_K . Specializing (7.12) with $|q| \leq R_*$ gives (7.13). Smooth continuity follows because $dC = \omega_g$ and $dV = \theta_g$ express every first derivative of the potentials as an explicit coefficient of P, Q , and every higher derivative as a derivative of those coefficients; on compact subsets of U the factor $1/y$ stays bounded, so finitely many C^{k-1} seminorms of g control the C^k seminorms of $\mathcal{N}g$, the zeroth order being (7.13). Applying all of this to differences of targets gives continuity. $\square$

Remark 7.9 (What the estimate does not assert). The constants L_K and R_* in (7.13) depend on the domain, on the chosen base point, and on the path family; nothing above gives a bound uniform over a family of domains, and in particular nothing is asserted about domains that degenerate, about a singular sphere collapsing onto the real axis, or about bases with a shrinking inradius. Likewise, Corollary 7.6 is a statement about the compact-open topology only: the splitting S is *not* asserted to be bounded in any weighted L^2 , Bergman or Hilbert-space norm on $\mathcal{AM}(\Omega_U)$. Sharper forms of both statements are recorded as open problems in Part XI, §13. The refinement $(|q| + R_*)/2$ in (7.12) reduces to the cruder LR_*M of several sources exactly when $|q| = R_*$; it is genuinely smaller for q nearer the origin.

7.4 Period-corrected Runge approximation

The error floor (7.10) says that a target with a nonzero period is inaccessible from the range. It does not say that the range plus a finite correction is inaccessible, and in fact adjoining the $2m$ modes of Part IV restores a full Runge theorem. Two distinct rational classes occur in the sources; they are *not* interchangeable, and the distinction is entirely a matter of the pole set.

Definition 7.10 (Two rational stem classes). (i) *One-sided (product-domain) class*. Let $U \subset \mathbb{C}^+$ satisfy the hypothesis of Definition 5.5 with omitted points $p_1, \dots, p_m$, one in each bounded component of the complement of U in the Riemann sphere. Let Rat^{one} be the $\mathbb{H}_{\mathbb{C}}$ -valued rational functions with poles only in $\{p_1, \dots, p_m, \infty\}$. These are *upper-stem* rational functions.

(ii) *Symmetric (slice-domain) class*. Let $D \subset \mathbb{C}$ be conjugation-invariant with the analogous finite topology, let $E = \{p_1, \dots, p_m, \bar{p}_1, \dots, \bar{p}_m\}$, and let Rat_E be the $\mathbb{H}_{\mathbb{C}}$ -valued rational R with poles only in $E \cup \{\infty\}$ satisfying the stem symmetry $R(\bar{z}) = \kappa(R(z))$, where $\kappa(A + \iota B) = A - \iota B$.

Theorem 7.11 (Period-corrected rational approximation). *Let $\text{Per}(g) = ((a_j, b_j))_{j=1}^m$ and let S be the section of Part IV.*

(i) (Product domain.) *For every $g \in \mathcal{AM}(\Omega_U)$ there are $R_n \in \text{Rat}^{\text{one}}$ with*

$$\Delta_4 \mathcal{I}(R_n) + S\text{Per}(g) \longrightarrow g \quad (7.14)$$

locally uniformly on Ω_U , together with every fixed order of real derivatives.

(ii) (Symmetric slice domain.) *The same conclusion holds on Ω_D with $R_n \in \text{Rat}_E$ and with the reflected modes G^{sym} of Proposition 6.7 used in S . Here a common denominator may be chosen to be a real polynomial with zeros in E , and the numerator then has right quaternionic coefficients; the R_n are genuine rational slice functions.*

In both cases every approximant has exactly the periods of g , and

$$\overline{\Delta_4 \mathcal{I}(\mathcal{R})}^{\text{loc}} = \Delta_4 \mathcal{SR} \quad \text{inside } \mathcal{AM}.$$

In particular, uncorrected approximation — a locally uniform limit of Fueter images of global slice-regular functions — is possible if and only if $\text{Per}(g) = 0$.

Proof. By Theorem 5.6, $g - S\text{Per}(g)$ has vanishing periods, hence equals $\Delta_4 \mathcal{I}(F)$ for a single-valued holomorphic stem F (on U in case (i), on D in case (ii)). Classical scalar Runge approximation, applied to the four complex components of F with the prescribed pole set — which meets every complementary component of the relevant planar set, including the one containing ∞ — produces rational R_n^0 converging to F uniformly on compact subsets.

In case (ii) replace R_n^0 by the symmetrization

$$R_n(z) = \frac{1}{2}(R_n^0(z) + \kappa(R_n^0(\bar{z}))), \quad (7.15)$$

which is still rational, moves no pole into D , satisfies $R_n(\bar{z}) = \kappa(R_n(z))$, and still converges locally uniformly on the symmetric compact sets. Taking the common denominator $d(z) = \prod_j (z - p_j)^{m_j} (z - \bar{p}_j)^{m_j}$, which has real coefficients and is therefore fixed by κ -symmetrization, the numerator $N = R_n d$ inherits $N(\bar{z}) = \kappa(N(z))$, and a polynomial with that property has all coefficients in $\mathbb{H}$.

Cauchy estimates on slightly larger compact sets upgrade uniform convergence of the stems to uniform convergence of all complex derivatives. Since compact subsets of Ω_U (respectively of Ω_D away from the axis, and, across the axis, via the convergent symmetric Taylor expansions of the stems at real points) keep y in a bounded range, the axial formula (2.49) of Part I converts this into convergence of $\Delta_4 \mathcal{I}(R_n)$ with all fixed real derivatives. Each $\Delta_4 \mathcal{I}(R_n)$ is a global

Fueter image and so has zero periods; adding the fixed field $S\text{Per}(g)$ therefore reproduces exactly the periods of g . The closure statement follows by applying the construction to a zero-period target for one inclusion and Corollary 7.6 for the other; the final equivalence combines that closure with (7.10). $\square$

Remark 7.12 (Why the two pole-set hypotheses must both be displayed). Read carelessly, the two halves of Theorem 7.11 contradict one another: one says the approximating slice functions need *not* be rational with coefficients in $\mathbb{H}$, the other that they are. Both are correct, and the difference is exactly the pole set. In case (i) the poles are *one-sided* — only the chosen upper points — so after reflection to the lower half-plane the induced slice function is defined piecewise and is not in general a single rational function with quaternionic coefficients on a connected symmetric planar domain. In case (ii) the pole set E is conjugation-symmetric, and it is precisely that symmetry that makes the argument of (7.15) yield a real denominator and an $\mathbb{H}$ -coefficient numerator. A merge that recorded only one of the two statements, or recorded both without their pole-set hypotheses, would be either incomplete or apparently self-contradictory.

Proposition 7.13 (Minimality of the correction space). *Under the finite-winding-basis hypothesis, let $W \subset \mathcal{AM}(\Omega_U)$ be any fixed finite-dimensional real subspace such that $\Delta_4\mathcal{SR}(\Omega_U) + W$ is dense in $\mathcal{AM}(\Omega_U)$ for locally uniform convergence. Then $\dim_{\mathbb{R}} W \geq 8m$. The $2m$ quaternionic modes of the section S therefore form a correction space of the smallest possible dimension.*

Proof. The period map $\text{Per} : \mathcal{AM}(\Omega_U) \rightarrow \mathbb{H}^{2m}$ is continuous (Proposition 7.4), surjective (Theorem 5.6), and annihilates $\Delta_4\mathcal{SR}(\Omega_U)$. Hence $\text{Per}(\Delta_4\mathcal{SR}(\Omega_U) + W) = \text{Per}(W)$, a finite-dimensional and therefore closed real subspace of $\mathbb{H}^{2m} \cong \mathbb{R}^{8m}$. Density of $\Delta_4\mathcal{SR} + W$ and continuity of Per force $\text{Per}(W) \supseteq \overline{\text{Per}(W)} \supseteq \text{Per}(\mathcal{AM}) = \mathbb{H}^{2m}$, so $\text{Per}(W) = \mathbb{H}^{2m}$ and $\dim_{\mathbb{R}} W \geq \dim_{\mathbb{R}} \mathbb{H}^{2m} = 8m$. $\square$

Remark 7.14 (Scope of the approximation statement). The only approximation theorem imported above is the classical scalar Runge theorem; nothing here is a new approximation result for monogenic functions. What is specifically quaternionic is the exact finite augmentation that the Fueter map requires, its exact preservation of every period at every stage of the approximation, and the sharp count $8m$ of Proposition 7.13. No density statement is made for bases with infinitely many holes, where Per takes values in an infinite product and Proposition 7.13 has no finite-dimensional content, although Corollary 7.6 continues to apply.

The reconstruction procedure implied by this part. Integrate the two explicit coefficient pairs of (7.1) numerically around one loop per hole; subtract $\sum_j G_{p_j; a_j, b_j}$ using the branch-free formulas (5.2)–(5.3); then integrate the now-exact forms along any convenient paths and assemble $f = qC + V$. The two bounds (7.9) serve as the numerical diagnostic in both directions: they certify that a computed period is too large to be quadrature error, and (7.10) certifies that the residual cannot be reduced below an explicit floor by any better choice of primitive.

8 Singularities along a nonreal sphere: normal forms and sharp thresholds

8.1 Geometry, and the affine spherical residue

Fix $p = u + \iota v \in \mathbb{C}^+$ with $v > 0$, let $S_p = u + v\mathbb{S}$ be the corresponding conjugacy two-sphere in $\mathbb{H}$, and choose $0 < R < v$. Write

$$T_R = \Omega_{\{|z-p|<R\}} = \{x + Iy : (x-u)^2 + (y-v)^2 < R^2, I \in \mathbb{S}\}, \quad \xi = x - u, \quad \eta = y - v, \quad (8.1)$$

and $\rho = |z - p| = (\xi^2 + \eta^2)^{1/2}$. The letters s and t are not used in this part: three of the source manuscripts gave the pair (s, t) three mutually exclusive meanings.

Lemma 8.1 (ρ is the Euclidean distance to the sphere). *For $q = x + Iy \in T_R$ one has $\text{dist}(q, S_p) = \sqrt{(x - u)^2 + (y - v)^2} = \rho$, the nearest point of S_p being $p(q) = u + Iv$, with the same imaginary unit as q .*

Proof. For $J \in \mathbb{S}$, $|q - (u + vJ)|^2 = (x - u)^2 + y^2 + v^2 - 2vy\langle I, J \rangle$, where $\langle \cdot, \cdot \rangle$ is the Euclidean inner product on imaginary quaternions. Since $y > 0$ and $v > 0$ this is minimized exactly at $J = I$, where it equals $(x - u)^2 + (y - v)^2$. $\square$

Thus a punctured planar disk in the base describes a genuine tubular puncture along a two-sphere in four real dimensions. The singular set has real *codimension two*, not four. Every threshold below is calibrated to that geometry and must not be compared with the isolated-point Cauchy–Fueter thresholds of Part I: the sharp isolated-point removability exponent there is $o(|q|^{-3})$ (Theorem 2.46), and the Cauchy–Fueter kernel $\bar{q}/(2\pi^2|q|^4)$ has size $|q|^{-3}$, whereas the sharp exponent here is 1.

Definition 8.2 (Affine spherical residue). For $g \in \mathcal{AM}(T_R \setminus S_p)$ put

$$\text{Res}_{S_p}^{\text{aff}}(g) = (a, b) = \left(\int_{|z-p|=\rho} \omega_g, \int_{|z-p|=\rho} \theta_g \right), \quad 0 < \rho < R, \quad (8.2)$$

the circle positively oriented. Closedness of both forms and Stokes’ theorem on an annulus make the pair independent of ρ .

The pair $(a, b) \in \mathbb{H}^2$ is by construction the affine monodromy coefficient pair of Part II, evaluated on the meridian circle. It is *not* the slice Laurent residue Res_c at a real isolated singularity, and it is *not* the three-dimensional Cauchy–Fueter flux residue Res^F of an isolated point (Theorem 2.35 and Definition 2.47); the three notions are kept apart throughout. Under a real translation $q \mapsto q - c$ one has $a \mapsto a$ and $b \mapsto b + ca$, so the pair $(a, ua + b)$ is translation-invariant, and every invariant below is a function of it.

8.2 (A) The normal form, with no hypothesis whatever

Theorem 8.3 (Full spherical normal form). *Every $g \in \mathcal{AM}(T_R \setminus S_p)$ has a unique representation*

$$g = g_{\text{reg}} + G_{p;a,b} + \Delta_4 \mathcal{I} \left(\sum_{k \geq 1} (z - p)^{-k} c_k \right), \quad (a, b) \in \mathbb{H}^2, \quad c_k \in \mathbb{H}_{\mathbb{C}}, \quad (8.3)$$

where g_{reg} extends axially monogenically through the whole of S_p , $(a, b) = \text{Res}_{S_p}^{\text{aff}}(g)$, and the stem series $\sum_{k \geq 1} (z - p)^{-k} c_k$ converges as a holomorphic $\mathbb{H}_{\mathbb{C}}$ -valued function, with all complex derivatives, uniformly on compact subannuli; the transformed series converges with all real derivatives on compact subannuli of the punctured tube. In particular g extends regularly through S_p if and only if $a = b = 0$ and every $c_k = 0$.

Proof. Let (a, b) be the residue pair and put $g_0 = g - G_{p;a,b}$. By the period normalization (5.5) of the logarithmic modes, g_0 has vanishing periods on the punctured disk, whose first homology is generated by one circle; by Theorem 3.7 there is a single-valued holomorphic stem F_0 on $0 < |z - p| < R$ with $\Delta_4 \mathcal{I}(F_0) = g_0$. Apply the scalar Laurent theorem to the four complex components of F_0 : $F_0 = F_+ + \sum_{k \geq 1} (z - p)^{-k} c_k$ with F_+ holomorphic on the full disk. Set $g_{\text{reg}} = \Delta_4 \mathcal{I}(F_+)$. Laurent series and all their derivatives converge normally on compact subannuli, and the axial formula for $\Delta_4 \mathcal{I}(\cdot)$ involves only F' , $\text{Im}_{\mathbb{C}} F$ and the factors $1/y, 1/y^2$, which are bounded there; so the transform may be taken termwise.

For uniqueness, the periods already determine (a, b) , since a field regular across S_p has closed forms on the full disk and hence zero circle periods. Suppose a negative stem series H has Fueter image regular across S_p . That extended field has a holomorphic primitive K on the full disk by Part II, and on the punctured disk $\Delta_4 \mathcal{I}(H - K) = 0$, so $H - K = za' + b'$ with $a', b' \in \mathbb{H}$ by Corollary 3.9. The right-hand side and K are holomorphic through p , so every negative Laurent coefficient of H vanishes. This gives uniqueness of all c_k and then of g_{reg} . $\square$

Remark 8.4 (The logarithmic sector is not an optional rewriting). Every term $\Delta_4 \mathcal{I}((z - p)^{-k} c_k)$ has a single-valued holomorphic stem on the punctured disk and therefore vanishing affine residues. No sum, finite or convergent, of such terms can represent a $G_{p;a,b}$ with $(a, b) \neq (0, 0)$ on an annulus carrying the period cycle. Equivalently: the logarithmic sector of (8.3) cannot be produced by any single-valued holomorphic source on the same punctured tube, and it must be removed *before* an ordinary Laurent expansion of the primitive is available. This is a classification of *axially monogenic* singularities along a nonreal sphere, expressed through their slice-regular primitives; it is not a claim to introduce quaternionic Laurent expansions in general.

8.3 (B) The L^1 theorem: the weakest hypothesis making the two residues complete

The classification under the pointwise critical bound $M_g^{\text{sph}}(\rho) = O(\rho^{-1})$ is a corollary of the theorem proved in this subsection under the strictly weaker hypothesis of local integrability across the sphere. We record the logical relation explicitly, because seven of the ten source manuscripts state only the pointwise version.

Lemma 8.5 (Reduction to the planar base). *For an axial $g = P + IQ$ on $T_R \setminus S_p$ and $1 \leq s < \infty$, the four-dimensional condition $g \in L^s(T_{R_0})$ is equivalent to the planar condition $P, Q \in L^s(\{|z - p| < R_0\})$, for every $R_0 < R$.*

Proof. The volume element is $dV = y^2 dx dy d\sigma(I)$ with $\int_{\mathbb{S}} d\sigma = 4\pi$, and on a small tube y lies between $v - R_0$ and $v + R_0$, both positive. The averages $P = (4\pi)^{-1} \int_{\mathbb{S}} g d\sigma$ and $Q = -(4\pi)^{-1} \int_{\mathbb{S}} Ig d\sigma$, together with Jensen's inequality, bound the planar norms by the four-dimensional one; the pointwise bound $|P + IQ| \leq |P| + |Q|$ bounds it in the other direction. $\square$

Theorem 8.6 (L^1 spherical normal form). *Let $g \in \mathcal{AM}(T_R \setminus S_p)$ be locally integrable across S_p , that is $\int_{T_{R_0}} |g| dV < \infty$ for some $R_0 < R$. Then every c_k in (8.3) vanishes, and*

$$\boxed{g = G_{p;a,b} + g_0} \tag{8.4}$$

with $(a, b) = \text{Res}_{S_p}^{\text{aff}}(g)$ and g_0 extending as a smooth axially monogenic field through the whole sphere. The coefficients are unique. In particular

$$g \text{ is removable across } S_p \iff a = b = 0, \tag{8.5}$$

and the space of locally L^1 axially monogenic germs at S_p , modulo removable germs, is a right quaternionic vector space of dimension exactly two, with coordinates (a, b) and real dimension eight.

Two genuinely different proofs are recorded. Neither is a variant of the other: the first removes the singularity of the *primitive* at the endpoint case of Sobolev regularity, the second computes the *distributional defect* of the field itself and shows it is a point mass. The second proof produces the surface source of Part VIII directly, rather than deducing it afterwards.

The first proof uses the two endpoint lemmas of Appendix A: Lemma A.5, that a puncture does not obstruct an integrable gradient in two dimensions, so a C^1 function off the origin of a

planar disk with L^1 gradient extends to $W_{\text{loc}}^{1,1}$ with the same weak gradient; and Lemma A.6, weak Cauchy–Riemann removability for $W_{\text{loc}}^{1,1}$ data. Neither is restated here.

First proof of Theorem 8.6 (the $W^{1,1}$ route). By Lemma 8.5 the planar coefficients P, Q are L^1 near p . The branch-free formulas (5.2)–(5.3) give $|P_G| + |Q_G| = O(\rho^{-1} + 1 + |\log \rho|)$ for the components of $G_{p;a,b}$, so those modes are themselves L^1 on a small tube. Subtract the residue-normalized mode and call the remainder g_1 ; it is L^1 and has both periods zero, so by Theorem 3.7 it has a single-valued holomorphic stem $F = A + \iota B$ on the punctured disk with $\Delta_4 \mathcal{I}(F) = g_1$, and the potentials $C = B/y$, $V = A - xB/y$ satisfy $dC = \omega_{g_1}$, $dV = \theta_{g_1}$. The coefficients of those two forms are, by (7.1), real-linear combinations of P, Q with coefficients 1, x and y , all bounded on a smaller disk; hence ∇C and ∇V are in L^1 . Apply Lemma A.5 to all eight real components. Therefore $A = V + xC$ and $B = yC$ extend to $W_{\text{loc}}^{1,1}$ functions across p , with weak derivatives equal to the original ones, so their weak Cauchy–Riemann equations hold on the full disk. Lemma A.6 extends F holomorphically through p ; its induced slice function and its Fueter image extend smoothly through the entire sphere. A smooth extension has zero periods by Stokes’ theorem on a meridian disk, which forces the coefficients of any such decomposition to be exactly (8.2) and gives uniqueness and (8.5). Every pair (a, b) is realized by the corresponding mode, which is L^1 , so the germ quotient has quaternionic dimension exactly two. $\square$

The second proof runs on the planar Vekua operator

$$\mathcal{L}(P, Q) = \left(P_x - Q_y - \frac{2Q}{y}, \quad Q_x + P_y \right), \quad (8.6)$$

so that $g = P + IQ$ is axially monogenic exactly when $\mathcal{L}(P, Q) = 0$; its coefficients, including $1/y$, are smooth and bounded on $\{|z - p| < R\}$ because $y \geq v - R > 0$ there. Three facts about $\mathcal{L}$ are proved in Appendix A and are used here without restatement. First, by Lemma A.1 a distribution supported at one point is a finite sum of derivatives of the point mass. Second, Lemma A.2: if $P, Q \in L^1$ near p and $\mathcal{L}(P, Q) = 0$ away from p , then

$$\mathcal{L}(P, Q) = (\sigma_0, \sigma_1) \delta_p \quad (8.7)$$

for unique $\sigma_0, \sigma_1 \in \mathbb{H}$, and *no* derivative of δ_p occurs — the scaling argument there is precisely where the L^1 hypothesis and the first order of $\mathcal{L}$ are both consumed, and it is what makes the residue classification exhaustive rather than a list of examples. Third, the mode defect (A.2),

$$\mathcal{L}(P_G, Q_G) = \left(\frac{2(ua + b)}{v}, \quad 2a \right) \delta_p \quad \text{for } G = G_{p;a,b}, \quad (8.8)$$

computed there by a single circle integral from the leading term (8.16) below together with the normalization $\bar{\partial}((z - p)^{-1}) = \pi \delta_p$. Finally the L^1 Weyl Lemma A.4, applied to each of the four real components together with the factorization $\bar{\mathcal{D}}\mathcal{D} = \Delta_4$, shows there that weak axial monogenicity is removable: an L^1 pair with $\mathcal{L}(P, Q) = 0$ distributionally on the *full* disk has a smooth axially monogenic representative on the full tube T_R .

Second proof of Theorem 8.6 (the distributional route). By Lemma 8.5 the planar coefficients are L^1 , and Lemma A.2 gives a defect $(\sigma_0, \sigma_1)\delta_p$ as in (8.7). Put

$$a = \frac{\sigma_1}{2}, \quad b = \frac{v\sigma_0 - u\sigma_1}{2}, \quad (8.9)$$

so that $2(ua + b)/v = \sigma_0$. By (8.8) the field $g - G_{p;a,b}$ has zero planar defect; it is still L^1 , so the L^1 Weyl Lemma A.4, in the axial form recalled above, provides the smooth monogenic extension g_0 . Read backwards, (8.9) identifies the defect coefficients with the affine surface

source of Part VIII, §9.1. Since a field regular on the full disk has closed forms there and hence zero circle periods, applying the period normalization of the modes shows that the coefficients in (8.4) are exactly $\text{Res}_{S_p}^{\text{aff}}(g)$, whence (8.9) also identifies the defect coefficients in terms of the residues. Uniqueness and the dimension count follow as before. $\square$

Remark 8.7 (L^1 versus $O(\rho^{-1})$: which hypothesis is used where). Local integrability across S_p is a formally *weaker* hypothesis than the pointwise critical bound $M_g^{\text{sph}}(\rho) = O(\rho^{-1})$: since the transverse area element is $\rho d\rho d\theta$ and y is bounded, $O(\rho^{-1})$ implies local integrability, while the converse is not available a priori. Theorem 8.6 therefore subsumes the critical-growth classification, which is recovered as Corollary 8.14 below. A posteriori the two classes coincide — Theorem 8.6 shows that every L^1 germ is a mode plus a regular field, and the modes are $O(\rho^{-1})$ — but that is a *consequence* of the theorem, not a hypothesis of it, and the implication must not be used to justify the weaker statement. The boundary is exactly where Proposition 8.17 places it: the field $\Delta_4 \mathcal{I}((z-p)^{-1}c)$ has zero periods, exact order ρ^{-2} , fails to be locally L^1 (its planar integral diverges like $\int_0 \rho^{-1} d\rho$), and is not removable.

Remark 8.8 (What the contour formulas do and do not require). Definition 8.2 makes sense for every $g \in \mathcal{AM}(T_R \setminus S_p)$, with no integrability or growth hypothesis at all; the pair (a, b) is always defined and always radius-independent. What Theorem 8.6 adds is *completeness*: in the L^1 class no further singular invariant survives modulo regular germs. Outside that class the residues remain defined but stop being a complete invariant, in both directions — upward through the filtration of Theorem 8.11, and downward through Proposition 8.17.

8.4 (C) The growth filtration

Lemma 8.9 (Growth of a single-valued primitive). *Let $g_1 \in \mathcal{AM}(T_R \setminus S_p)$ have vanishing periods and satisfy $M_{g_1}^{\text{ax}}(\rho) = O(\rho^{-N})$ for an integer $N \geq 1$. Then every single-valued holomorphic primitive stem F_1 on $0 < |z-p| < R$ satisfies, uniformly in the polar angle,*

$$\|F_1(z)\|_{\mathbb{H}_{\mathbb{C}}} = \begin{cases} O(1 + |\log \rho|), & N = 1, \\ O(\rho^{1-N}), & N \geq 2. \end{cases} \quad (8.10)$$

Proof. Use the global potentials C, V of Definition 3.1. On a small disk $|z|$ is bounded, so the pointwise bounds of Proposition 7.4 give $|\nabla C| \leq c_1 \rho^{-N}$ and $|\nabla V| \leq c_2 \rho^{-N}$. Fix an outer circle $|z-p| = \rho_0 < R$. To reach $p + \rho e^{i\theta}$, follow the outer circle from a base point to angle θ — a uniformly bounded contribution — and then the radial segment inward, contributing at most a constant times $\int_{\rho}^{\rho_0} \varsigma^{-N} d\varsigma$, which is $O(1 + |\log \rho|)$ for $N = 1$ and $O(\rho^{1-N})$ for $N \geq 2$. Since $F_1 = zC + V$ with z bounded, (8.10) follows; adding an element of $\text{Aff}_{\mathbb{H}}$ does not change it. $\square$

Lemma 8.10 (Growth of the holomorphic invariant). *If $M_g^{\text{sph}}(\rho) = O(\rho^{-N})$ for an integer $N \geq 1$, then the invariant $\mathcal{H}_g$ of Definition 4.1 satisfies $\mathcal{H}_g(z) = O(\rho^{-N-1})$, so $\mathcal{H}_g$ is meromorphic at p with pole order at most $N + 1$.*

Proof. Every real component of g is harmonic, since $\overline{\partial} \mathcal{D}g = \Delta_4 g = 0$. A ball of radius $\rho/2$ centred at a point at distance ρ from S_p stays in the punctured tube, with distance to S_p between $\rho/2$ and $3\rho/2$ throughout. The interior gradient estimate for harmonic functions — which follows from the volume mean-value property and the divergence theorem in the form $|\partial_{\mu} h(w)| \leq (4/r) \sup_{B(w,r)} |h|$ — therefore gives $|\nabla g| = O(\rho^{-N-1})$ uniformly in I . Evaluating on a fixed pair of opposite slices recovers P, Q and their planar derivatives from g and its derivatives with bounded constants, and y stays away from zero, so the defining formula $\mathcal{H}_g = -\frac{1}{2}(P + yP_y) - \frac{iy}{2}P_x$ of Definition 4.1 inherits the bound. The Laurent coefficient estimate then kills every coefficient below degree $-N - 1$. $\square$

Theorem 8.11 (Finite-order principal parts and their exact dimension). *Let $N \geq 1$ be an integer and $g \in \mathcal{AM}(T_R \setminus S_p)$. The following are equivalent.*

- (i) $M_g^{\text{sph}}(\rho) = O(\rho^{-N})$ uniformly as $\rho \downarrow 0$.
- (ii) $M_g^{\text{ax}}(\rho) = O(\rho^{-N})$ uniformly as $\rho \downarrow 0$.
- (iii) The normal form (8.3) of g terminates:

$$g = g_{\text{reg}} + G_{p;a,b} + \Delta_4 \mathcal{I} \left(\sum_{k=1}^{N-1} (z-p)^{-k} c_k \right), \quad (8.11)$$

the sum being empty for $N = 1$, with $a, b \in \mathbb{H}$ and $c_k \in \mathbb{H}_{\mathbb{C}}$ arbitrary and mutually independent.

- (iv) $\mathcal{H}_g$ is meromorphic at p with pole order at most $N + 1$.

Writing $\mathcal{M}_N$ for the space of germs at S_p satisfying these conditions and $\mathcal{M}_0$ for the regular germs,

$$\mathcal{M}_N / \mathcal{M}_0 \cong \mathbb{H}^2 \oplus \mathbb{H}_{\mathbb{C}}^{N-1}, \quad \dim_{\mathbb{H}}(\mathcal{M}_N / \mathcal{M}_0) = 2N, \quad \dim_{\mathbb{R}}(\mathcal{M}_N / \mathcal{M}_0) = 8N. \quad (8.12)$$

Proof. (i) $\Leftrightarrow$ (ii) is the norm equivalence (7.6), which changes the implied constant by at most $\sqrt{2}$ and no exponent.

(ii) $\Rightarrow$ (iii). The branch-free formulas of Part IV give $|G_{p;a,b}| = O(\rho^{-1})$: the numerators ξ, η of the ρ^{-2} terms are $O(\rho)$, y and $xa + b$ remain bounded, and $|\log \rho| = O(\rho^{-1})$. Subtracting the mode with the actual residue pair preserves the bound and produces a zero-period remainder g_1 ; choose a single-valued primitive stem F_1 . By Lemma 8.9 and the Cauchy coefficient estimate, $\|c_k\|_{\mathbb{H}_{\mathbb{C}}} \leq \rho^k \max_{|z-p|=\rho} \|F_1\|_{\mathbb{H}_{\mathbb{C}}}$, which tends to zero for every $k \geq 1$ when $N = 1$, and for every $k \geq N$ when $N \geq 2$. Only the coefficients displayed in (8.11) survive.

(iii) $\Rightarrow$ (i). For $F_k = (z-p)^{-k} c_k$ both stem components are $O(\rho^{-k})$ and their first derivatives $O(\rho^{-k-1})$; the axial formula (2.49), with y bounded away from zero, gives $\Delta_4 \mathcal{I}(F_k) = O(\rho^{-k-1}) = O(\rho^{-N})$ for $k \leq N-1$. With the $O(\rho^{-1})$ mode and the bounded regular term this proves (i).

(i) $\Rightarrow$ (iv) is Lemma 8.10. (iv) $\Rightarrow$ (iii): subtract the mode determined by the two periods; by the residue formulas (4.9) this cancels the entire polar part of $\mathcal{H}_g$ of order ≤ 2 , so the remainder g_1 has zero periods and $\mathcal{H}_g$ with a pole of order at most $N + 1$. Its single-valued primitive stem $F_1 = \sum_n d_n (z-p)^n$ has $F_1'' = \mathcal{H}_g$ (Theorem 4.3), and the coefficient of $(z-p)^{n-2}$ in F_1'' is $n(n-1)d_n$, nonzero for $n < 0$; the pole-order bound therefore forces $d_n = 0$ for $n < -(N-1)$, which is (8.11).

Finally, the data $((a, b), (c_k)_{k < N})$ map right $\mathbb{H}$ -linearly onto $\mathcal{M}_N / \mathcal{M}_0$; injectivity is the uniqueness in Theorem 8.3 and surjectivity is (8.11). Since $\dim_{\mathbb{R}} \mathbb{H}^2 = 8$ and $\dim_{\mathbb{R}} \mathbb{H}_{\mathbb{C}} = 8$, the real dimension is $8 + 8(N-1) = 8N$. $\square$

Remark 8.12 (Two proofs, and what the count is not). The implications (ii) $\Rightarrow$ (iii) and (iv) $\Rightarrow$ (iii) are two independent proofs of the vanishing of the high Laurent coefficients. The first bounds the *primitive* and applies Cauchy's coefficient estimate; the second bypasses the primitive entirely, bounding the target-derived invariant $\mathcal{H}_g$ by the harmonic interior gradient estimate. The second route is available because each real component of an axially monogenic field is harmonic, so it uses no period theory at all.

The count in (8.12) is $8N$: eight real dimensions from the affine residue pair $(a, b) \in \mathbb{H}^2$, plus $8(N-1)$ from the complexified Laurent coefficients $c_1, \dots, c_{N-1} \in \mathbb{H}_{\mathbb{C}}$. It is emphatically not $4N$ — the Laurent coefficients live in $\mathbb{H}_{\mathbb{C}}$, not in $\mathbb{H}$ — and not $8N + 8$, because the residue pair is not an additional datum beyond the $N = 1$ level but is precisely that level.

Corollary 8.13 (An explicit real basis and exact orders). *With $e_0 = 1$, $e_1 = \mathbf{i}$, $e_2 = \mathbf{j}$, $e_3 = \mathbf{k}$, a real basis of $\mathcal{M}_N/\mathcal{M}_0$ is*

$$G_{p;e_\ell,0}, \quad G_{p;0,e_\ell} \quad (0 \leq \ell \leq 3), \quad \Delta_4 \mathcal{I}((z-p)^{-k} e_\ell), \quad \Delta_4 \mathcal{I}((z-p)^{-k} \iota e_\ell) \quad (1 \leq k \leq N-1, 0 \leq \ell \leq 3).$$

Every nonzero critical residue class has exact uniform order ρ^{-1} , and every class whose highest Laurent index is k has exact uniform order ρ^{-k-1} : the Fueter map loses exactly one power of the distance at a nonreal sphere.

Proof. The basis is (8.12) written out. The exact orders are the positive limits of Theorem 8.21(ii) and Corollary 8.23 below. $\square$

8.5 (D) Sharp transverse removability

Corollary 8.14 (Critical growth and the sharp threshold). *Let $g \in \mathcal{AM}(T_R \setminus S_p)$.*

- (i) *If $M_g^{\text{sph}}(\rho) = O(\rho^{-1})$ then $g = G_{p;a,b} + g_{\text{reg}}$, and g is removable if and only if $\text{Res}_{S_p}^{\text{aff}}(g) = (0, 0)$.*
- (ii) *If $M_g^{\text{sph}}(\rho) = o(\rho^{-1})$ uniformly, then g extends axially monogenically through S_p , and the extension is unique.*
- (iii) *The little- o in (ii) cannot be weakened to big- O : any $G_{p;a,b}$ with $(a, b) \neq (0, 0)$ satisfies the big- O bound and is not removable.*

Quantitatively, on the circle $|z - p| = \rho$,

$$|a| \leq \pi \rho M_g^{\text{sph}}(\rho), \quad |b| \leq \pi \rho (|p| + \rho) M_g^{\text{sph}}(\rho). \quad (8.13)$$

Proof. Part (i) is the case $N = 1$ of Theorem 8.11 together with the uniqueness of the normal form. For (ii), the meridian circle has length $2\pi\rho$ and $\max_{|z-p|=\rho} |z| \leq |p| + \rho$; substituting into (7.9), and using $\|g\|_{\text{ax}} \leq |g|_{\text{sph}}$, gives (8.13). The two periods are independent of ρ , so the little- o hypothesis forces them to vanish; part (i) then supplies the extension, and uniqueness is continuity. Part (iii) is the period normalization of the modes. $\square$

Note the codimension. The exponent 1 here reflects a singular set of real codimension two together with the axial structure. It is not the exponent 3 of the isolated-point Cauchy–Fueter removability theorem of Part I, where the distance is to a single point of $\mathbb{H}$ and the kernel has size $|q|^{-3}$. Mixing the two geometries produces an incorrect comparison of thresholds and of residues.

8.6 (E) L^2 removability

Theorem 8.15 (Sharp L^2 removability across a nonreal sphere). *Let g be Cauchy–Fueter regular on a punctured neighbourhood of S_p and locally square integrable across S_p . Then g extends uniquely to a smooth Cauchy–Fueter-regular field through S_p . Axiality is not needed for this statement; if g is axial, so is the extension. The exponent 2 is sharp: for every $1 \leq s < 2$ the nonzero modes $G_{p;a,b}$ are locally L^s and not removable.*

Proof. Assign arbitrary values on the measure-zero set S_p , so that $g \in L^2_{\text{loc}}$ on the full tube. Choose a logarithmic cutoff χ_ε equal to 0 for $\rho \leq \varepsilon^2$, to 1 for $\rho \geq \varepsilon$, and affine in $\log \rho$ in between, smoothed without changing the bounds

$$|\nabla \chi_\varepsilon| \leq \frac{C}{\rho \log(1/\varepsilon)}, \quad \|\nabla \chi_\varepsilon\|_{L^2}^2 \leq \frac{C'}{\log(1/\varepsilon)} \longrightarrow 0,$$

the second using the codimension-two volume element $O(\rho d\rho d\theta dS)$, which makes $\int \rho^{-2} \log(1/\varepsilon)^{-2} \rho d\rho$ a multiple of $1/\log(1/\varepsilon)$. Test $\mathcal{D}g = 0$ on the punctured tube against $\chi_\varepsilon \phi$ with ϕ smooth and compactly supported. The term carrying $\chi_\varepsilon \nabla \phi$ converges to the desired distributional pairing, and the cutoff error is at most $C_\phi \|g\|_{L^2} \|\nabla \chi_\varepsilon\|_{L^2} \rightarrow 0$. Hence $\mathcal{D}g = 0$ distributionally on the full tube; applying $\bar{\mathcal{D}}$ and Weyl's lemma gives a smooth harmonic representative, which then satisfies the first-order equation pointwise. Two continuous extensions agree on the dense complement of S_p , giving uniqueness, and axially passes to the limit. For sharpness, $G_{p;a,b} = O(\rho^{-1})$ and the transverse polar element $\rho d\rho$ give $\int_0^R \rho^{1-s} d\rho < \infty$ for $s < 2$, while a nonzero residue pair obstructs removability by (8.5). $\square$

Remark 8.16 (Three routes to the same theorem). The cutoff proof above is adopted because it is the shortest, needs no classification machinery, no growth bound and — as several sources independently observe — no axially; it is a standard first-order elliptic capacity argument, and the new content here is not the mechanism but its exact relation to the two periods and to the energy coefficient in (9.16). Two alternatives are recorded. (a) A *linear* shell cutoff, vanishing within distance ε and equal to one beyond 2ε with $|\nabla \chi_\varepsilon| \leq C/\varepsilon$: a compact portion of a codimension-two submanifold has shell volume $O(\varepsilon^2)$, so $\|\nabla \chi_\varepsilon\|_{L^2} \leq C$ is merely bounded, and the error still tends to zero, this time by absolute continuity of $\int |g|^2$ over the shrinking shells. (b) A route through the classification: local L^2 implies, by the harmonic mean-value property and Cauchy–Schwarz, the bound $|g| = O(\rho^{-2})$, hence the case $N = 2$ of Theorem 8.11; the surviving Laurent coefficient is then killed because a nonzero Laurent mode cannot have finite local L^2 energy (compare Proposition 9.15), and the residues are killed by the energy law (9.16). Route (b) is correct but needs the whole $8N$ classification and is restricted to axial fields, so it is not taken as primary. Within the L^1 class the chain of equivalences

$$\text{removability} \iff (a, b) = (0, 0) \iff \text{finite local } L^2 \text{ mass}$$

holds, the last equivalence being the energy law (9.16).

8.7 (F) The sharp L^p trichotomy

Proposition 8.17 (Sharpness of the L^1 hypothesis). *For every $0 < s < 1$ there is an axially monogenic field near S_p , locally in L^s , with $\text{Res}_{S_p}^{\text{aff}} = (0, 0)$, that is not removable. One may take $g = \Delta_4 \mathcal{I}(F)$ with the single-valued stem $F(z) = (z - p)^{-1}c$, $0 \neq c \in \mathbb{H}_{\mathbb{C}}$.*

Proof. The stem is single valued on the punctured disk, so g is a global Fueter image there, both periods vanish, and g has a global single-valued slice-regular primitive on the punctured tube. The explicit components and the exact size are computed in Example 11.1 of Part X: for $c = 1$ one finds $|P|^2 + |Q|^2 = 4/(v^2 \rho^4) + O(\rho^{-3})$, so $M_g^{\text{ax}}(\rho) \sim 2/(v \rho^2)$ uniformly in the meridian angle and in $I \in \mathbb{S}$, with no exceptional quaternionic direction. Hence g is locally L^s exactly when $\int_0 \rho^{1-2s} d\rho < \infty$, that is for $s < 1$, and g is *not* locally L^1 . It is not removable: otherwise the local inverse theorem on the full meridian disk would give a holomorphic stem $F_\sharp$ with $\Delta_4 \mathcal{I}(F_\sharp) = g$ there, and Corollary 3.9 would force $F - F_\sharp = za' + b'$ on the punctured disk, impossible since F has a pole. This last step needs no pointwise lower bound in a possibly exceptional quaternionic direction. $\square$

Corollary 8.18 (Sharp L^p trichotomy at a nonreal sphere). *For locally axially monogenic germs along S_p :*

- (i) *if $1 \leq s < 2$, the locally L^s germs modulo removable germs form a right quaternionic vector space of dimension two, with coordinates (a, b) — real dimension eight;*
- (ii) *if $2 \leq s \leq \infty$, every locally L^s germ is removable;*

(iii) for every $0 < s < 1$ the residues are not a complete invariant modulo removable germs.

Proof. The modes are locally L^s for $s < 2$ by the computation in Theorem 8.15. For $s \geq 1$, local L^s implies local L^1 on a finite-volume tube by Hölder, so Theorem 8.6 applies and gives (i). For $s \geq 2$, local L^s implies local L^2 and Theorem 8.15 gives (ii). Part (iii) is Proposition 8.17. $\square$

Remark 8.19 (Two different thresholds). The exponents 1 and 2 are two distinct sharp thresholds describing two distinct phenomena, and neither is a consequence of the other. At L^1 the pair of periods *becomes a complete invariant* of the singularity; at L^2 the periods are *forced to vanish*. Below L^1 the periods survive as invariants but no longer classify. The example separating L^1 from L^2 is the mode itself, which is L^s for all $s < 2$ and never removable; the example separating L^s , $s < 1$, from L^1 is Proposition 8.17, which shows in addition that a global single-valued primitive on a punctured domain is not the same thing as removability through the missing sphere.

8.8 (G) Exact leading constants, in both norms

The classification decides whether a singularity is present. We now determine its exact size, which is where the sphere $\mathbb{S}$ of imaginary units must be handled explicitly rather than replaced by one complex section. Recall the residue norm and pair norm

$$\mathcal{R}_p(a, b)^2 = |ua+b|^2 + v^2|a|^2 = \|pa+b\|_{\mathbb{H}\mathbb{C}}^2, \quad \mathbf{n}(\alpha, \beta) = \max_{I \in \mathbb{S}} |\beta + I\alpha| = (|\alpha|^2 + |\beta|^2 + 2|\operatorname{Im}_{\mathbb{H}}(\beta\bar{\alpha})|)^{1/2}, \quad (8.14)$$

the second evaluation being (7.5). By (7.6), $(|\alpha|^2 + |\beta|^2)^{1/2} \leq \mathbf{n}(\alpha, \beta) \leq \sqrt{2}(|\alpha|^2 + |\beta|^2)^{1/2}$, so $\mathbf{n}$ is a genuine norm on the eight-real-dimensional pair space. $\mathcal{R}_p$ is positive definite because $v > 0$, and it is unchanged by a real translation of the origin, since $ua + b$ is.

Lemma 8.20 (The uniform leading term). *Let g have residue pair (a, b) and satisfy the critical bound of Corollary 8.14(i), or merely be locally L^1 across S_p . Put $c_0 = ua + b$ and $c_1 = va$, so that $pa + b = c_0 + \iota c_1$, and write $z = p + \rho e^{\iota\theta}$. Then, uniformly in θ and in $I \in \mathbb{S}$,*

$$P = \frac{\cos \theta c_0 + \sin \theta c_1}{\pi v \rho} + O(1 + |\log \rho|), \quad Q = \frac{-\sin \theta c_0 + \cos \theta c_1}{\pi v \rho} + O(1 + |\log \rho|), \quad (8.15)$$

equivalently

$$P + \iota Q = \frac{pa + b}{\pi v(z - p)} + O(1 + |\log \rho|), \quad g(x + Iy) = \frac{e^{-I\theta}((ua + b) + Iva)}{\pi v \rho} + O(1 + |\log \rho|). \quad (8.16)$$

The quaternionic coefficients remain on the right throughout; no commutation with I is implicit.

Proof. By Theorem 8.6 (or Corollary 8.14(i)) the remainder g_0 is bounded on a smaller tube, so only $G_{p;a,b}$ contributes. Take the local stem $F = (za + b)\ell_p$ with $\ell_p = (2\pi\iota)^{-1} \log(z - p)$ on a branch whose argument is bounded at the point being estimated; a change of branch adds an element of $\operatorname{Aff}_{\mathbb{H}}$ and does not change $\Delta_4 \mathcal{I}(F)$. Then

$$F' = a\ell_p + \frac{za + b}{2\pi\iota(z - p)} = \frac{c_0 + \iota c_1}{2\pi\iota\rho e^{\iota\theta}} + O(1 + |\log \rho|), \quad \|F\|_{\mathbb{H}\mathbb{C}} = O(1 + |\log \rho|).$$

The axial Fueter formula (2.49) of Part I, written on stems as $P = -\frac{2}{y} \operatorname{Im}_{\mathbb{C}} F'$ and $Q = \frac{2}{y} \operatorname{Re}_{\mathbb{C}} F' - \frac{2}{y^2} \operatorname{Im}_{\mathbb{C}} F$, together with $y = v + O(\rho)$ and $(\iota e^{\iota\theta})^{-1} = -\sin \theta - \iota \cos \theta$, gives (8.15). Assembling $P + \iota Q$ produces $(c_0 + \iota c_1)/(\pi v \rho e^{\iota\theta})$, which is the first formula in (8.16), and $P + IQ = e^{-I\theta}(c_0 + Ic_1)/(\pi v \rho)$ by expanding $e^{-I\theta} = \cos \theta - I \sin \theta$. One may also read (8.15) off the branch-free formulas (5.2)–(5.3) by substituting $\xi = \rho \cos \theta$, $\eta = \rho \sin \theta$, $y = v + O(\rho)$ and $xa + b = c_0 + O(\rho)$, the logarithmic terms being $O(1 + |\log \rho|)$. $\square$

Theorem 8.21 (Exact leading size at a critical singular sphere). *Let g be as in Lemma 8.20, with residue pair (a, b) .*

(i) (Axial pair norm.) *Uniformly in θ ,*

$$\rho \|g\|_{\text{ax}}(p + \rho e^{i\theta}) \longrightarrow \frac{\mathcal{R}_p(a, b)}{\pi v}, \quad \text{hence} \quad \rho M_g^{\text{ax}}(\rho) \longrightarrow \frac{\mathcal{R}_p(a, b)}{\pi v}. \quad (8.17)$$

(ii) (Spherical supremum.)

$$\rho M_g^{\text{sph}}(\rho) \longrightarrow \frac{1}{\pi v} \sqrt{\mathcal{R}_p(a, b)^2 + 2v |\text{Im}_{\mathbb{H}}(b\bar{a})|} = \frac{1}{\pi} \mathbf{n}\left(a, \frac{ua + b}{v}\right). \quad (8.18)$$

Both limits are strictly positive for every nonzero residue pair. The two constants are different: they agree if and only if $\text{Im}_{\mathbb{H}}(b\bar{a}) = 0$, and their ratio never exceeds $\sqrt{2}$.

Proof. Write $c_0 = ua + b$, $c_1 = va$. The pair in (8.15) is obtained from (c_0, c_1) by the real rotation of angle θ , scaled by $(\pi v \rho)^{-1}$; rotations preserve the sum of squared quaternionic norms, so

$$|P_*|^2 + |Q_*|^2 = \frac{|c_0|^2 + |c_1|^2}{\pi^2 v^2 \rho^2} = \frac{\mathcal{R}_p(a, b)^2}{\pi^2 v^2 \rho^2},$$

independently of θ . Since the error in (8.15) is $O(1 + |\log \rho|) = o(\rho^{-1})$, multiplying by ρ and letting $\rho \downarrow 0$ gives (8.17), uniformly in θ and hence also for the maximum over the circle.

For (ii) apply (7.5) to the leading pair. Using (7.7) with $X = c_0$, $Y = c_1$, $\alpha = \cos \theta$ and $\beta = -\sin \theta$, the cross term of the rotated pair is

$$|\text{Im}_{\mathbb{H}}(P_* \overline{Q_*})| = \frac{|\text{Im}_{\mathbb{H}}(c_0 \bar{c}_1)|}{\pi^2 v^2 \rho^2} = \frac{v |\text{Im}_{\mathbb{H}}((ua + b)\bar{a})|}{\pi^2 v^2 \rho^2} = \frac{v |\text{Im}_{\mathbb{H}}(b\bar{a})|}{\pi^2 v^2 \rho^2},$$

because $ua\bar{a} = u|a|^2$ is real and drops out of $\text{Im}_{\mathbb{H}}$. Again the result is independent of θ . Substituting into (7.5) gives

$$\max_{I \in \mathbb{S}} |P_* + IQ_*|^2 = \frac{\mathcal{R}_p(a, b)^2 + 2v |\text{Im}_{\mathbb{H}}(b\bar{a})|}{\pi^2 v^2 \rho^2},$$

and the uniform $O(1 + |\log \rho|)$ error gives (8.18). The second expression in (8.18) is the same quantity rewritten with $\beta_0 = (ua + b)/v$, since $\mathbf{n}(a, \beta_0)^2 = |a|^2 + |\beta_0|^2 + 2|\text{Im}_{\mathbb{H}}(\beta_0 \bar{a})| = v^{-2}(\mathcal{R}_p(a, b)^2 + 2v |\text{Im}_{\mathbb{H}}(b\bar{a})|)$. Positivity holds because $\mathcal{R}_p$ is positive definite for $v > 0$, and the comparison of the two constants is (7.6). $\square$

Remark 8.22 (The correction is governed by $\text{Im}_{\mathbb{H}}(b\bar{a})$, not by commutativity). The extra term $2v |\text{Im}_{\mathbb{H}}(b\bar{a})|$ in (8.18) measures the mismatch between the average of $|g|^2$ over a conjugacy sphere and its largest value on that sphere. It vanishes exactly when $\text{Im}_{\mathbb{H}}(b\bar{a}) = 0$, which is *not* the same as $ab = ba$. It is therefore never legitimate to replace the four-dimensional supremum by evaluation on one preselected complex section: this is the single point at which the two laws quoted in the literature under the same symbol M_g genuinely diverge, and the two constants must be reported with their respective norms.

The worked instance is Example 11.2 of Part X: $p = i$, $a = 1$, $b = \mathbf{i}$ give $\mathcal{R}_p(a, b)^2 = 2$ and $|\text{Im}_{\mathbb{H}}(b\bar{a})| = 1$, hence the two different constants $2/\pi$ and $\sqrt{2}/\pi$ in (8.18) and (8.17) — for a pair a, b that *commutes*. There $g(\rho + I) = (\mathbf{i} + I)/(\pi \rho) + O(1 + |\log \rho|)$ vanishes at $I = -\mathbf{i}$ and is maximal at $I = \mathbf{i}$, so pointwise cancellation along a selected approach to the sphere does not make the spherical singularity removable, and a numerical probe restricted to one complex slice can report the wrong constant or no singularity at all.

Corollary 8.23 (Exact leading size above the critical order). *Let $g \in \mathcal{M}_N \setminus \mathcal{M}_{N-1}$ with $N \geq 2$, and let $c_{N-1} = \alpha + \iota\beta \neq 0$ be the top Laurent coefficient in (8.11), $\alpha, \beta \in \mathbb{H}$. Then, with $k = N - 1$,*

$$\rho^{k+1} M_g^{\text{ax}}(\rho) \longrightarrow \frac{2k}{v} \|c_k\|_{\mathbb{H}_{\mathbb{C}}} = \frac{2k}{v} (|\alpha|^2 + |\beta|^2)^{1/2}, \quad \rho^{k+1} M_g^{\text{sph}}(\rho) \longrightarrow \frac{2k}{v} \mathbf{n}(\alpha, \beta). \quad (8.19)$$

Equivalently, in terms of the Laurent coefficients h_n of the invariant $\mathcal{H}_g$ of Part III, where $c_k = h_{-k-2}/(k(k+1))$, the invariant $\mathcal{H}_g$ has a pole of order exactly $N + 1$ and

$$M_g^{\text{ax}}(\rho) \sim \frac{2\|h_{-N-1}\|_{\mathbb{H}_{\mathbb{C}}}}{vN} \rho^{-N}. \quad (8.20)$$

Proof. Put $w = z - p$ and $F_k = w^{-k} c_k$. With $\phi = (k+1)\theta$, the components of $F'_k = -kw^{-k-1}c_k$ are $A_x = -k\rho^{-k-1}(\cos\phi\alpha + \sin\phi\beta)$ and $B_x = -k\rho^{-k-1}(\cos\phi\beta - \sin\phi\alpha)$; using $A_y = -B_x$, $B_y = A_x$ and $y = v + O(\rho)$ in the axial Fueter formula gives

$$\Delta_4 \mathcal{I}(F_k)(x + Iy) = \frac{2k}{v\rho^{k+1}} e^{-I(k+1)\theta} (\beta - I\alpha) + O(\rho^{-k}), \quad (8.21)$$

uniformly in θ and I . Equivalently, in complexified form, $Q - \iota P = -\frac{2k}{v} w^{-k-1} c_k + O(\rho^{-k})$, where replacing y by v and the term $\text{Im}_{\mathbb{C}} F_k/y^2$ each cost only $O(\rho^{-k})$. All lower polar terms, the logarithmic mode (which is $O(\rho^{-1})$) and the regular term contribute $O(\rho^{-k})$. Since $\|Q - \iota P\|_{\mathbb{H}_{\mathbb{C}}} = \|g\|_{\text{ax}}$ and central complex multiplication scales $\|\cdot\|_{\mathbb{H}_{\mathbb{C}}}$, the first limit in (8.19) follows; the second follows from (8.21) and (7.5), since $e^{-I(k+1)\theta}$ is a unit quaternion and $\max_I |\beta - I\alpha| = \mathbf{n}(\alpha, \beta)$, the maximum being unchanged by the sign of $I\alpha$. Both limits are positive by (7.6) and $c_k \neq 0$. The statement about $\mathcal{H}_g$ follows from the multipole extraction formula $c_k = h_{-k-2}/(k(k+1))$ of (4.10) with $k = N - 1$, which gives $h_{-N-1} = N(N-1)c_{N-1}$ and hence $2\|h_{-N-1}\|_{\mathbb{H}_{\mathbb{C}}}/(vN) = 2(N-1)\|c_{N-1}\|_{\mathbb{H}_{\mathbb{C}}}/v$. $\square$

Corollary 8.24 (No fractional singularity orders). *Let $\alpha \geq 0$ be a real exponent. A bound $M_g^{\text{ax}}(\rho) = O(\rho^{-\alpha})$ — equivalently, by (7.6), $M_g^{\text{sph}}(\rho) = O(\rho^{-\alpha})$ — allows precisely the principal parts of (8.11) with $N = \lfloor \alpha \rfloor$, where $N = 0$ means no principal part at all. The corresponding singular quotient $\mathcal{M}_{\lfloor \alpha \rfloor}/\mathcal{M}_0$ has real dimension $8\lfloor \alpha \rfloor$. In particular every nonremovable singularity of finite power growth has a positive integer growth order, and the possible uniform orders are exactly $\rho^{-1}, \rho^{-2}, \rho^{-3}, \dots$*

Proof. Apply Theorem 8.11 with the integer $\max(1, \lceil \alpha \rceil)$ to obtain a finite normal form. If the surviving principal part had least integer order $N > \alpha$, then Theorem 8.21 (for $N = 1$) or Corollary 8.23 (for $N \geq 2$) would give a strictly positive limit for $\rho^N M_g^{\text{ax}}(\rho)$, contradicting $M_g^{\text{ax}}(\rho) = O(\rho^{-\alpha})$ with $\alpha < N$. Conversely each such normal form does satisfy the bound, by the same theorems, and regular germs are bounded. The dimension is (8.12). $\square$

Remark 8.25 (What zero residues do and do not say, and the limits of this classification). Three notions are separated by the results above and must not be conflated: existence of a global single-valued slice-regular primitive on the punctured tube; vanishing of the affine spherical residues; and removability through S_p . The first two are equivalent on a punctured disk, by Theorem 3.7. Neither implies the third without a hypothesis: for $0 \neq c \in \mathbb{H}_{\mathbb{C}}$ the field $\Delta_4 \mathcal{I}((z - p)^{-1}c)$ has zero periods, a single-valued primitive, exact uniform order ρ^{-2} , and is not removable (Proposition 8.17 and Example 11.1); and $\Delta_4 \mathcal{I}(\exp((z - p)^{-1}))$ has zero periods, an essential singularity of its invariant $\mathcal{H}_g$, and admits no finite power bound at all, so it lies outside every $\mathcal{M}_N$ (Example 11.4). What supplies the missing hypothesis is local integrability (Theorem 8.6) or a finite growth order (Theorem 8.11), and in the presence of either the residue pair is a complete or a partial invariant respectively.

Finally, the scope. Every classification statement in this part is within the *axial* class; general Cauchy–Fueter-regular fields near a two-sphere are not classified here, and only the L^2 removability Theorem 8.15 is free of the axiomatic hypothesis. All constants above contain a factor $1/v$ or $1/v^2$ and therefore degenerate as the sphere collapses onto the real axis; no uniform statement as $v \downarrow 0$ is claimed, and a real point is a different local problem with different thresholds. Nothing in this part asserts a classification at a singular set that is not a single conjugacy sphere.

9 The surface source, the conserved fluxes, and the logarithmic energy law

Parts II–VII produced the affine spherical residue $\text{Res}_{S_p}^{\text{aff}}(g) = (a, b) \in \mathbb{H}^2$ in three guises: as a pair of periods of the two closed forms ω_g and θ_g , as a pair of complex moments of the holomorphic invariant $\mathcal{H}_g$, and as the coefficients of the logarithmic mode in the local normal form. All three are contour constructions in the planar base. The present part identifies the same eight real parameters by three four-dimensional measurements made on the field itself: a distributional charge carried by the singular sphere, two conserved three-dimensional boundary fluxes, and the exact coefficient of a logarithmic energy divergence. None of the three requires a primitive, a branch, or a choice of loop.

Throughout, $p = u + \iota v$ with $v > 0$, $S_p = u + v\mathbb{S}$, and

$$T_\rho(S_p) = \{q \in \mathbb{H} : \text{dist}(q, S_p) < \rho\}, \quad T_{\varepsilon, R} = \{q : \varepsilon < \text{dist}(q, S_p) < R\}, \quad 0 < \varepsilon < R < v.$$

For $q = x + Iy$ the transverse distance is $\text{dist}(q, S_p) = |z - p| = \rho$ with $z = x + \iota y$. We use repeatedly the two exact measure identities for the tube,

$$dS_{\partial T_\rho} = y^2 \rho \, d\theta \, d\sigma(I), \quad dV = y^2 \, dx \, dy \, d\sigma(I) = y^2 \rho \, d\rho \, d\theta \, d\sigma(I), \quad (9.1)$$

where $x = u + \rho \cos \theta$, $y = v + \rho \sin \theta$, and σ is the ordinary surface measure on $\mathbb{S}$ with $\sigma(\mathbb{S}) = 4\pi$. Part VII’s uniform leading term is quoted so often below that we display it once, in the form it will be used:

$$g(x + Iy) = \frac{e^{-I\theta}(c_0 + Ic_1)}{\pi v \rho} + O(1 + |\log \rho|), \quad c_0 = ua + b, \quad c_1 = va, \quad (9.2)$$

uniformly in θ and in $I \in \mathbb{S}$; equivalently $P + \iota Q = (pa + b)/(\pi v(z - p)) + O(1 + |\log \rho|)$ in $\mathbb{H}_{\mathbb{C}}$. The quaternionic coefficients stand on the right and no commutation with I is implicit. The residue norm of the contract is

$$\mathcal{R}_p(a, b)^2 = |ua + b|^2 + v^2|a|^2 = \|pa + b\|_{\mathbb{H}_{\mathbb{C}}}^2 = |c_0|^2 + |c_1|^2, \quad (9.3)$$

positive definite in (a, b) precisely because $v > 0$.

9.1 The affine surface source

For a compact smooth surface $S \subset \mathbb{H}$ and a quaternion-valued density j on it, $j \delta_S$ denotes the $\mathbb{H}$ -valued distribution $\langle j \delta_S, \varphi \rangle = \int_S j(q) \varphi(q) dS(q)$ acting on real scalar test functions; the operator $\mathcal{D}$ carries its quaternionic units on the left, in the distributional pairing as in the classical one.

Theorem 9.1 (The singular sphere as an affine Cauchy–Fueter source). *Let g be axially monogenic on a punctured tube about S_p and locally integrable across S_p , with affine spherical*

residue $\text{Res}_{S_p}^{\text{aff}}(g) = (a, b)$. Extend g arbitrarily over the measure-zero set S_p . Then, on a sufficiently small full tube,

$$\mathcal{D}g = \sigma_p \delta_{S_p}, \quad \sigma_p(s) = \frac{2}{v}(sa + b) \quad (s \in S_p), \quad (9.4)$$

in the sense of distributions. Written out at $s = u + Iv$,

$$\sigma_p(u + Iv) = \frac{2}{v}(c_0 + Ic_1) = 2 \left(\frac{ua + b}{v} + Ia \right). \quad (9.5)$$

Conversely, the map $(a, b) \mapsto \sigma_p$ is injective, and its image is exactly the space of densities that are affine in the point of the sphere: one constant angular mode and one mode linear in I .

Proof. Let φ be a real test function supported in the tube. Because g is locally integrable, integration by parts on the complement of the shrinking tube $T_\rho(S_p)$ is legitimate and gives

$$\langle \mathcal{D}g, \varphi \rangle = \lim_{\rho \downarrow 0} \int_{\partial T_\rho} n_\rho g \varphi \, dS, \quad n_\rho = \cos \theta + I \sin \theta = e^{I\theta}.$$

The sign is positive because the outward normal of the punctured region along its inner boundary is $-n_\rho$. Multiplying (9.2) on the left by n_ρ cancels $e^{-I\theta}$ exactly, so

$$n_\rho g = \frac{c_0 + Ic_1}{\pi v \rho} + O(1 + |\log \rho|).$$

By (9.1) the surface element carries a factor $y^2 \rho = v^2 \rho + O(\rho^2)$, so the remainder contributes $O(\rho(1 + |\log \rho|)) \rightarrow 0$, while $\varphi(x + Iy) \rightarrow \varphi(u + Iv)$ uniformly. The limit is

$$\frac{v}{\pi} \int_{\mathbb{S}} \int_0^{2\pi} (c_0 + Ic_1) \varphi(u + Iv) \, d\theta \, d\sigma(I) = 2v \int_{\mathbb{S}} (c_0 + Ic_1) \varphi(u + Iv) \, d\sigma(I),$$

and since $dS_{S_p} = v^2 d\sigma(I)$ this equals $\int_{S_p} (2/v)(sa + b) \varphi(s) \, dS(s)$, which is (9.4). Injectivity: subtracting the values of σ_p at $u + Iv$ and $u - Iv$ gives $4a$, so $a = 0$, and then $b = 0$. Surjectivity onto the affine densities is the identity (9.5) read backwards. $\square$

Remark 9.2 (Where axiality has content). The hypothesis used in Theorem 9.1 is the weakest of the family: local integrability across S_p , not $M_g^{\text{ax}}(\rho) = O(\rho^{-1})$. The conclusion is nevertheless a finite-dimensional restriction on what a sphere can carry. An arbitrary surface density on S_p produces a locally integrable field with $\mathcal{D}$ -source supported on S_p ; the axially monogenic ones use only the eight-real-dimensional affine subspace $\{s \mapsto (2/v)(sa + b)\}$, which is exactly the trace on S_p of the affine kernel $\text{Aff}_{\mathbb{H}}$. Fields with general densities exist and are not classified here.

9.2 The Cauchy–Fueter layer representation

Let $E(q) = \bar{q}/(2\pi^2|q|^4)$ be the normalized Cauchy–Fueter kernel of Part I, so that $\mathcal{D}E = \delta_0$ in distributions; the normalization is fixed by $nE = 1/(2\pi^2 r^3)$ on the sphere $|q| = r$, whose area is $2\pi^2 r^3$, so the total flux is 1.

Theorem 9.3 (Canonical layer representation, with uniqueness). *For all $a, b \in \mathbb{H}$ and $q \notin S_p$,*

$$G_{p;a,b}(q) = \frac{2}{v} \int_{S_p} E(q - s) (sa + b) \, dS(s). \quad (9.6)$$

The right-hand side is the unique locally integrable field on $\mathbb{H}$ that tends to zero at infinity and whose distributional Cauchy–Fueter source is $\sigma_p \delta_{S_p}$. The uniqueness step does not assume that a competing field is axial.

Proof. Call the right-hand side K . Differentiation under the integral sign makes K smooth and monogenic off S_p . It is locally integrable in four variables: by Fubini, $\int_{\text{cpt}} |K| dV$ is bounded by a constant times $\int_{S_p} \int_{\text{cpt}} |q - s|^{-3} dV_q dS(s)$, finite and uniform in s because $|q|^{-3}$ is integrable in $\mathbb{R}^4$. Convolving the fundamental solution with the compactly supported surface density gives $\mathcal{D}K = \sigma_p \delta_{S_p}$, and the bound for E gives $K(q) = O(|q|^{-3})$ at infinity.

By Part IV the left-hand side $G_{p;a,b}$ is smooth on $\mathbb{H} \setminus S_p$ and, by Part VII, locally integrable across S_p ; Theorem 9.1 gives it the same source, and Proposition 10.2 below gives it the same decay. Hence $G_{p;a,b} - K$ is globally L^1_{loc} , distributionally monogenic across S_p , and tends to zero at infinity. Applying $\overline{\mathcal{D}}$ makes every real component weakly harmonic, so Weyl's lemma (Appendix A) produces a smooth entire harmonic representative. It is bounded, hence constant by harmonic Liouville, and the constant is zero by the decay. The same three steps applied to the difference of *any* two locally integrable fields with the given source and decay prove uniqueness; nowhere was axiomaticity used. $\square$

Remark 9.4. Theorem 9.3 is what turns the abstract period obstruction into a concrete object: the mode $G_{p;a,b}$ that Part IV manufactured from a multivalued logarithm is, without any correction term, a single-layer Cauchy–Fueter potential with an affine density. It also explains the count: the layer has two free quaternionic parameters because the admissible density has one constant angular mode and one mode linear in I . The relation to the sphere-supported Cauchy kernels of the literature is recorded in Part XI; see also the sphere-average computation of the $\mathbb{H} \setminus \mathbb{S}$ counterexample in Part V, which is this theorem's simplest instance computed by hand.

9.3 Two conserved fluxes that recover both residues

The ordinary Cauchy–Fueter flux of g through a tube boundary is conserved but blind to half of the residue.

Proposition 9.5 (Total flux, and what it misses). *Let g be as in Theorem 9.1. For every sufficiently small ρ ,*

$$\Phi_0 := \int_{\partial T_\rho(S_p)} n_\rho g dS = 8\pi v (ua + b), \quad (9.7)$$

independently of ρ . Consequently a nonremovable singularity can have zero total flux: take $a \neq 0$ and $b = -ua$.

Proof. The componentwise divergence theorem gives $\int_{\partial V} n g dS = \int_V \mathcal{D}g dV$, which vanishes on an annular tube where g is regular; hence Φ_0 is ρ -independent. Its small-radius limit is the total mass of the source (9.4), and $\int_{\mathbb{S}} Id\sigma(I) = 0$ leaves $2v \cdot 4\pi c_0 = 8\pi v c_0$. For the last claim, $a \neq 0$ and $b = -ua$ give $c_0 = 0$ but $c_1 = va \neq 0$, so $\mathcal{R}_\rho(a, b) \neq 0$ and the germ is not removable by Part VII. $\square$

The missing component is recovered by a *weighted* flux. The weight must be chosen with care, and this is not a cosmetic point.

Theorem 9.6 (Two conserved fluxes). *Put*

$$H_u(q) = 3(\text{Re } q - u) + \text{Im } q, \quad (9.8)$$

and define, for small ρ ,

$$\Phi_0(\rho) = \int_{\partial T_\rho} n_\rho g dS, \quad \Phi_1(\rho) = \int_{\partial T_\rho} H_u(q) n_\rho g dS. \quad (9.9)$$

Both are independent of ρ , and

$$\Phi_0 = 8\pi v (ua + b), \quad \Phi_1 = -8\pi v^3 a, \quad \text{so} \quad a = -\frac{\Phi_1}{8\pi v^3}, \quad b = \frac{\Phi_0}{8\pi v} - ua. \quad (9.10)$$

Removability of an L^1 axial germ along S_p is therefore equivalent to the simultaneous vanishing of these two conserved three-dimensional boundary integrals.

Proof. The mechanism is the Green identity for a *right* Fueter-regular weight. If H is smooth with $\sum_{\mu=0}^3 (\partial_{x_\mu} H) e_\mu = 0$, where $e_0 = 1$, $e_1 = \mathbf{i}$, $e_2 = \mathbf{j}$, $e_3 = \mathbf{k}$, then on any region V where $\mathcal{D}g = 0$,

$$\int_{\partial V} H n g \, dS = \int_V \left(\sum_{\mu} (\partial_{x_\mu} H) e_\mu \right) g \, dV + \int_V H \mathcal{D}g \, dV = 0. \quad (9.11)$$

The weight (9.8) qualifies: writing $q - u = x_0 - u + x_1 \mathbf{i} + x_2 \mathbf{j} + x_3 \mathbf{k}$ one has $H_u = 3(x_0 - u) + x_1 \mathbf{i} + x_2 \mathbf{j} + x_3 \mathbf{k}$, whence

$$\sum_{\mu} (\partial_{x_\mu} H_u) e_\mu = 3 \cdot 1 + \mathbf{i} \cdot \mathbf{i} + \mathbf{j} \cdot \mathbf{j} + \mathbf{k} \cdot \mathbf{k} = 3 - 3 = 0.$$

Applying (9.11) to an annular tube, with $H = 1$ and then with $H = H_u$, makes both integrals in (9.9) independent of ρ .

The values are computed by the small-radius limit of the proof of Theorem 9.1. For $H = 1$ this is Proposition 9.5. For $H = H_u$ note that on the source sphere itself $\operatorname{Re}(u + Iv) = u$ and $\operatorname{Im}(u + Iv) = Iv$, so $H_u(u + Iv) = Iv$, and therefore

$$\Phi_1 = 2v \int_{\mathbb{S}} Iv (c_0 + Ic_1) d\sigma(I) = 2v^2 \int_{\mathbb{S}} (Ic_0 - c_1) d\sigma(I) = -8\pi v^2 c_1 = -8\pi v^3 a,$$

using $I^2 = -1$ and $\int_{\mathbb{S}} I d\sigma = 0$. Solving the two linear relations gives (9.10); vanishing of both fluxes forces $a = b = 0$, which by Part VII is removability in the L^1 class. $\square$

Remark 9.7 (The weight cannot be simplified). On S_p one has $H_u(q) = q - u$, so (9.10) looks like the first angular moment $M_1 = \int_{S_p} (q - u) \sigma_p(q) dS_q = -8\pi v^3 a$ of the surface density. The two numbers agree, but the two formulas are not interchangeable. The moment presupposes that one already knows the density — that is, the residue — whereas Φ_1 is a boundary integral of g itself, computable at any finite radius. Moreover the naive weight $q - u$ does *not* give a conserved flux, because it is not right Fueter regular:

$$\sum_{\mu=0}^3 \partial_{x_\mu} (q - u) e_\mu = 1 + \mathbf{i} \mathbf{i} + \mathbf{j} \mathbf{j} + \mathbf{k} \mathbf{k} = -2 \neq 0.$$

Only the coefficient 3 in (9.8) repairs this. A presentation of the angular moment as a computable flux would therefore be incorrect.

Remark 9.8 (The weight is the first entire axially monogenic polynomial). With $\mathcal{P}_k$ as in Lemma 10.1 below,

$$H_u(q) = -\frac{1}{4} \Delta_4((q - u)^3) = 3 \mathcal{P}_1(q - u), \quad (9.12)$$

because $\Delta_4(q^3) = -4(3 \operatorname{Re} q + \operatorname{Im} q)$ and $\mathcal{P}_1(q) = -\Delta_4(q^3)/12$. The constant weight 1 is $\mathcal{P}_0$. Thus the two residues are recovered by pairing g against the two lowest entire axially monogenic polynomials, $\mathcal{P}_0$ and $\mathcal{P}_1$ — precisely the two that also span the kernel of the period map at infinity in Part IX. This identification links the flux calculus of this part to the polynomial calculus of the next; it is not present in any single source manuscript.

9.4 The exact logarithmic energy law

Here and throughout, *energy* means the squared L^2 norm $\int |g|^2 dV$ of the field, never a Dirichlet integral $\int |\nabla g|^2$. That distinction is what makes the critical exponent 2 and the divergence logarithmic. Set

$$\mathcal{E}_g(\varepsilon, R) = \int_{T_{\varepsilon, R}} |g(q)|^2 dV, \quad J_g(\rho) = \int_{\partial T_\rho} |g(q)|^2 dS. \quad (9.13)$$

The spherical mean identity of Part VI, $\frac{1}{4\pi} \int_{\mathbb{S}} |P + IQ|^2 d\sigma(I) = |P|^2 + |Q|^2$, converts both into planar integrals exactly:

$$\mathcal{E}_g(\varepsilon, R) = 4\pi \int_{\varepsilon}^R \rho \int_0^{2\pi} y^2 (|P|^2 + |Q|^2) d\theta d\rho, \quad J_g(\rho) = 4\pi \rho \int_0^{2\pi} y^2 (|P|^2 + |Q|^2) d\theta. \quad (9.14)$$

Theorem 9.9 (Exact energy law with a finite renormalized constant). *Let g be axially monogenic on a punctured tube about S_p and locally integrable across S_p , with residue (a, b) . Then the shell density has the exact trace*

$$\lim_{\rho \downarrow 0} \rho J_g(\rho) = 8 \mathcal{R}_p(a, b)^2, \quad (9.15)$$

and for every fixed sufficiently small R ,

$$\mathcal{E}_g(\varepsilon, R) = 8 \mathcal{R}_p(a, b)^2 \log \frac{R}{\varepsilon} + E_g^{\text{ren}}(R) + o(1) \quad (\varepsilon \downarrow 0), \quad (9.16)$$

with a finite real constant given in closed form by the convergent integral

$$E_g^{\text{ren}}(R) = \int_0^R \left(J_g(s) - \frac{8 \mathcal{R}_p(a, b)^2}{s} \right) ds. \quad (9.17)$$

Subtracting the logarithm therefore leaves an actual limit, not merely a bounded remainder. The coefficient admits three further descriptions:

$$8 \mathcal{R}_p(a, b)^2 = 32\pi^2 \|h_{-2}\|_{\mathbb{H}_{\mathbb{C}}}^2 \quad (\text{Laurent form, with } h_{-2} \text{ the coefficient of Part III}), \quad (9.18)$$

$$8 \mathcal{R}_p(a, b)^2 = \frac{1}{2\pi} \int_{S_p} |\sigma_p(s)|^2 dS(s) \quad (\text{geometric form}), \quad (9.19)$$

$$8 \mathcal{R}_p(a, b)^2 = \frac{1}{8\pi^2} \left(\frac{|\Phi_0|^2}{v^2} + \frac{|\Phi_1|^2}{v^4} \right) \quad (\text{flux form}). \quad (9.20)$$

In particular the coefficient is strictly positive exactly when the singularity is nonremovable.

Proof. By the L^1 normal form of Part VII the hypothesis puts $g = G_{p; a, b} + g_0$ with g_0 axially monogenic across S_p , hence bounded on a smaller tube; in particular the leading estimate (9.2) is available. Writing g_* for the first term of (9.2) and using the real rotation

$$P_* = \frac{\cos \theta c_0 + \sin \theta c_1}{\pi v \rho}, \quad Q_* = \frac{-\sin \theta c_0 + \cos \theta c_1}{\pi v \rho},$$

one gets, with no commutation hypothesis on c_0, c_1 because the cancellation uses only the real inner product,

$$|P_*|^2 + |Q_*|^2 = \frac{|c_0|^2 + |c_1|^2}{\pi^2 v^2 \rho^2} = \frac{\mathcal{R}_p(a, b)^2}{\pi^2 v^2 \rho^2}, \quad (9.21)$$

uniformly in θ . Since $y^2 = v^2 + O(\rho)$, substitution into (9.14) gives

$$J_g(\rho) = \frac{8 \mathcal{R}_p(a, b)^2}{\rho} + O(1 + |\log \rho| + \rho(1 + |\log \rho|)^2), \quad (9.22)$$

where the three error terms come, in order, from $y^2 - v^2$, from the cross term $2|g_*||g - g_*| = O(\rho^{-1}(1 + |\log \rho|))$, and from $|g - g_*|^2 = O((1 + |\log \rho|)^2)$, each multiplied by the normal area factor ρ . This proves (9.15). The error in (9.22) is absolutely integrable at $\rho = 0$; integrating in ρ therefore yields (9.16) with (9.17), the integral converging precisely because the subtracted singular term matches the trace.

For (9.18), Part III gives $pa + b = -2\pi i h_{-2}$, so $\mathcal{R}_p(a, b)^2 = \|pa + b\|_{\mathbb{H}_C}^2 = 4\pi^2 \|h_{-2}\|_{\mathbb{H}_C}^2$. For (9.19), $\sigma_p(u + Iv) = 2(c_0 + Ic_1)/v$ and $dS = v^2 d\sigma(I)$, so $\int_{S_p} |\sigma_p|^2 dS = 4 \int_{\mathbb{S}} |c_0 + Ic_1|^2 d\sigma = 16\pi(|c_0|^2 + |c_1|^2) = 16\pi \mathcal{R}_p^2$, using the spherical mean identity again. For (9.20), substitute $\Phi_0 = 8\pi v c_0$ and $\Phi_1 = -8\pi v^2 c_1$ from Theorem 9.6: $|\Phi_0|^2/v^2 + |\Phi_1|^2/v^4 = 64\pi^2(|c_0|^2 + |c_1|^2)$. Positivity is $v > 0$ in (9.3). $\square$

Remark 9.10 (Hypothesis, and what is weaker than what). Theorem 9.9 is stated under local integrability across S_p , which is strictly weaker as a hypothesis than the pointwise condition $M_g^{\text{ax}}(\rho) = O(\rho^{-1})$ under which several of the source manuscripts stated the law: $O(\rho^{-1})$ implies L^1 because $dV \sim \rho d\rho$, but not conversely *a priori*. That the two classes coincide *a posteriori* is a consequence of the L^1 normal form of Part VII, not an available hypothesis. Separately, the assertion that the remainder converges — rather than being merely $O(1)$ — is genuinely stronger than a bounded-remainder statement, and is what makes (9.17) meaningful as a number attached to g . We do not assert that $E_g^{\text{ren}}(R)$ has a closed form in (a, b, p, R) for a general g ; (9.17) is an explicit convergent integral, not an evaluation.

Remark 9.11 (Numerical corroboration). Two of the source manuscripts evaluated the coefficient $8\mathcal{R}_p(a, b)^2$ against an independent quadrature of $\int |g|^2 dV$ rather than against its own formula. Those certificates, and three further ones, are reconciled in the single table of Part X; all five reproduce from (9.3).

9.5 A growth-free lower bound with the sharp leading constant

The energy law above assumes enough regularity to have a normal form. The following bound assumes nothing at all beyond smoothness off the sphere, and still sees the exact constant.

Theorem 9.12 (Sharp lower bound, no growth hypothesis). *Let g be any smooth axially monogenic field on the punctured tube, with period pair (a, b) ; no growth bound and no integrability hypothesis is assumed. For $0 < \varepsilon < R < v$,*

$$\mathcal{E}_g(\varepsilon, R) \geq 8 \int_{\varepsilon}^R \frac{(v - \rho)^2}{\rho} \left(|a|^2 + \frac{|ua + b|^2}{v^2 + \rho^2} \right) d\rho. \quad (9.23)$$

Consequently

$$\liminf_{\varepsilon \downarrow 0} \frac{\mathcal{E}_g(\varepsilon, R)}{\log(R/\varepsilon)} \geq 8 \mathcal{R}_p(a, b)^2, \quad (9.24)$$

and the constant $8\mathcal{R}_p(a, b)^2$ is sharp for every residue pair. In particular, if $\mathcal{E}_g(0, R) < \infty$ for one R , then $a = b = 0$, with no pointwise hypothesis whatever.

Proof. Parameterize the transverse circle of radius ρ by $x = u + \rho \cos \varphi$, $y = v + \rho \sin \varphi$ and substitute $dx = -\rho \sin \varphi d\varphi$, $dy = \rho \cos \varphi d\varphi$ into the two period integrals of Part II. The first form gives directly

$$a = \frac{\rho}{2} \int_0^{2\pi} (P \sin \varphi + Q \cos \varphi) d\varphi. \quad (9.25)$$

For the second, form $\oint (\theta_g + u \omega_g)$, whose value is $ua + b$; expanding the integrand the terms linear in ρP cancel and the coefficient of P becomes $v \cos \varphi$, that of Q becomes $-v \sin \varphi - \rho$:

$$ua + b = \frac{\rho}{2} \int_0^{2\pi} (vP \cos \varphi - (v \sin \varphi + \rho)Q) d\varphi. \quad (9.26)$$

Regard (P, Q) as an element of $L^2([0, 2\pi]; \mathbb{R}^2)$ in each of the four real quaternionic components. The two coefficient vectors

$$f_1(\varphi) = (\sin \varphi, \cos \varphi), \quad f_2(\varphi) = (v \cos \varphi, -v \sin \varphi - \rho)$$

are *exactly orthogonal*: $\int_0^{2\pi} (v \sin \varphi \cos \varphi - v \sin \varphi \cos \varphi - \rho \cos \varphi) d\varphi = 0$. Their squared norms are 2π and $2\pi(v^2 + \rho^2)$, so the functionals in (9.25)–(9.26) have squared norms $\pi\rho^2/2$ and $\pi\rho^2(v^2 + \rho^2)/2$. Bessel's inequality applied componentwise and summed over the four real components gives

$$\int_0^{2\pi} (|P|^2 + |Q|^2) d\varphi \geq \frac{2}{\pi\rho^2} \left(|a|^2 + \frac{|ua + b|^2}{v^2 + \rho^2} \right). \quad (9.27)$$

Insert this into (9.14) with $y^2 \geq (v - \rho)^2$ and integrate in ρ ; the factor $4\pi\rho \cdot 2/(\pi\rho^2) = 8/\rho$ produces (9.23). As $\rho \downarrow 0$ the bracket in (9.23) tends to $|a|^2 + |ua + b|^2/v^2$ and the prefactor to v^2 , so the radial integrand differs from $8\mathcal{R}_p(a, b)^2/\rho$ by $O(1)$, which is integrable; this gives (9.24). Sharpness is Theorem 9.9 applied to $G_{p,a,b}$ itself, whose proof uses only the explicit mode formulas and not the present bound, so there is no circularity. Finally, if the energy were finite then (9.24) would force $\mathcal{R}_p(a, b) = 0$, hence $a = b = 0$ because $v > 0$. $\square$

Corollary 9.13 (An explicit finite-radius form). *With $W = \mathcal{R}_p(a, b)^2$ and $0 < \varepsilon < R < v$,*

$$\mathcal{E}_g(\varepsilon, R) \geq 8W \int_\varepsilon^R \left(\frac{v - \rho}{v + \rho} \right)^2 \frac{d\rho}{\rho} = 8W \left[\log \frac{R}{\varepsilon} + \frac{4v}{v + R} - \frac{4v}{v + \varepsilon} \right]. \quad (9.28)$$

No optimality is claimed for the finite-radius correction in (9.28); only its leading logarithmic coefficient is sharp.

Proof. Since $(v + \rho)^2 > v^2 + \rho^2$ for $v, \rho > 0$, one has $v^2/(v + \rho)^2 < 1$ and $1/(v + \rho)^2 < 1/(v^2 + \rho^2)$, so the integrand of (9.28) is pointwise dominated by that of (9.23); the corollary is therefore weaker than the theorem at every radius. The closed form follows from the elementary identity

$$\frac{1}{\rho} \left(\frac{v - \rho}{v + \rho} \right)^2 = \frac{1}{\rho} - \frac{4v}{(v + \rho)^2},$$

integrated between ε and R . $\square$

Remark 9.14 (Two routes to the same liminf, and one that is not sharp). Inequality (9.23) is obtained from an exact orthogonal decomposition and Bessel; (9.28) is obtained by bounding the operator norm of the 2×2 matrix relating $(va, ua + b)$ to (yP, yQ) by $(v + \rho)/(v - \rho)$ and applying Cauchy–Schwarz, which discards the orthogonality. Both give the sharp liminf; the first dominates the second at every radius. A third route — bounding the two periods separately by Cauchy–Schwarz on the circle and taking the maximum — yields a shell bound with leading constant $8v^2 \max\{|a|^2, |b|^2/|p|^2\}$, which is strictly smaller than $8\mathcal{R}_p(a, b)^2$ whenever a and b are both nonzero (with $u = 0$, $v = 1$, $a = b = 1$ it gives 8 against the sharp 16). That route is not adopted here.

9.6 The energy scale of a higher principal part

Above the critical order the divergence is a power, and its exponent and coefficient are completely determined.

Proposition 9.15 (Leading L^2 scale of a rational principal part). *Let g be axially monogenic on the punctured tube with the normal form of Part VII, and suppose $k \geq 1$ is the largest index with $c_k \neq 0$ in its principal part $\Delta_4 \mathcal{I}(\sum_{k \geq 1} (z - p)^{-k} c_k)$. Then*

$$\mathcal{E}_g(\varepsilon, R) = 16\pi^2 k \|c_k\|_{\mathbb{H}_\mathbb{C}}^2 \varepsilon^{-2k} + o(\varepsilon^{-2k}). \quad (9.29)$$

Thus logarithmic defects and rational principal parts have completely specified and mutually distinct leading L^2 scales: $\log(1/\varepsilon)$ for the former and ε^{-2k} for the latter.

Proof. For a holomorphic stem $F = A + \iota B$ the complexified axial pair of its Fueter image is

$$P + \iota Q = \frac{2\iota}{y} F'(z) - \frac{2\iota}{y^2} B(z), \quad (9.30)$$

which is the identity of Part II. The top pole of the stem is $(z - p)^{-k} c_k$, whose derivative is $-k(z - p)^{-k-1} c_k$; the term $2\iota B/y^2$ and every lower pole are of strictly lower order, and the logarithmic sector contributes $O(\rho^{-1} \log \rho)$. Since $\|P + \iota Q\|_{\mathbb{H}_{\mathbb{C}}}^2 = |P|^2 + |Q|^2$ and $y = v + O(\rho)$,

$$|P|^2 + |Q|^2 = \frac{4k^2}{v^2} \|c_k\|_{\mathbb{H}_{\mathbb{C}}}^2 \rho^{-2k-2} (1 + O(\rho))$$

uniformly in the angle. Substituting into (9.14) with $y^2 = v^2 + O(\rho)$ gives

$$32\pi^2 k^2 \|c_k\|^2 \int_{\varepsilon}^R \rho^{-2k-1} d\rho = 16\pi^2 k \|c_k\|^2 \varepsilon^{-2k} + O(1),$$

and the lower-order contributions are $o(\varepsilon^{-2k})$ by splitting the radial integral at a fixed small radius and then letting that radius tend to zero. $\square$

Remark 9.16 (Why the meromorphic sector alone already excludes finite energy). Proposition 9.15 is quantitative; the following qualitative statement isolates the same mechanism for a single mode and is worth recording because it separates the meromorphic sector from the logarithmic one. If $F = (z - p)^{-m} c$ with $c \in \mathbb{H}_{\mathbb{C}}$ and $m \geq 1$, then by (9.30) and the spherical average $\frac{1}{4\pi} \int_{\mathbb{S}} |\mathcal{I}(e^{\iota t} c)(I)|^2 d\sigma(I) = \|c\|_{\mathbb{H}_{\mathbb{C}}}^2$, the shell integral of $|\Delta_4 \mathcal{I}(F)|^2$ is a positive multiple of ρ^{-2m-1} , so the local energy is a positive multiple of $\int_0 \rho^{-2m-1} d\rho = \infty$. Thus no nonzero Laurent mode has finite local L^2 energy. This is an alternative to the cutoff proof of L^2 removability in Part VII; it is not needed for that theorem, but it is the only argument in the family that quantifies the exclusion of the meromorphic part separately from the logarithmic part.

9.7 Invariance, and an audit of the constant

Remark 9.17 (Coordinate invariance of the coefficient). Under a real translation $q' = q - t$, $t \in \mathbb{R}$, the affine increment $qa + b$ becomes $q'a + (b + ta)$ and the centre becomes $u' = u - t$. Hence the slope residue is unchanged and $u'a + b' = ua + b$, so both $\mathcal{R}_p(a, b)^2$ and the two fluxes (9.10) are unchanged. The invariant pair is $(a, ua + b)$; neither a nor b separately is a translation-invariant scalar, and $\mathcal{R}_p$ is not an artifact of the choice of real origin. Equation (9.19) gives the coefficient an intrinsic geometric meaning without choosing period coordinates at all.

Remark 9.18 (Where the constant 8 comes from). It is worth isolating the arithmetic, because the same letter 8 appears in the cokernel count $8m$ of Part IV for an entirely unrelated reason. In (9.14) the factor 4π is the total mass of σ on $\mathbb{S}$, the angular integration over θ contributes 2π , and the leading squared norm (9.21) carries $1/\pi^2$ after the factors v^2 from y^2 and $1/v^2$ from (9.2) cancel. Thus

$$8 = \frac{(4\pi) \times (2\pi)}{\pi^2},$$

a constant of the codimension-two normal-circle geometry. The $2\pi^2$ of the point Cauchy–Fueter kernel E belongs to the three-sphere geometry of an isolated point in $\mathbb{R}^4$ and does not appear here. The same caution applies to the critical exponents: the threshold exponent 1 at a nonreal sphere reflects a singular set of real codimension two and must not be confused with the

isolated-point exponent 3 of Part I. Further sign and normalization conventions — $\mathcal{D}$ with no factor $\frac{1}{2}$, $\Delta_4 q^2 = -4$, coefficients on the right, normal circle length $2\pi\rho$, sphere area $4\pi v^2$, tube boundary element $y^2\rho d\theta d\sigma$, and the derived constants $2/v$, $8\pi v$, 8 , $16\pi^2 k$ — are collected in Appendix C.

10 Global classification: prescribed data, realization, and decay

The local data of Part VII — an affine residue pair and a finite or infinite Laurent principal part at each singular sphere — are now assembled into global statements. Three questions are answered: which global configurations of local data actually occur (a Mittag–Leffler realization theorem), what the resulting global function space looks like when the transverse orders and the growth at infinity are prescribed (a normal form with an exact dimension), and how the local residues are visible from infinity (a moment hierarchy). The entire axially monogenic polynomials appear at both ends of the story, as the growth sector of the normal form and as the flux weights of Part VIII.

10.1 Entire axially monogenic polynomials

Lemma 10.1 (Polynomial growth and the normalized basis). *Define*

$$\mathcal{P}_k(q) = -\frac{\Delta_4(q^{k+2})}{2(k+1)(k+2)}, \quad k \geq 0. \quad (10.1)$$

Then $\mathcal{P}_k$ is entire axially monogenic with real-axis trace $\mathcal{P}_k(x) = x^k$. If h is entire axially monogenic and $|h(q)| \leq C(1 + |q|)^\alpha$ for some $\alpha \geq 0$, then h is a polynomial in the four real coordinates of degree at most $N = \lfloor \alpha \rfloor$, and there are unique $c_0, \dots, c_N \in \mathbb{H}$ with

$$h(q) = \sum_{k=0}^N \mathcal{P}_k(q) c_k. \quad (10.2)$$

That space has real dimension $4(N+1)$.

Proof. Since $\overline{\mathcal{D}}\mathcal{D} = \Delta_4$, every real component of h is entire harmonic. The interior derivative estimate for a harmonic u on a ball, $|\partial^\nu u(q_0)| \leq C_\nu R^{-|\nu|} \sup_{B(q_0, R)} |u|$ — obtained by writing u as its convolution with a smooth radial averaging kernel supported in the ball and transferring the derivatives to the kernel, so that C_ν depends on neither u nor R — gives, on letting $R \rightarrow \infty$, that every derivative of order $|\nu| > \alpha$ vanishes. Real analyticity makes h a polynomial of degree at most N .

For the basis, let F be an entire holomorphic stem with $\Delta_4 \mathcal{I}(F) = h$; one exists because the upper half-plane is simply connected and the period criterion of Part II is vacuous there, and it extends across the real axis by the reflection theorem of Part V. The real-trace identity

$$(\Delta_4 \mathcal{I}(F))(x) = -2F''(x), \quad x \in \mathbb{R}, \quad (10.3)$$

follows from Taylor expansion at a real point, $A(x, y) = F(x) - y^2 F''(x)/2 + O(y^4)$ and $B(x, y) = yF'(x) - y^3 F'''(x)/6 + O(y^5)$, inserted into the Fueter formula for Δ_4 on stems. Writing the real-axis polynomial as $h(x) = \sum_{k \leq N} x^k c_k$ gives $F''(z) = -\frac{1}{2} \sum_k z^k c_k$ by the holomorphic identity theorem applied to the four central-complex coordinates; integrating twice and applying $\Delta_4 \mathcal{I}$ proves (10.2). The same computation gives $\Delta_4(q^{k+2})|_{\mathbb{R}} = -2(k+2)(k+1)x^k$, hence $\mathcal{P}_k(x) = x^k$, and the linear independence of the real powers over $\mathbb{H}$ gives uniqueness and the dimension. $\square$

The Liouville-type conclusion is standard harmonic analysis; the normalized family (10.1) is recorded to make the classification below explicit, and is not offered as a new family of polynomials. Its two lowest members are the two flux weights of Part VIII: $\mathcal{P}_0 = 1$ and $3\mathcal{P}_1(q - u) = H_u(q)$ by (9.12).

10.2 Exterior expansions of the modes

On $\mathbb{H}$ itself the relevant modes are the reflected ones $G_{p;a,b}^{\text{sym}}$ of Part V, built from the symmetric logarithm $\ell_p^s(z) = \frac{1}{2\pi\iota} \log \frac{z-p}{z-\bar{p}}$, which is smooth across the real axis; near S_p it differs from $G_{p;a,b}$ by a field regular there, so it carries the same affine residue (a, b) . Likewise the reflection-compatible rational stems

$$R_{p,k,c}(z) = \frac{c}{(z-p)^k} + \frac{\bar{c}}{(z-\bar{p})^k}, \quad c \in \mathbb{H}_{\mathbb{C}}, \quad k \geq 1, \quad (10.4)$$

(the bar being complex-stem conjugation only) satisfy $R_{p,k,c}(\bar{z}) = \overline{R_{p,k,c}(z)}$, so they are genuine stems on $\mathbb{C} \setminus \{p, \bar{p}\}$, with upper principal part exactly $(z-p)^{-k}c$; the reflected summand is holomorphic near p .

Proposition 10.2 (Exterior expansion and uniform decay). *For $|q| > |p|$,*

$$G_{p;a,b}^{\text{sym}}(q) = \sum_{n \geq 1} \Delta_4(q^{-n}) d_n(p; a, b), \quad d_n(p; a, b) = -\frac{1}{\pi} \left(\frac{\text{Im}_{\mathbb{C}}(p^{n+1})}{n+1} a + \frac{\text{Im}_{\mathbb{C}}(p^n)}{n} b \right), \quad (10.5)$$

the coefficients $\text{Im}_{\mathbb{C}}(p^n)$ being ordinary real scalars. The series and all its derivatives converge uniformly on compact subsets of $\{|q| > |p|\}$. Each of $G_{p;a,b}^{\text{sym}}$ and $\Delta_4 \mathcal{I}(R_{p,k,c})$ is $O(|q|^{-3})$ at infinity, uniformly in the quaternionic direction, including directions arbitrarily close to the real axis where the axial formulas have apparent denominators.

Proof. Near infinity choose the branch of the symmetric logarithm vanishing there:

$$\ell_p^s(z) = \frac{1}{2\pi\iota} (\log(1 - p/z) - \log(1 - \bar{p}/z)) = -\sum_{n \geq 1} \frac{p^n - \bar{p}^n}{2\pi\iota n} z^{-n} = -\sum_{n \geq 1} \frac{\text{Im}_{\mathbb{C}}(p^n)}{\pi n} z^{-n},$$

whose coefficients are real. Multiplying by $za + b$ produces the constant term $-\text{Im}_{\mathbb{C}}(p)a/\pi = -va/\pi$, which is real-affine and is killed by Δ_4 , followed by the coefficients (10.5). The reflection-compatible rational stem likewise has a convergent exterior Laurent series $\sum_{n \geq 1} z^{-n} a_n$ with $a_n \in \mathbb{H}$.

For the uniform decay, apply $\Delta_4 \mathcal{I}$ term by term. Each $\Delta_4(q^{-n})$ is homogeneous of degree $-n-2$ and smooth on the unit three-sphere. Concretely, for $q \neq 0$ the first and second real derivatives of q^{-1} are bounded by $C|q|^{-2}$ and $C|q|^{-3}$, so the product rule bounds second derivatives of q^{-n} by $Cn^2|q|^{-n-2}$ uniformly on spheres; the Laurent coefficients have geometric bounds, so the differentiated series converges uniformly outside any larger ball. The constant term has zero Laplacian, so the first surviving term is $O(|q|^{-3})$. Homogeneity, not the axial formula, is what makes the estimate uniform near the real axis. $\square$

We record the identification that anchors the leading term: $\Delta_4(q^{-1}) = -4\bar{q}/|q|^4 = -8\pi^2 E(q)$, the kernel link of Part I.

10.3 A Mittag–Leffler realization theorem

Parts IV–VII compute what can occur and what the obstruction is. The following converse — the only one in the family — says that every formally admissible global singular configuration is actually realized, with no compatibility conditions between spheres.

Theorem 10.3 (Realization of locally finite spherical data). *Let $V \subset \mathbb{C}^+$ be a simply connected open set and let $Z = \{p_j\}_{j \in A} \subset V$ be discrete and locally finite in V . At each p_j prescribe, independently and with no relation to the data at any other point,*

$$a_j, b_j \in \mathbb{H}, \quad R_j(z) = \sum_{m=1}^{M_j} (z - p_j)^{-m} c_{jm}, \quad c_{jm} \in \mathbb{H}_{\mathbb{C}},$$

with $M_j \geq 0$ finite and the empty sum allowed. Then there exists $g \in \mathcal{AM}(\Omega_{V \setminus Z})$ whose local normal form at every S_{p_j} has exactly the affine residue (a_j, b_j) and exactly the Laurent principal part R_j , and which has no other singularity in Ω_V . Two such functions differ by a field in $\mathcal{AM}(\Omega_V)$, and g admits a single-valued slice-regular Fueter primitive on $\Omega_{V \setminus Z}$ if and only if $a_j = b_j = 0$ for every j .

Proof. The construction prescribes the holomorphic invariant, not the field. At p_j prescribe the meromorphic principal part

$$\mathcal{H}_g^{(j)}(z) = \sum_{m=1}^{M_j} m(m+1)(z - p_j)^{-m-2} c_{jm} + \frac{a_j}{2\pi\iota(z - p_j)} - \frac{p_j a_j + b_j}{2\pi\iota(z - p_j)^2}. \quad (10.6)$$

Applied to the four central-complex coordinates, the classical scalar Mittag–Leffler theorem produces a meromorphic $H : V \rightarrow \mathbb{H}_{\mathbb{C}}$ with precisely these principal parts and no other poles; no absolute convergence of the uncorrected sum is required, the theorem supplying the holomorphic corrections.

For any closed curve γ in $V \setminus Z$ the residue theorem — legitimate because V is simply connected and Z has no accumulation point in V — gives

$$\int_{\gamma} H(z) dz = \sum_j \text{Ind}(\gamma, p_j) a_j, \quad - \int_{\gamma} z H(z) dz = \sum_j \text{Ind}(\gamma, p_j) b_j, \quad (10.7)$$

only finitely many terms being nonzero for a fixed γ . Indeed R_j'' and zR_j'' have no simple pole, the simple-pole coefficient of $\mathcal{H}_g^{(j)}$ is $a_j/(2\pi\iota)$, and that of $z\mathcal{H}_g^{(j)}$ is $(p_j a_j - (p_j a_j + b_j))/(2\pi\iota) = -b_j/(2\pi\iota)$. Both integrals in (10.7) therefore lie in $\mathbb{H}$, so H belongs to the moment-real class $\mathcal{O}_{\text{mom}}(V \setminus Z)$ of Part III. By the surjectivity half of Part III's range theorem, $H = \mathcal{H}_g$ for some $g \in \mathcal{AM}(\Omega_{V \setminus Z})$: integrate $H dz$ and $-zH dz$ on the universal cover to obtain C and V , set $F = zC + V$ and descend $\Delta_4 \mathcal{I}(F)$, which is single-valued because the monodromy of F lies in the affine kernel $\text{Aff}_{\mathbb{H}}$. Its affine periods are exactly (10.7).

It remains to check the *complete* local data, not merely the periods. Near p_j consider the multivalued holomorphic expression $F_j = R_j + \frac{za_j + b_j}{2\pi\iota} \log(z - p_j)$. A direct differentiation shows $F_j'' = \mathcal{H}_g^{(j)}$, so $H - F_j''$ extends holomorphically to a disk about p_j ; choose a holomorphic double primitive K_j of that difference on the disk. On a branch over the punctured disk $(F - F_j - K_j)'' = 0$, so the difference is complex-affine, and its image under $\Delta_4 \mathcal{I}$ is regular on the full disk because the disk stays away from the real axis. Hence $g - \Delta_4 \mathcal{I}(R_j) - G_{p_j; a_j, b_j}$ extends across S_{p_j} , which is exactly the assertion about the normal form. Away from Z the construction is locally $\Delta_4 \mathcal{I}$ of a holomorphic stem, so there are no extra singularities.

If two fields have the same data, the uniqueness in Part VII's normal form makes their difference removable at every p_j ; local finiteness lets these local extensions be combined into a member of $\mathcal{AM}(\Omega_V)$. Finally, if every pair vanishes then both integrals in (10.7) vanish on every loop, so Part II gives a global primitive; conversely a small loop about p_j detects its pair, which must then vanish. $\square$

Remark 10.4 (Why the second derivative is the right thing to prescribe). The formal sum $\sum_j (R_j + \frac{za_j + b_j}{2\pi\iota} \log(z - p_j))$ need not converge, and it cannot be repaired by silently choosing

global branches of the logarithms. Prescribing (10.6) avoids both problems at once: the polar data are single-valued meromorphic data, classical Mittag–Leffler supplies the convergence-forcing holomorphic corrections, and the two residue moments guarantee that the surviving monodromy of the primitive lies in exactly the real-affine kernel $\text{Aff}_{\mathbb{H}}$. This is why arbitrary residue pairs at arbitrarily many spheres can coexist without producing a multivalued target. Simple connectivity of V is used twice, for the residue theorem and to exclude periods of H coming from the topology of V itself; over a base with holes the realization statement would have to be combined with the split range theorem of Part IV.

Remark 10.5 (The prescribed data automatically satisfy the reality constraint). Part III proves that the two polar coefficients of $\mathcal{H}_g$ at a nonreal p are not independent: $h_{-1} = (\iota/v) \text{Re}_{\mathbb{C}}(h_{-2})$. The data (10.6) comply automatically, and it is worth seeing why, since that is what makes the prescription consistent. Here $h_{-1} = a/(2\pi\iota) = -\iota a/(2\pi)$ and $h_{-2} = -(pa + b)/(2\pi\iota) = \iota(pa + b)/(2\pi) = (-va + \iota(ua + b))/(2\pi)$, so $\text{Re}_{\mathbb{C}}(h_{-2}) = -va/(2\pi)$ and $(\iota/v) \text{Re}_{\mathbb{C}}(h_{-2}) = -\iota a/(2\pi) = h_{-1}$. Equivalently, with $h_{-2} = c + \iota d$ one recovers $a = -(2\pi/v)c$ and $b = 2\pi d + (2\pi u/v)c$. The free datum at each sphere is therefore the single $\mathbb{H}_{\mathbb{C}}$ coefficient h_{-2} , which carries all eight real obstruction coordinates, together with the higher coefficients c_{jm} ; there is no independent choice of h_{-1} to be made, and no pole of order exactly one can be prescribed.

Example 10.6 (Independent defects and poles at three spheres). Over a simply connected upper base containing distinct p_1, p_2, p_3 , prescribe the first-period defect $(\mathbf{i}, 0)$ at p_1 , the second-period defect $(0, \mathbf{j})$ at p_2 , and the zero-period principal part $(z - p_3)^{-2}(1 + \iota \mathbf{k})$ at p_3 . Theorem 10.3 produces a single axially monogenic target with exactly these classes. By Part VII the first two spheres carry exact transverse order ρ^{-1} and the third exact order ρ^{-3} ; after subtracting $G_{p_1; \mathbf{i}, 0} + G_{p_2; 0, \mathbf{j}}$ the remainder has zero periods and hence a global primitive on the punctured base, although it is still singular at S_{p_3} . This is the sharpest available illustration that global invertibility on a punctured domain is not removability through the missing sphere.

10.4 The global normal form with mixed transverse orders

We now fix finitely many spheres in $\mathbb{H}$ and prescribe both the transverse orders and the growth at infinity.

Theorem 10.7 (Rational–logarithmic global classification). *Let $p_1, \dots, p_n \in \mathbb{C}^+$ be distinct, $S_j = u_j + v_j \mathbb{S}$, and choose integers $m_j \geq 1$ and $d \geq 0$. Let $\mathcal{M}(\mathbf{m}, d)$ be the real vector space of smooth axially monogenic fields on $\mathbb{H} \setminus \bigcup_j S_j$ with transverse growth $M_g^{\text{ax}}(\rho_j) = O(\rho_j^{-m_j})$ near each S_j and $|g(q)| = O(|q|^d)$ at infinity. Every $g \in \mathcal{M}(\mathbf{m}, d)$ has a unique representation*

$$g(q) = \sum_{k=0}^d \mathcal{P}_k(q) c_k + \sum_{j=1}^n G_{p_j; a_j, b_j}^{\text{sym}}(q) + \sum_{j=1}^n \sum_{i=1}^{m_j-1} \Delta_4 \mathcal{I}(R_{p_j, i, c_{j,i}})(q), \quad (10.8)$$

with $c_k, a_j, b_j \in \mathbb{H}$ and $c_{j,i} \in \mathbb{H}_{\mathbb{C}}$, and all choices of these coefficients occur. Consequently

$$\dim_{\mathbb{R}} \mathcal{M}(\mathbf{m}, d) = 4(d+1) + 8 \sum_{j=1}^n m_j. \quad (10.9)$$

A field in this class has a global slice-regular Fueter primitive on $\mathbb{H} \setminus \bigcup_j S_j$ exactly when every a_j and every b_j vanishes.

Proof. Remove the local singular classes. Apply Part VII’s normal form near each p_j ; it yields unique (a_j, b_j) and unique $c_{j,1}, \dots, c_{j,m_j-1}$. Near p_j the symmetric logarithm differs from ℓ_{p_j} by a function holomorphic up to a harmless real branch constant, so $G_{p_j; a_j, b_j}^{\text{sym}}$ has the same local singular class as $G_{p_j; a_j, b_j}$; and $R_{p_j, i, c}$ has exactly the required upper principal part by (10.4).

The terms attached to the other spheres are regular near S_j . Subtracting every singular term in (10.8) leaves a field g_0 that extends axially and monogenically across every sphere, hence to all of $\mathbb{H}$.

Use the growth at infinity. By Proposition 10.2 the subtracted terms are $O(|q|^{-3})$, so $g_0 = O(|q|^d)$. Lemma 10.1 makes $g_0 = \sum_{k \leq d} \mathcal{P}_k c_k$ with unique c_k .

Uniqueness, converse, and count. The local theorem determines (a_j, b_j) and all $c_{j,i}$ uniquely; the polynomial coefficients are unique because $\Delta_4(q^{k+2})|_{\mathbb{R}} = -2(k+2)(k+1)x^k$ and the real powers are independent over $\mathbb{H}$. Conversely every term in (10.8) has the stated regularity and local growth, and Proposition 10.2 supplies the exterior bound. Counting: $4(d+1)$ from the $c_k \in \mathbb{H}$, 8 from each pair $(a_j, b_j) \in \mathbb{H}^2$, and $8(m_j - 1)$ from the $c_{j,i} \in \mathbb{H}_{\mathbb{C}}$, giving (10.9). Finally the only terms with nonzero periods are the logarithmic ones, so the reflection form of the period criterion (Part V) gives the last assertion. $\square$

Remark 10.8 (Which hypothesis each statement needs). Theorem 10.7 is stated under the pointwise transverse hypothesis $M_g^{\text{ax}}(\rho_j) = O(\rho_j^{-m_j})$, and that is genuinely what it needs: for $m_j \geq 2$ the higher Laurent coefficients are extracted by a growth argument and no integrability hypothesis replaces it. At the bottom of the filtration, $m_j = 1$, the hypothesis can be weakened to local integrability across S_j , which is strictly weaker; that is the content of the next corollary, and it is the only place in this part where the L^1 theorem of Part VII is used in preference to a growth bound. We do not assert that the mixed-order theorem holds under an L^1 -type hypothesis at the higher-order spheres. Note also that the two norms M^{ax} and M^{sph} define the same $O(\cdot)$ classes here, since $\|g\|_{\text{ax}} \leq M_g^{\text{sph}} \leq \sqrt{2}\|g\|_{\text{ax}}$; the constants in the exact asymptotics of Part VII do differ between them, and are not used in this section.

10.5 The decaying class, the layer inverse, and additivity

Corollary 10.9 (Decaying fields with critical spherical growth). *Let $\Sigma = \bigcup_{j=1}^n S_{p_j}$ with $p_j = u_j + v_j$ distinct, and let $\mathcal{M}_{1,0}(\Sigma)$ be the right quaternionic vector space of smooth axially monogenic fields on $\mathbb{H} \setminus \Sigma$ that are locally integrable across every sphere and satisfy $|g(q)| \rightarrow 0$ as $|q| \rightarrow \infty$. Then the residue map is an isomorphism*

$$\text{Res}^{\text{aff}} : \mathcal{M}_{1,0}(\Sigma) \xrightarrow{\sim} \mathbb{H}^{2n}, \quad g \longmapsto ((a_j, b_j))_{j \leq n}, \quad (10.10)$$

of real dimension $8n$, with explicit inverse

$$g(q) = \sum_{j=1}^n G_{p_j; a_j, b_j}^{\text{sym}}(q) = \sum_{j=1}^n \frac{2}{v_j} \int_{S_{p_j}} E(q-s) (sa_j + b_j) dS(s). \quad (10.11)$$

Every such field decays at least as $O(|q|^{-3})$, and more precisely

$$g(q) = 8\pi E(q) \sum_{j=1}^n v_j (u_j a_j + b_j) + O(|q|^{-4}) = \frac{4}{\pi} \frac{\bar{q}}{|q|^4} \sum_{j=1}^n v_j (u_j a_j + b_j) + O(|q|^{-4}). \quad (10.12)$$

The only field in this class with a global slice-regular Fueter primitive is 0. Dropping the decay hypothesis, every smooth axially monogenic field on $\mathbb{H} \setminus \Sigma$ locally integrable across the spheres is uniquely $g_{\text{ent}} + \sum_j G_{p_j; a_j, b_j}^{\text{sym}}$ with g_{ent} entire axially monogenic.

Proof. Extract unique residues in disjoint small tubes by the L^1 normal form of Part VII, and subtract the corresponding finite sum of reflected modes. Near each sphere the other modes are smooth and the singular part at that sphere has been removed, so the remainder extends to an entire axially monogenic field; by Proposition 10.2 it still tends to zero at infinity, so harmonic Liouville makes it zero. Without the decay hypothesis the same subtraction leaves an entire field, which is the last assertion. Conversely every finite mode sum lies in the class

and has the prescribed residue vector, which gives the isomorphism; the layer form of (10.11) is Theorem 9.3, and the asymptotic (10.12) is the $n = 1$ term of (10.5) summed over j , using $D_1 = -\pi^{-1} \sum_j v_j(u_j a_j + b_j)$ and $\Delta_4(q^{-1}) = -4\bar{q}/|q|^4 = -8\pi^2 E(q)$. A global primitive forces all periods to vanish, hence $g = 0$; and if all pairs vanish in the non-decaying version the field is entire, its upper base is the simply connected half-plane, and Parts II and V give a primitive on all of $\mathbb{H}$. $\square$

Remark 10.10 (Local integrability at finite points, not global L^1). The hypothesis in Corollary 10.9 is local integrability across each sphere together with decay at infinity. It is not an assertion that $g \in L^1(\mathbb{H})$, and none is used.

Remark 10.11 (Two finite-dimensional objects, two different questions). The split exact sequence of Part IV computes a *cokernel*: the obstruction to inverting Δ_4 on a fixed punctured domain, of real dimension $8m$. Corollary 10.9 computes a *space of actual fields*, of real dimension $8n$. They are linked by the same explicit representatives $G_{p_j; a_j, b_j}^{\text{sym}}$, and the numerical coincidence of the two counts is not accidental, but they answer different questions and neither implies the other.

Remark 10.12 (Real punctures are excluded on purpose). Theorem 10.7 and Corollary 10.9 remove only nonreal spheres. Real punctures carry no additional *inversion* obstruction — that is the content of Part V's corollary on $\mathbb{H}$ minus finitely many real points, where the codimension stays $8m$ independently of the number of deleted reals — but they do carry independent singular principal parts, which would have to be listed in any classification theorem that also deleted real points. The distinction is exactly the one Part V insists on: q^{-1} is a global primitive of a nonremovable singularity.

10.6 Far-field moments: what a charge at infinity can and cannot see

Theorem 10.13 (Exact moment conditions for faster decay). *In the decaying class of Corollary 10.9 set*

$$D_N = -\frac{1}{\pi} \sum_{j=1}^n \left(\frac{\text{Im}_{\mathbb{C}}(p_j^{N+1})}{N+1} a_j + \frac{\text{Im}_{\mathbb{C}}(p_j^N)}{N} b_j \right), \quad N \geq 1. \quad (10.13)$$

Then for $|q| > \max_j |p_j|$,

$$g(q) = \sum_{N \geq 1} \Delta_4(q^{-N}) D_N, \quad (10.14)$$

and for every integer $K \geq 1$,

$$g(q) = O(|q|^{-K-3}) \text{ uniformly at infinity} \iff D_1 = \dots = D_K = 0. \quad (10.15)$$

Proof. Summing the finitely many expansions (10.5) gives (10.14). If the first K coefficients vanish, then homogeneity of $\Delta_4(q^{-N})$ (degree $-N-2$) and the uniform convergence of Proposition 10.2 give the stated uniform decay. Conversely evaluate (10.14) on the positive real axis: by the real-trace identity (10.3) applied to z^{-N} ,

$$\Delta_4(q^{-N})|_{q=x} = -2N(N+1)x^{-N-2}.$$

A first nonvanishing D_N with $N \leq K$ would therefore produce a nonzero term of order x^{-N-2} , larger than x^{-K-3} , contradicting the assumed decay; uniqueness of Laurent coefficients (or successive multiplication by powers of x and passage to the limit) makes the argument componentwise rigorous. $\square$

Corollary 10.14 (A single charge at infinity cannot separate the two residues). *There exist fields in the decaying class with nonzero affine spherical residue and vanishing first far-field*

moment. For $p = \iota$, the mode $G_{\iota;1,0}^{\text{sym}}$ has residue $(1, 0)$ and $D_1 = -\pi^{-1}v(ua + b) = 0$ since $u = 0$ and $b = 0$, so it decays as $O(|q|^{-4})$, whereas $G_{\iota;0,1}^{\text{sym}}$ has residue $(0, 1)$, $D_1 = -1/\pi \neq 0$, and a genuine $|q|^{-3}$ term.

This is the far-field counterpart of the flux statement of Part VIII: Φ_0 alone misses the slope residue a , and D_1 , being $-\pi^{-1}\sum_j v_j(u_j a_j + b_j)$, is $-\Phi_0^{\text{tot}}/(8\pi^2)$ up to the obvious normalization and misses exactly the same information. Both are cured in the same way: by pairing against $\mathcal{P}_1$ as well as $\mathcal{P}_0$.

10.7 Additivity of the singular energy

Corollary 10.15 (Energy additivity over disjoint tubes). *Let g be axially monogenic on $\mathbb{H} \setminus \Sigma$ and locally integrable across every S_{p_j} , with residues (a_j, b_j) . For pairwise disjoint sufficiently small tubes, with fixed outer radii R_j ,*

$$\sum_{j=1}^n \int_{\varepsilon < \text{dist}(q, S_{p_j}) < R_j} |g(q)|^2 dV = 8 \sum_{j=1}^n \mathcal{R}_{p_j}(a_j, b_j)^2 \log \frac{1}{\varepsilon} + C + o(1) \quad (\varepsilon \downarrow 0), \quad (10.16)$$

with C a finite real constant. In particular the total logarithmic energy coefficient of a configuration is the sum of the individual residue norms, with no interaction term.

Proof. Near each sphere the other modes, and any entire remainder, are smooth and therefore contribute a bounded amount to the shell density. Theorem 9.9 applies inside each tube separately; adding the n laws and absorbing the fixed terms $8\mathcal{R}_{p_j}^2 \log R_j$ and the n renormalized constants into C gives (10.16). There is no cross term because the tubes are disjoint. $\square$

Remark 10.16 (What the global statements do not assert). Three limitations are worth stating explicitly, since the merged statements above are deliberately sharp. First, every classification here is within the *axial* class; the uniqueness step in Theorem 9.3 is the only statement in Parts VIII–IX that is proved without axiality, and nonaxial fields with the same surface source are not classified. Second, all constants degenerate as $v \rightarrow 0$: they contain $1/v$, $1/v^3$ and $1/v^4$, so nothing here survives a sphere collapsing onto the real axis, and no uniform statement over such a degeneration is claimed. Third, Theorem 10.3 needs simple connectivity of the base, and we do not assert a realization theorem over a base with holes; over such a base the prescribed data must in addition be compatible with the period lattice computed by the split range theorem of Part IV.

11 Examples and separations

The nine research manuscripts merged here carried, between them, some forty worked examples, most of them duplicates of one another. The eight collected below are exactly the ones that *separate* concepts: each is the unique reason that some hypothesis in Parts I–IX cannot be dropped, or that two notions which look interchangeable are not. Each is proved where it is first needed; this section does not reprove any of them. What it adds is the comparison — which hypothesis each one kills, and which pairs of notions each one pulls apart — together with a closing table that reads the whole catalogue against the theorems.

Throughout, $p = u + \iota v$ with $v > 0$, $S_p = u + v\mathbb{S}$, $\xi = x - u$, $\eta = y - v$, $\rho = |z - p| = \text{dist}(q, S_p)$, and $G_{p;a,b} = \Delta_4 \mathcal{I}((za + b)(2\pi\iota)^{-1} \log(z - p))$ is the logarithmic mode of (5.1), with period pair (a, b) by (5.5).

11.1 $G_{p;0,1}$ against $G_{p;1,0}$: the two obstructions are independent

Example 5.8 exhibits the pair $\text{Per}(G_{p;0,1}) = (0, 1)$ and $\text{Per}(G_{p;1,0}) = (1, 0)$ on $U = \mathbb{C}^+ \setminus \{p\}$.

What it separates. It separates the *first* integration from the *second*. For $G_{p;0,1}$ the form ω_g is exact, so the slope potential $C = B/y$ does have a global single-valued primitive and the first of the two classical integrations succeeds — yet there is no global slice-regular primitive, because θ_g is not exact. A construction that verified only the exactness of the first closed form would therefore report success and be wrong. For $G_{p;1,0}$ the failure is already at the first step. Neither test implies the other, and by Proposition 3.4 the second is not an artifact of a badly chosen first primitive: θ_g is computed from the target alone. Right multiplication by $1, \mathbf{i}, \mathbf{j}, \mathbf{k}$ turns each family into four real directions, which is how the count $8m$ of (5.10) becomes visible one coordinate at a time.

A second separation carried by the same pair. The obstruction is global and leaves no local trace on $\mathbb{R}$: by (6.9) the real-axis traces of $G_{\iota;0,1}$ and $G_{\iota;1,0}$ are the perfectly regular functions $4x/(\pi(1+x^2)^2)$ and $-4/(\pi(1+x^2)^2)$. Both fields have $\mathcal{R}_p(a, b)^2 = 1$, so by (8.18) and (9.16) they obey the *same* leading law $\rho M_g^{\text{sph}}(\rho) \rightarrow 1/\pi$ and the same energy law with coefficient 8. The two obstructions are independent but isometric, and no size measurement can tell them apart — only the pair of periods can.

11.2 The stem $F = \iota$: the kernel is $\text{Aff}_{\mathbb{H}}$, not the complex-affine stems

The stem $F \equiv \iota$ has $A = 0$, $B = 1$, hence $f(x + Iy) = I$ and, by the Fueter formula (2.49), $\Delta_4 \mathcal{I}(\iota) = -2I/y^2 \neq 0$; more generally $\Delta_4 \mathcal{I}(\iota(za + b)) = -2a/y - 2I(xa + b)/y^2$, which is (2.51), and which reappears as the kernel of the holomorphic invariant in Remark 4.7.

What it separates. Three things at once.

- (i) It separates $\text{Aff}_{\mathbb{H}} = \{qa + b : a, b \in \mathbb{H}\}$ from the stems affine over $\mathbb{H}_{\mathbb{C}}$. Replacing $\mathbb{H}$ by $\mathbb{H}_{\mathbb{C}}$ would double the kernel and erase the obstruction entirely, since the period pair takes values in $\mathbb{H}^2$ precisely because the kernel does. This is the single most frequently repeated warning in the source manuscripts, and this one-line computation is the whole of its content.
- (ii) It separates the period criterion from the holomorphic invariant. The field (2.51) has zero periods on every loop — it is a global Fueter image — and it spans $\ker \mathcal{H}_g$ by Remark 4.7. So $\mathcal{H}_g$ is not injective on a product domain, and “ $\mathcal{H}_g$ extends” is a statement about g only modulo this eight-dimensional space.
- (iii) It separates product domains from slice domains meeting $\mathbb{R}$. The factor y^{-2} blows up at the axis, so smooth continuation across a real interval forces $a = b = 0$; $\ker \mathcal{H}_g$ is eight-real-dimensional on a product domain and vacuous on a symmetric slice domain. Both statements are needed, because Part III states the range theorem in both settings.

11.3 $\Delta_4 \mathcal{I}((z - p)^{-1}c)$: zero periods without removability

Example 11.1 (Exact order and exact integrability of the rational mode). Let $0 \neq c \in \mathbb{H}_{\mathbb{C}}$ and let $g = \Delta_4 \mathcal{I}(F)$ with $F(z) = (z - p)^{-1}c$, on the punctured tube $T_R \setminus S_p$. Since F is single valued there, g is a global Fueter image, so both of its periods vanish and it has a global single-valued slice-regular primitive; its holomorphic invariant is $\mathcal{H}_g = F'' = 2(z - p)^{-3}c$, a pole of order exactly three with $h_{-1} = h_{-2} = 0$. For $c = 1$ the components are explicit: from $(\xi + \iota\eta)^{-1} = (\xi - \iota\eta)/\rho^2$ one has $A = \xi/\rho^2$, $B = -\eta/\rho^2$, and the Fueter formula gives

$$P = -\frac{4\xi\eta}{y\rho^4}, \quad Q = \frac{2(\eta^2 - \xi^2)}{y\rho^4} + \frac{2\eta}{y^2\rho^2}. \quad (11.1)$$

Because $4\xi^2\eta^2 + (\eta^2 - \xi^2)^2 = (\xi^2 + \eta^2)^2 = \rho^4$, the leading squared pair norm is

$$|P|^2 + |Q|^2 = \frac{16\xi^2\eta^2 + 4(\eta^2 - \xi^2)^2}{v^2\rho^8} + O(\rho^{-3}) = \frac{4}{v^2\rho^4} + O(\rho^{-3}), \quad (11.2)$$

so $M_g^{\text{ax}}(\rho) \sim 2/(v\rho^2)$: the order ρ^{-2} is attained *uniformly*, with no dependence on the meridian angle and no exceptional direction $I \in \mathbb{S}$ along which the field is smaller. Consequently g is locally L^s exactly for $0 < s < 1$, since $\int_0 \rho^{1-2s} d\rho < \infty$ only then, and in particular g is *not* locally L^1 : its planar integral diverges like $\int_0 \rho^{-1} d\rho$. It is nonetheless not removable, and this needs no quantitative input: a removable g would have a stem primitive $F_\sharp$ on the full disk, and the affine-kernel theorem would force $F - F_\sharp = za' + b'$ on the punctured disk, impossible because F has a pole.

What it separates. It pulls apart three notions that the L^1 theorem identifies and that nothing else does:

$$\text{a global primitive on } T_R \setminus S_p, \quad \text{Res}_{S_p}^{\text{aff}}(g) = (0, 0), \quad \text{removability through } S_p.$$

The first two are equivalent on a punctured tube for *every* axially monogenic g , with no hypothesis — that is the content of the period criterion. Neither implies the third. This is exactly why the classification theorem of Part VII is stated with the local integrability hypothesis (Theorem 8.6) and not with the vanishing of the periods, and it is the reason that theorem's two sharp thresholds are two *different* phenomena (Remark 8.19): at L^1 the periods become a complete invariant, at L^2 they must vanish, and the first threshold is not a consequence of the second. The field above sits immediately below the first threshold, its planar integral diverging like $\int_0 \rho^{-1} d\rho$.

Why the order matters. The leading squared pair norm $4/(v^2\rho^4)$ is *independent of the meridian angle and of $I \in \mathbb{S}$* . So the order ρ^{-2} is attained uniformly, with no exceptional quaternionic direction along which the field is smaller — which is what rules out any attempt to rescue removability by evaluating on a well-chosen slice.

11.4 $\Delta_4\mathcal{I}(\exp((z-p)^{-1}))$: an essential spherical singularity

The same construction with $F(z) = \exp((z-p)^{-1})$ gives zero periods, an essential holomorphic encoding $\mathcal{H}_g = F''$, and therefore no bound $M_g^{\text{sph}}(\rho) = O(\rho^{-N})$ for any N .

What it separates. It separates the hypothesis-free normal form $g = g_{\text{reg}} + G_{p;a,b} + \Delta_4\mathcal{I}(\sum_{k \geq 1} (z-p)^{-k} c_k)$ of Theorem 8.3 from the finite-growth normal form of Theorem 8.11: the series really can have infinitely many nonzero coefficients, so the filtration by N does not exhaust the axially monogenic germs at a sphere. Together with Example 11.3 it shows that the growth filtration is a genuine filtration in both directions — there are germs above every finite level, and germs below L^1 that the residues do not see.

11.5 The shell and the deleted real points: surjectivity on multiply connected domains

Example 6.10 treats $\Omega_D = \{1 < |q| < 2\}$, where D is the planar annulus: its single hole is conjugation invariant, the upper meridian is simply connected, and the Fueter map is *onto*, although $b_1(D) = 1$. Corollary 6.11 treats $\Omega = \mathbb{H} \setminus (\bigcup_j S_{p_j} \cup \{c_1, \dots, c_\ell\})$ with real c_k , where the cokernel has real dimension $8m$ independent of ℓ .

What it separates. It separates the obstruction from the first Betti number. With s conjugation-fixed holes and m exchanged pairs one has $b_1(D) = s + 2m$ and $\dim_{\mathbb{R}} \text{coker } \Delta_4 = 8m$: not $8b_1(D)$ and not $8(s + 2m)$. The shell is the case $s = 1, m = 0$, where the naive count predicts an eight-dimensional obstruction and the truth is zero; the deleted real points are the

same phenomenon with ℓ degenerate fixed holes. The bridge that makes the two indices of Part V agree — exchanged pairs on the one hand, winding basis of the upper meridian on the other — is Lemma 6.5, and these two examples are where it is felt.

What it does not say. Corollary 6.11 is about *inversion*, not removability. Real-point singularities keep their own principal parts: by (6.8), for real c one has $\Delta_4((q - c)^{-1}) = -4(q - c)/|q - c|^4 = -8\pi^2 E(q - c)$, so $q \mapsto (q - c)^{-1}$ is, up to a constant, a global slice-regular primitive of a field with a genuinely nonremovable isolated singularity at c . A primitive exists; the singularity does not go away. Dropping this caveat turns a true inversion theorem into a false removability theorem.

11.6 $p = \iota$, $a = 1$, $b = \mathbf{i}$: the correction is $\text{Im}_{\mathbb{H}}(b\bar{a})$, not commutativity

Example 11.2 (A commuting residue pair with a nonzero correction). Take $p = \iota$, so $u = 0$ and $v = 1$, and let $a = 1$, $b = \mathbf{i}$. Then

$$\mathcal{R}_p(a, b)^2 = |ua + b|^2 + v^2|a|^2 = 1 + 1 = 2, \quad \text{Im}_{\mathbb{H}}(b\bar{a}) = \text{Im}_{\mathbb{H}}(\mathbf{i}) = \mathbf{i}, \quad |\text{Im}_{\mathbb{H}}(b\bar{a})| = 1,$$

so the two leading-size laws of Part VII give two *different* constants,

$$\lim_{\rho \downarrow 0} \rho M_g^{\text{ax}}(\rho) = \frac{\mathcal{R}_p(a, b)}{\pi v} = \frac{\sqrt{2}}{\pi}, \quad \lim_{\rho \downarrow 0} \rho M_g^{\text{sph}}(\rho) = \frac{\sqrt{\mathcal{R}_p(a, b)^2 + 2v|\text{Im}_{\mathbb{H}}(b\bar{a})|}}{\pi v} = \frac{2}{\pi}. \quad (11.3)$$

The mechanism is visible pointwise. Approaching the sphere along the meridian direction $\varphi = 0$, the uniform leading term of Part VII reads

$$g(\rho + I) = \frac{\mathbf{i} + I}{\pi \rho} + O(1 + |\log \rho|), \quad (11.4)$$

which *vanishes* at $I = -\mathbf{i}$ and attains the modulus $2/(\pi\rho)$ at $I = \mathbf{i}$. Here $a = 1$ and $b = \mathbf{i}$ commute, so the correction term is not governed by $ab - ba$.

What it separates. Two things.

- (i) It separates the axial pair norm from the spherical supremum. These are different functions of ρ with different limits, and the source manuscripts of this merge wrote both with the same symbol M_g , which makes their statements contradictory as printed. Here $a = 1$ and $b = \mathbf{i}$ *commute*, so the vanishing condition for the correction term is not $ab = ba$; it is $\text{Im}_{\mathbb{H}}(b\bar{a}) = 0$ (Remark 8.22). Any merged statement that quietly identifies the two norms, or that offers commutativity as the criterion, is wrong on this example.
- (ii) It separates pointwise cancellation from removability. The leading term vanishes identically along the ray $\{q = \rho - \mathbf{i}\}$, that is, on an entire one-parameter family of points of the four-dimensional domain, and the singularity is nonetheless a nonremovable critical one with a nonzero residue pair. Evaluating an axially monogenic field on one preselected complex slice is therefore not a substitute for the supremum over $\mathbb{S}$, and a numerical test that sampled a single slice would report a smaller order than the true one.

11.7 $a \neq 0$, $b = -ua$: a nonremovable singularity of zero total flux

By (9.9) this data gives $\Phi_0 = 8\pi v(ua + b) = 0$ at every radius, while $\Phi_1 = -8\pi v^3 a \neq 0$; the singularity is nonremovable, its germ carries all eight real obstruction coordinates, and its energy diverges logarithmically with coefficient $8v^2|a|^2$. For $p = \iota$, $a = 1$, $b = 0$ this is $G_{\iota, 1, 0}$, whose surface source $\sigma_p(s) = \frac{2}{v}(sa + b) = 2I$ on $\mathbb{S}$ has *zero total mass* and nonzero L^2 norm.

What it separates. It separates the ordinary quaternionic flux from the affine spherical residue. A single scalar flux is a one-quaternion functional and cannot recover a two-quaternion

invariant; the second, conserved flux of Theorem 9.6 is what completes it. The choice of weight there is not cosmetic: the naive $q - u$ satisfies $\sum_{\mu} \partial_{x_{\mu}}(q - u)e_{\mu} = -2 \neq 0$, so it is not right-Fueter-regular and its flux is not radius-independent, whereas $H_u(q) = 3(\operatorname{Re} q - u) + \operatorname{Im} q$ satisfies $\sum_{\mu} \partial_{x_{\mu}} H_u e_{\mu} = 3 - 3 = 0$ (Remark 9.7). The two expressions agree numerically because H_u restricted to S_p equals $q - u$, but the moment presupposes knowledge of the density while the flux does not.

A unification worth repeating. By Remark 9.8 that weight is exactly $-\frac{1}{4}\Delta_4((q - u)^3) = 3\mathcal{P}_1(q - u)$, the degree-one entire axially monogenic polynomial of (10.1). So the two residues are recovered by pairing the field against the two *lowest entire axially monogenic polynomials*, $\mathcal{P}_0 = 1$ and $\mathcal{P}_1$ — which is the reason the flux pair and the period pair carry the same information.

11.8 The classical catalogue, and why it is still binding

Four failures from the expository trunk of Part I are invoked repeatedly above and are collected here with the statements they delimit.

- (a) *Regular factors do not list roots* (Example 2.20): $(q - \mathbf{i}) * (q - \mathbf{j})$ vanishes at $\mathbf{i}$ but not at $\mathbf{j}$, while $(q - \mathbf{i}) * (q + \mathbf{i}) = q^2 + 1$ vanishes on the whole sphere $\mathbb{S}$; the two share the symmetrization $(q^2 + 1)^2$. The symmetrization sees a conjugacy class, not a point. *Binding on:* the fact that the singular set of Parts VII–IX is an entire two-sphere, never an isolated nonreal point, and that the transverse geometry is codimension two.
- (b) *The unqualified open mapping theorem fails* (Example 2.28): q^2 is not open at $\mathbf{i}$, because $\operatorname{Im}(q^2) = 2x \operatorname{Im} q$ forces $|q - \mathbf{i}| \geq 1$ whenever $q^2 = -1 + t\mathbf{j}$ with $t \neq 0$. *Binding on:* the need to remove the *closure* of the degenerate set in the restricted open-map statement of Part I, and on the general principle that a quaternionic theorem must name which regularity it uses.
- (c) *Fueter regularity kills identity and open mapping* (Example 2.50): $L(q) = x_1 - \mathbf{i}x_0$ is Fueter regular, vanishes on the plane $\{x_0 = x_1 = 0\}$ without being zero, and has image inside one slice. *Binding on:* every uniqueness argument in Parts VII–IX, which must therefore run through *axial* data and the planar Vekua system, and never through an accumulation-point identity theorem for $\mathcal{D}$.
- (d) *A zero flux residue without removability* (Example 2.49): $\partial_{x_0} E$ is regular off the origin, homogeneous of degree -4 , and even, while the outward normal is odd, so $\operatorname{Res}_0^F(\partial_{x_0} E) = 0$ although the singularity is not removable. *Binding on:* Example 11.7, of which it is the exact isolated-point ancestor, and on the standing rule that the three residue notions of this document — the slice Laurent residue Res_c at a real isolated singularity (Part I), the Cauchy–Fueter flux residue Res^F (Part I), and the affine spherical residue $\operatorname{Res}_{S_p}^{\operatorname{aff}}$ (Definition 8.2) — are three different objects and are never to be identified. Their removability exponents belong to different geometries: $o(\rho^{-1})$ at a nonreal sphere of codimension two, against $o(r^{-3})$ at an isolated point in four dimensions.

11.9 The catalogue read against the theorems

Example	Separates	Would make true
$G_{p;0,1}$ vs. $G_{p;1,0}$	first integration / second	“ ω_g exact” as a criterion
$F = \iota$	$\mathbb{H}$ - / $\mathbb{H}_{\mathbb{C}}$ -affine kernel	“ $\ker \mathcal{H}_g = 0$ ”
$\Delta_4 \mathcal{I}((z-p)^{-1}c)$	zero periods / removability	Thm. 8.6 without L^1
$\Delta_4 \mathcal{I}(\exp((z-p)^{-1}))$	finite / infinite principal part	Thm. 8.11 for all germs
shell, deleted real points	obstruction / first Betti number	“ $\dim \text{coker} = 8b_1$ ”
$p = \iota$, $a = 1$, $b = \mathbf{i}$	$M^{\text{ax}} / M^{\text{sp}^{\text{h}}}$; $\text{Im}_{\mathbb{H}}(b\bar{a}) / ab - ba$	one constant for both norms
$a \neq 0$, $b = -ua$	total flux / affine residue	“ Φ_0 classifies”
$L(q) = x_1 - \mathbf{i}x_0$	Fueter / slice regularity	an identity theorem for $\mathcal{D}$
$\partial_{x_0} E$	zero flux residue / removability	“ $\text{Res}^F = 0$ implies removable”

The third column names the statement each example refutes: every one of those nine sentences is a plausible simplification of a theorem in Parts II–IX, and each is false for exactly one reason, recorded above.

12 The consolidated verification account

The ten source manuscripts each carried a computational supplement. They overlap almost completely, and they are consolidated here into a single account, together with a single reconciliation of the numerical certificates and a single disclaimer. Nothing below is offered as a proof; the proofs are in Parts II–IX.

12.1 What is checked symbolically

The following identities are finite algebraic identities in the real components and can be, and have been, verified by exact symbolic differentiation with symbolic real x, y, u, v and symbolic quaternionic parameters. Because every displayed identity is real-linear in each quaternionic component separately, scalar parameters suffice for those checks in which no product of two quaternionic unknowns occurs; where a product does occur — in the layer formula, in the spherical supremum and in the energy identities — genuinely noncommuting data are used.

- (S1) The two axial differential identities for $\mathcal{D}(A + IB)$ and $\Delta_4(A + IB)$, hence the Vekua system $P_x - Q_y = 2Q/y$, $Q_x + P_y = 0$, and the Fueter formula $\Delta_4 \mathcal{I}(F) = 2A_y/y + I(2B_y/y - 2B/y^2)$.
- (S2) Both Vekua equations for the branch-free mode components (5.2)–(5.3), for both specializations $(1, 0)$ and $(0, 1)$ and for the reflected modes.
- (S3) Closedness of ω_g and θ_g , i.e. vanishing of the $dx \wedge dy$ coefficients $(Q_x + P_y)/2$ and $\frac{1}{2}\{y(P_x - Q_y) - 2Q - x(Q_x + P_y)\}$.
- (S4) The four potential derivative identities $C_x = -P/2$, $C_y = Q/2$, $V_x = (xP + yQ)/2$, $V_y = (yP - xQ)/2$, for a holomorphic test stem with genuinely complex $\mathbb{H}_{\mathbb{C}}$ coefficients.
- (S5) The elimination identity $\eta_C - d(xC) = \theta_g$.
- (S6) The second-derivative identity $\mathcal{H}_g = F''$ and the two residue moments $a_\gamma = \int_\gamma \mathcal{H}_g dz$, $b_\gamma = -\int_\gamma z \mathcal{H}_g dz$, together with the equality of the two printed forms of $\mathcal{H}_g$ under $Q_x = -P_y$.
- (S7) The vanishing of $\mathcal{H}_g$ on the complex-affine kernel (2.51).

- (S8) The leading angular energy identity and the two diagonal Gram entries $\pi\rho^2/2$ and $\pi\rho^2(v^2 + \rho^2)/2$, together with the vanishing off-diagonal entry, which is what makes the growth-free energy lower bound of Part VIII sharp.
- (S9) The real-axis limits (6.9) and the identity $\Delta_4(q^3) = -4(3x_0 + \text{Im } q)$, hence $H_u = -\frac{1}{4}\Delta_4((q-u)^3) = 3\mathcal{P}_1(q-u)$.

12.2 What is checked numerically

- (N1) *Periods by contour integration.* Both forms are integrated around circles in the planar base by periodic quadrature. The independent runs behind this merge used a periodic trapezoidal rule with 4 096, 16 384 and 32 768 nodes and, in one case, 45-decimal working precision. Circles both enclosing and not enclosing the singular point were used, as were circles of several radii about the same point (to test radius-independence), reflected modes (to test smoothness across $\mathbb{R}$), and a two-hole superposition with different noncommuting quaternionic data at each hole (to test the full 16-entry period matrix of a two-hole basis). Observed maximum componentwise discrepancies from the predicted pairs $(1, 0)$, $(0, 1)$, $(0, 0)$ ranged from $\sim 10^{-16}$ to $\sim 10^{-13}$ in double precision and below 10^{-46} in the high-precision run.
- (N2) *The layer formula.* The four-dimensional sphere integral $\frac{2}{v} \int_{S_p} E(q-s)(sa+b)dS(s)$ is evaluated by independent spherical quadrature implementing the Hamilton product — not by replacing quaternionic products with scalar multiplication — and compared against the explicit mode at real and nonreal test points with noncommuting (a, b) ; observed component errors were of order 10^{-15} to 10^{-12} .
- (N3) *The two conserved fluxes.* Φ_0 and Φ_1 are evaluated on tubes of several radii with the spherical average taken analytically and the meridian integral by quadrature; observed radius-independence and agreement with $8\pi v(ua+b)$ and $-8\pi v^3 a$ to about 10^{-12} .
- (N4) *Shell energy.* The shell density $J_g(\rho) = \int_{\partial T_\rho} |g|^2 dS$ and the truncated energy $\mathcal{E}_g(\varepsilon, R)$ are computed by Gauss–Legendre or midpoint rules in the meridian angle and in $\log \rho$, using the exact spherical average $\frac{1}{4\pi} \int_{\mathbb{S}} |P + IQ|^2 d\sigma = |P|^2 + |Q|^2$ rather than sampling $\mathbb{S}$. Both $\rho J_g(\rho) \rightarrow 8\mathcal{R}_p(a, b)^2$ and the convergence of $\mathcal{E}_g(\varepsilon, R) - 8\mathcal{R}_p^2 \log(R/\varepsilon)$ to a finite limit were observed.
- (N5) *Higher-order energy scale.* The leading shell energy for top Laurent coefficients of orders $k = 1, 2, 3$ with genuinely complexified $\mathbb{H}_{\mathbb{C}}$ coefficients was compared with $16\pi^2 k \|c_k\|_{\mathbb{H}_{\mathbb{C}}}^2 \varepsilon^{-2k}$; observed relative errors below 5×10^{-10} at $\rho = 10^{-4}$.
- (N6) *Supremum constant.* $\rho M_g^{\text{sph}}(\rho)$ is computed using the exact algebraic formula $\max_I |P + IQ|^2 = |P|^2 + |Q|^2 + 2|\text{Im}_{\mathbb{H}}(P\bar{Q})|$, so that only the meridian angle is sampled and no sampling error over $\mathbb{S}$ is incurred.

12.3 Reconciliation of the numerical certificates

Six independent parameter sets were used across the source manuscripts, with no two the same. They are reconciled here: every one of them reproduces from the two formulas

$$8\mathcal{R}_p(a, b)^2 = 8(|ua+b|^2 + v^2|a|^2), \quad \lim_{\rho \downarrow 0} \rho M_g^{\text{sph}}(\rho) = \frac{1}{\pi v} \sqrt{\mathcal{R}_p(a, b)^2 + 2v|\text{Im}_{\mathbb{H}}(b\bar{a})|}.$$

(u, v)	(a, b)	$8\mathcal{R}_p(a, b)^2$	$\lim_{\rho \downarrow 0} \rho M_g^{\text{sph}}(\rho)$
(0.3, 1.2)	$a = 1 - 2\mathbf{i} + \frac{1}{2}\mathbf{j} + \mathbf{k}$ $b = \frac{1}{2} + \mathbf{i} - \mathbf{j} + 2\mathbf{k}$	$253/2 = 126.5$	$1.4724221755937\dots$
(0.3, 1.7)	$a = 1 - 2\mathbf{i} + \frac{1}{2}\mathbf{j} + 3\mathbf{k}$ $b = -0.4 + \mathbf{i} + 2\mathbf{j} - \mathbf{k}$	367.88	$1.611890081993\dots$
(0.3, 1.7)	$a = 1 + \frac{1}{2}\mathbf{i} - \frac{1}{4}\mathbf{j} + \frac{3}{4}\mathbf{k}$ $b = 0.2 - 0.4\mathbf{i} + 0.3\mathbf{j} - 0.1\mathbf{k}$	46.38	—
(0.5, 2)	$a = 1 + 2\mathbf{i} - \mathbf{j}$ $b = -\mathbf{i} + 3\mathbf{j} + \mathbf{k}$	252	—
(0.4, 1.3)	$a = 1 + 2\mathbf{i} - \mathbf{j} + \frac{1}{2}\mathbf{k}$ $b = \frac{1}{4} - \mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$	181	—
(0.4, 1.7)	$a = 1 + 0.2\mathbf{i} - 0.3\mathbf{j} + 0.7\mathbf{k}$ $b = -0.4 + 0.5\mathbf{i} + 0.9\mathbf{j} - 1.1\mathbf{k}$	50.392	—

Two of these deserve comment. In the first row, $\mathcal{R}_p^2 = 253/16$ and $\text{Im}_{\mathbb{H}}(b\bar{a}) = 4\mathbf{i} + 3.75\mathbf{j} + 3\mathbf{k}$ has modulus 6.25, so the spherical constant is $\sqrt{253/16 + 15}/(1.2\pi) = 1.47242\dots$, whereas the *pair-norm* constant for the same data is $\mathcal{R}_p/(\pi v) = 1.05480\dots$; the two differ by nearly forty percent and are limits of different quantities. In the second row, $\mathcal{R}_p^2 = 45.985$ and $|\text{Im}_{\mathbb{H}}(b\bar{a})| = \sqrt{68.42} = 8.27164\dots$, giving $\sqrt{45.985 + 28.12357}/(1.7\pi) = 1.61189\dots$. These two certificates were produced independently, by manuscripts that did not share a formula; that they reproduce from the *same* spherical-supremum expression is the strongest available evidence that the exact leading-size law of Part VII is correctly normalized.

The observed convergence is of the expected order. For the fifth row (with $R = 0.3$) the recorded shell density and renormalized remainder were

ε	$\mathcal{E}_g(\varepsilon, R)$	$\mathcal{E}_g(\varepsilon, R) - 181 \log(R/\varepsilon)$	$\rho J_g(\rho)$ at $\rho = \varepsilon$
10^{-2}	626.9031484	11.28642233	181.2151049
10^{-3}	1043.8028211	11.41819314	181.0049681
10^{-4}	1460.5735515	11.42102173	181.0000892
10^{-5}	1877.3415023	11.42107075	181.0000014

The middle column is the numerical appearance of E^{ren} : it stabilizes, rather than merely staying bounded, which is exactly the distinction between the weaker $O(1)$ form of the energy law and the finite-limit form adopted in Part VIII. For the first row the corresponding table of $\rho M_g^{\text{sph}}(\rho)$ and $\rho J_g(\rho)$ reads 1.49118, 1.47469, 1.47269, 1.47245 and 126.669, 126.504, 126.5001, 126.50000 at $\rho = 10^{-2}, \dots, 10^{-5}$, approaching the proved limits 1.4724221756 and 126.5 at the predicted rate $O(\rho \log(1/\rho))$.

Remark 12.1 (What the computations do and do not establish). These are checks of formulas, signs, normalizations and multiplication order. They are not proofs. In particular they do not establish the topological statements (connectedness of the upper meridian, the identification of the two hole counts, the exactness of the range sequence), the classification theorems (the hypothesis-free normal form, the L^1 theorem, the growth filtration), the realization theorem, or any priority claim; all of those rest on the written arguments of Parts II–IX. Quadrature discrepancies reported above are *observed* floating-point errors, not certified interval bounds. Symbolic checks verify finite algebraic identities only. No proof assistant was used on any statement in this document.

Two specific failure modes are worth recording as excluded by the checks rather than by the prose. First, the central unit ι of $\mathbb{H}_{\mathbb{C}}$ appearing in a holomorphic logarithm is never identified with a variable quaternionic $I \in \mathbb{S}$; the period tests use quaternionic data whose imaginary parts are not collinear, so a computation that silently restricted all coefficients to one complex

slice would fail them. Second, all quaternionic coefficients multiply on the right, and the layer quadrature implements the Hamilton product; a computation that replaced quaternionic multiplication by scalar multiplication would agree on the commuting test data and fail on these.

13 Scope, literature boundary, and open problems

This section states once, in one editorial voice, what this document claims, what it declines to claim, and what it leaves open. The ten source manuscripts each carried a version of it; they did not all take the same position, and the reconciliation below is deliberate rather than a composite.

13.1 The contribution ledger

Established background, for which no priority is claimed. The stem formalism for slice functions and the representation formula; Fueter’s mapping theorem $\mathcal{D}(\Delta_4 f) = 0$ and the biharmonicity $\Delta_4^2 f = 0$; the affine kernel $\ker(\Delta_4|_{\mathcal{SR}}) = \text{Aff}_{\mathbb{H}}$; the axial Vekua system; the local inverse obtained by two successive integrations on a simply connected base, together with its own period caveat; the spherical Cauchy and arctangent kernels attached to $p = \iota$; classical one-variable Runge [11] and Mittag–Leffler approximation and interpolation; Weyl’s lemma; and the standard cutoff/capacity method for L^2 removability of a first-order elliptic system. The entire classical apparatus assembled in Part I — the slice Cauchy formula, real-centred Taylor and Laurent theory, the regular product and symmetrization, the zero geometry and the argument principle, Schwarz–Pick and the intrinsic Riemann mapping theorem, the Cauchy–Fueter Green identity and its consequences — is likewise expository synthesis of the cited sources.

Proposed contributions. The connected package is:

- (a) the two *simultaneously* defined, primitive-independent closed one-forms ω_g and θ_g , computed from the target alone, together with the elimination identity $\eta_C - d(xC) = \theta_g$ that shows the second obstruction is not a residual of the first;
- (b) the same-domain solvability criterion with *no* topological and no boundary-regularity hypothesis, and its identification with the affine monodromy homomorphism $H_1(U; \mathbb{Z}) \rightarrow \mathbb{H} \oplus \mathbb{H}$;
- (c) the split range theorem with its explicit section, its $8m$ cokernel, and — via the reflection analysis — the correct hole count, together with the lemma that every conjugation-invariant hole meets the real axis, which is what reconciles the winding-basis index with the exchanged-pair index;
- (d) the holomorphic invariant $\mathcal{H}_g$, its exact image $\mathcal{O}_{\text{mom}}(U)$, its exact kernel, the recovery of every multipole by an ordinary contour integral, and the reality constraint $h_{-1} = (\iota/v) \text{Re}_{\mathbb{C}}(h_{-2})$ with its consequences (no isolated simple pole of $\mathcal{H}_g$ at a nonreal p ; the sharp criterion $h_{-2} = 0$);
- (e) the local classification: the hypothesis-free normal form, the L^1 theorem with two independent proofs, the growth filtration with its $8N$ count, the sharp removability thresholds, the L^p trichotomy, and the exact leading constants in both norms;
- (f) the surface source $\mathcal{D}g = \frac{2}{v}(qa + b)\delta_{S_p}$, the layer representation with uniqueness, the two conserved three-dimensional fluxes, and the energy law $\mathcal{E}_g(\varepsilon, R) = 8\mathcal{R}_p(a, b)^2 \log(R/\varepsilon) + E^{\text{ren}} + o(1)$ with its sharp growth-free lower bound;
- (g) the realization (Mittag–Leffler) theorem and the global normal form with mixed transverse orders and polynomial growth.

13.2 The literature boundary, and a single editorial position

The literature audit behind this document was carried out on 21 September 2026 and was targeted rather than exhaustive. Inspected primary sources include the slice-function framework of Ghiloni–Perotti [3]; the inverse Fueter mapping theorem of Colombo–Sabadini–Sommen [2] and its integral form using spherical monogenics; the integral-formula paper of Colombo–Peña Peña–Sabadini–Sommen, whose Theorem 3 is stated on a *rectangular* axial base and whose Example 2 exhibits the spherical kernels; the contour surjectivity formulation of Dong–Qian; Perotti’s local Cauchy integral formula, whose Theorem 2 asserts surjectivity under the hypothesis that every component of the symmetric planar set is simply connected; Kraußhar–Perotti on eigenvalue problems for slice functions; and the 2025–2026 preprints of Colombo–De Martino–Sabadini and De Martino–Pinton, which concern Laurent expansions of axially harmonic and polyanalytic classes and a polyanalytic Cauchy–Kovalevskaya extension rather than the ordinary four-dimensional Laplacian studied here. Common–Sommen’s axial construction and the 2016 inversion paper of Dong–Kou–Qian–Sabadini were bibliographic leads whose complete texts were not inspected. The absence of a matching statement from a bounded search is not a proof of absence from the published literature, and a specialist review remains necessary before any priority claim.

Remark 13.1 (The position taken on published surjectivity statements). The source manuscripts of this merge did not agree on this point, and a single position is adopted here.

- (i) What this document replaces is an *unqualified same-domain surjectivity assertion*: the statement that every axially monogenic function on an axially symmetric domain is the Fueter image of a single-valued slice-regular function on that same domain. Part V exhibits explicit obstructed fields on $\mathbb{H} \setminus \mathbb{S}$, so such an assertion is false as stated, and Parts II and IV supply the necessary-and-sufficient criterion and the exact size of the failure.
- (ii) The simply-connected sufficient condition is correct as hypothesized. What is removed here is the hypothesis, not the theorem.
- (iii) No claim is made that any particular published theorem, stated with its own domain, locality and branch hypotheses, is incorrect. Statements called “inverse” or “surjectivity” theorems in this literature occur with different domain and continuation conventions — rectangular bases, contour formulations, local integral formulas — and neither the word “inverse” nor the existence of a local integral formula by itself removes monodromy. Adjudicating any individual published sentence would require auditing its own domain, locality and branch hypotheses, which this document does not do.

Remark 13.2 (Attribution of the logarithmic modes). The two-parameter family $G_{p;a,b}$ is not claimed as new. For $p = \iota$ one has $L'_\iota(z) = \pi^{-1}(1 + z^2)^{-1}$, so a local primitive is $\pi^{-1} \arctan z$ up to an additive real constant, and the fields with local primitive stems $(2\pi)^{-1} \arctan z$ and $(2\pi)^{-1} z \arctan z$ are the spherical Cauchy kernels N^+ and N^- ; in that normalization $G_{\iota;0,1} = 2N^+$ and $G_{\iota;1,0} = 2N^-$. This identification is internally consistent with the normalizations used throughout this document and is stated *with* them, because it is normalization-dependent. The attribution itself is unsettled in the source material: some of the merged manuscripts cite Colombo–Sabadini–Sommen (2011), Theorem 5.1, for the arctangent primitives and sphere-integrated Cauchy kernels, and others cite Colombo–Peña Peña–Sabadini–Sommen (2014), Example 2. At most one of these is the primary locus. The bibliography of this document should not print either attribution without direct verification against both papers; until that is done, the safe statement is that these kernels are known and that what is developed here is their *period normalization*, their use as a complete obstruction basis, and the classification, source, flux and energy laws attached to them.

13.3 Boundaries: what the theorems do not assert

- (B1) *Axiality.* Every classification statement in Parts VII and IX is in the *axial* class. It is not asserted that an arbitrary Cauchy–Fueter-regular field is the Laplacian of a slice-regular function, nor that a general monogenic field with a spherical singularity is classified by two quaternions. A nonaxial field can carry far richer tangential data along the sphere; the admissible source density there need not be affine, and an infinite-dimensional analysis is to be expected. The exceptions, explicitly noted where they occur, are the L^2 removability theorem of Part VII and the uniqueness half of the layer representation of Part VIII, neither of which uses axiality.
- (B2) *Positive radius.* All spherical results concern a nonreal sphere S_p of radius $v > 0$. The constants $2/v$, $8\pi v$, $8\pi v^3$ and the residue norm $\mathcal{R}_p(a, b)^2 = |ua + b|^2 + v^2|a|^2$ all degenerate as $v \downarrow 0$, and none of the statements is uniform in that limit. The collapse of a two-sphere to a real point is a different local problem, with a different removability exponent; it cannot be read off by setting $v = 0$ in any formula here.
- (B3) *Real points and nonaxial fields are different problems.* Corollary 6.11 says real punctures create no *inversion* obstruction. It does not say their singularities are removable, and it does not classify their principal parts.
- (B4) *Finite topology.* The period criterion of Part II requires no finite-topology hypothesis of any kind, and the closed-range statement of Part VI holds on every connected base, even with infinitely many holes. The finite-dimensional cokernel statements, by contrast, use the explicit finite winding basis of Part IV; no finite-dimensional cokernel is asserted for a base with infinitely many independent holes.
- (B5) *Topology of the splitting.* The projection $\Pi = \text{id} - S\text{Per}$ and the normalized inverse are continuous for compact-open convergence, with domain- and path-dependent constants. They are *not* asserted to be bounded in any unspecified weighted, Hardy, Bergman, Sobolev or Hilbert norm, and the direct sum $\mathcal{AM} = \Delta_4\mathcal{SR} \oplus S(\mathbb{H}^{2m})$ is not asserted to be orthogonal in any such norm. No uniform conditioning is claimed when holes collide or approach the real axis.
- (B6) *Approximation.* The Runge statement of Part VI concerns locally uniform convergence with all derivatives on compact subsets, not interpolation on a contractible patch, and the two settings have genuinely different conclusions about the approximants: on a product domain with a one-sided pole set the reflected slice functions need not be rational with coefficients in $\mathbb{H}$, whereas on a symmetric slice domain with a conjugation-symmetric pole set they are.
- (B7) *Energy.* “Energy” means the squared L^2 norm $\int |g|^2 dV$, not the Dirichlet integral $\int |\nabla g|^2$. The distinction matters for the critical exponent and for the logarithm.
- (B8) *The constant 8 is codimension-two geometry.* It arises as the angular length 2π of the transverse circle times the area 4π of $\mathbb{S}$, divided by the π^2 of the logarithmic normalization $1/(2\pi\iota)$. The $2\pi^2$ of the point Cauchy–Fueter kernel belongs to a different geometry and does not appear in any of these constants.

13.4 Open problems

- (O1) *Uniform estimates under degeneration.* The path-dependent constants in the continuous inverse of Part VI, and the $1/v$ powers in the spherical constants, both obstruct any uniform statement for a family of parameter domains whose holes collide, or for a sphere collapsing

onto the real axis. Find the correct rescaled limits. The exact leading asymptotics of Part VII, rather than the upper bounds, are the natural input.

- (O2) *Infinitely connected bases.* The criterion holds and the range is closed for every connected base, but the splitting of Part IV is constructed from a finite winding basis. Is there a convergent kernel splitting, with sharp norm control, on a base with infinitely many holes? The naive sum of logarithmic modes diverges — which is exactly the difficulty the Mittag–Leffler realization theorem of Part IX resolves in the discrete locally finite case by prescribing the meromorphic second derivative instead — so the question is whether an analogous device controls norms and not merely convergence.
- (O3) *A weighted Hilbert theory.* The energy law identifies a canonical positive-definite quadratic form $8\mathcal{R}_p(a, b)^2$ on the obstruction space, with the geometric and flux expressions $8\mathcal{R}_p^2 = \frac{1}{2\pi} \|\sigma_p\|_{L^2(S_p)}^2 = \frac{1}{8\pi^2} (|\Phi_0|^2/v^2 + |\Phi_1|^2/v^4)$. Is there a weighted Hilbert space of axially monogenic fields in which the critical energy norm controls a canonical inverse after period subtraction, so that Π becomes an orthogonal projection and the normalized inverse a bounded operator? Nothing proved here implies such boundedness.
- (O4) *Higher order: the Fueter–Sce map.* For Fueter–Sce operators in a higher odd number of spatial variables the local kernel is a polynomial space of higher degree. That suggests a corresponding higher polynomial monodromy — one period coordinate for each kernel coefficient — and a family of closed forms $\omega_g^{(0)}, \dots, \omega_g^{(N)}$ generalizing the pair (ω_g, θ_g) , together with a higher spherical residue and a higher energy law. The proofs here indicate the shape of the invariant, but none of the higher-dimensional formulas is proved, and none is needed for any statement in this document.
- (O5) *Beyond L^1 .* Outside the locally integrable class the affine residues stop being a complete invariant (§11.3). Classify the higher distributional derivatives of spherical sources under weaker integrability or prescribed growth, and compare the resulting hierarchy with the growth filtration of Part VII.

A Distributional toolkit

Part VII proves its sharpest local theorem twice, by two genuinely different mechanisms: a distributional route on the target, and a $W^{1,1}$ route on the primitive. The general lemmas both routes need are collected here, so that no part of the document repeats them. A third subsection records the removability cutoffs, of which Part VII uses one and the sources supplied three.

A.1 The distributional route

Lemma A.1 (Distributions supported at one point). *A distribution T on $\mathbb{R}^n$ supported at 0 is a finite linear combination of derivatives of δ_0 .*

Proof. Continuity of a distribution gives, for a compact neighbourhood K of 0, an order r and a constant C with $|\langle T, \phi \rangle| \leq C \max_{|\alpha| \leq r} \sup_K |\partial^\alpha \phi|$ for all test functions supported in K . Let χ be a smooth cutoff equal to one near 0 and put $\chi_\varepsilon(x) = \chi(x/\varepsilon)$; since $\text{supp } T = \{0\}$ we have $\langle T, \phi \rangle = \langle T, \chi_\varepsilon \phi \rangle$. If all derivatives of ϕ through order r vanish at 0, Taylor’s theorem applied to each derivative gives $\sup_{|x| \leq C_0 \varepsilon} |\partial^\beta \phi(x)| = o(\varepsilon^{r-|\beta|})$ for $|\beta| \leq r$; the product rule and the bounds $|\partial^\alpha \chi_\varepsilon| \leq C \varepsilon^{-|\alpha|}$ then give $\max_{|\alpha| \leq r} \sup |\partial^\alpha (\chi_\varepsilon \phi)| = o(1)$, whence $\langle T, \phi \rangle = 0$. Therefore T depends only on the r -jet of its argument at 0, and linear functionals on that finite-dimensional jet space are exactly linear combinations of derivative evaluations. The $\mathbb{H}$ - and $\mathbb{H}_{\mathbb{C}}$ -valued statements follow componentwise over $\mathbb{R}$. $\square$

The structure theorem alone is not enough. It permits derivatives of δ_p of every order, whereas the classification of Part VII needs *only* a point mass. The missing input is a scaling argument, and it is exactly here that the L^1 hypothesis and the first order of the operator are both consumed.

Lemma A.2 (An L^1 puncture creates only a point mass). *Let*

$$\mathcal{L}(P, Q) = \left(P_x - Q_y - \frac{2Q}{y}, \quad Q_x + P_y \right)$$

be the planar first-order Vekua operator on a disk $B_p(R) \subset \mathbb{C}^+$, so that $1/y$ is smooth and bounded there. If $P, Q \in L^1(B_p(R); \mathbb{H})$ and $\mathcal{L}(P, Q) = 0$ away from p , then there are unique $s_0, s_1 \in \mathbb{H}$ with

$$\mathcal{L}(P, Q) = (s_0, s_1) \delta_p \tag{A.1}$$

in the sense of distributions. No derivative of δ_p occurs.

Proof. The defect is supported at p , so by Lemma A.1 it is a finite sum of derivatives of δ_p . Suppose the highest nonvanishing order were $r \geq 1$. Choose a test function ϕ whose r -jet at p detects a nonzero top-order coefficient and set $\phi_\varepsilon(z) = \varepsilon^r \phi((z - p)/\varepsilon)$. Pairing the defect against ϕ_ε tends to a nonzero constant, all terms of lower order tending to 0. On the other hand $\mathcal{L}$ is *first* order with bounded coefficients, so moving its derivatives onto the test function gives

$$|\langle \mathcal{L}(P, Q), \phi_\varepsilon \rangle| \leq C(\varepsilon^{r-1} + \varepsilon^r) \int_{|z-p| < C'\varepsilon} (|P| + |Q|) dx dy,$$

and the right-hand side tends to 0 by absolute continuity of the integral of an L^1 function — *including* at $r = 1$, where the prefactor is $O(1)$ and all of the decay comes from the shrinking domain of integration. This contradiction leaves only the point mass, and uniqueness is immediate. $\square$

Remark A.3 (What is used, and where). It is the conjunction of first order and L^1 that does the work: either alone would leave the case $r = 1$ open. This is the step that makes the residue classification of Part VII exhaustive rather than a list of examples, and it is the exact point at which the integrability hypothesis of that theorem is consumed. The $W^{1,1}$ route below consumes the same hypothesis at a different place — on the primitive rather than on the target — which is why the distributional route produces the surface source of Theorem 9.1 directly while the other deduces it afterwards.

The coefficient of the point mass is computed by one circle integral. Write $W = P + \iota Q$ and $\bar{\partial} = \frac{1}{2}(\partial_x + \iota \partial_y)$, so that $\mathcal{L}(P, Q) = 0$ becomes $2\bar{\partial}W = 2Q/y$. For the mode $G = G_{p;a,b}$ the uniform leading term of Part VII gives

$$W = \frac{(ua + b) + \iota va}{\pi v(z - p)} + O(1 + |\log \rho|).$$

The boundary functional for $2\bar{\partial}$ on the circle $z - p = \varepsilon e^{i\varphi}$ is $\int (n_x + \iota n_y) W ds$; its principal part integrates to $2((ua + b) + \iota va)/v$, the logarithmic remainder contributes $O(\varepsilon(1 + |\log \varepsilon|))$, and the lower-order term $2Q/y$ contributes nothing over a shrinking disk because it is L^1 . Equivalently this is the elementary normalization $\bar{\partial}((z - p)^{-1}) = \pi \delta_p$, verified by the same computation. Separating central real and imaginary parts gives the defect of the canonical modes,

$$\mathcal{L}(P_G, Q_G) = \left(\frac{2(ua + b)}{v}, \quad 2a \right) \delta_p. \tag{A.2}$$

Lifted through the volume element $y^2 dx dy d\sigma(I)$ and $dS = v^2 d\sigma(I)$ on S_p , (A.2) is exactly the affine surface source $\mathcal{D}g = \frac{2}{v}(qa + b) \delta_{S_p}$ of Theorem 9.1.

Lemma A.4 (An L^1 Weyl lemma). *If $w \in L^1_{\text{loc}}(V)$ on an open $V \subset \mathbb{R}^n$ and $\Delta w = 0$ distributionally, then w has a smooth harmonic representative.*

Proof. Mollify: $w_\varepsilon = w * \psi_\varepsilon$ is smooth and harmonic on slightly smaller subdomains, with local L^1 norms bounded uniformly by those of w . For a smooth harmonic h , averaging the spherical mean-value identities against a fixed smooth radial weight of total integral one supported in a ball of radius r writes h , on a concentric smaller ball B' , as a convolution with a smooth kernel; differentiating the kernel gives

$$\sup_{B'} |\partial^\alpha h| \leq C_{\alpha,r} \int_B |h| dx$$

for every multi-index α , with $B' \Subset B$ and the kernel supports fitting inside B . Applying this to w_ε converts the uniform L^1 bounds into uniform bounds on all derivatives on compact subsets. Arzelà–Ascoli extraction gives a subsequence converging smoothly on compact sets to a harmonic h , and $w_\varepsilon \rightarrow w$ in L^1_{loc} identifies h as a representative of w . Local representatives agree almost everywhere and hence, being continuous, everywhere. $\square$

For $\mathbb{H}$ -valued fields the lemma is applied to each of the four real components. Together with the factorization $\overline{\mathcal{D}}\mathcal{D} = \Delta_4$ it is the only elliptic regularity input used in Part VII: it upgrades a distributionally monogenic L^1 field to a smooth monogenic one, and a distributionally monogenic L^2 field to a smooth one in the removability theorem. The axial form used there — that weak axial monogenicity across the missing sphere is removable — follows by mollifying P, Q in the planar variables, since pairing the lifted field with a four-dimensional test function amounts to pairing the planar coefficients with y^2 times its spherical averages, and those are again smooth planar test functions.

A.2 The $W^{1,1}$ route

This route never mentions distributions supported at a point. Its content is that in two dimensions a puncture is too small to obstruct an integrable gradient.

Lemma A.5 (A puncture does not obstruct an integrable gradient). *Let $B_R \subset \mathbb{R}^2$ be a disk centred at 0 and $w \in C^1(B_R \setminus \{0\}; \mathbb{R}^N)$. If $\nabla w \in L^1(B_R)$, then w extends, up to its value at the origin, to a function in $W^{1,1}_{\text{loc}}(B_R)$ whose weak gradient is its pointwise gradient off 0.*

Proof. Fix $0 < R_0 < R$. In polar coordinates (ρ, φ) , radial integration gives $|w(\rho, \varphi)| \leq |w(R_0, \varphi)| + \int_\rho^{R_0} |w_t(t, \varphi)| dt$. Multiplying by ρ and integrating in $0 < \rho < R_0$ and in φ shows $w \in L^1(B_{R_0})$, the contribution of w_t being at most

$$\frac{1}{2} \int_0^{2\pi} \int_0^{R_0} t^2 |w_t(t, \varphi)| dt d\varphi \leq \frac{R_0}{2} \int_{B_{R_0}} |\nabla w| dx dy.$$

Moreover, with $H(t) = \int_0^{2\pi} |w_t(t, \varphi)| d\varphi$,

$$\rho \int_0^{2\pi} |w(\rho, \varphi)| d\varphi \leq O(\rho) + \int_\rho^{R_0} \frac{\rho}{t} tH(t) dt \longrightarrow 0 \quad (\rho \downarrow 0),$$

by dominated convergence, since $tH(t)$ is integrable. Integrating by parts against a test function on $B_{R_0} \setminus B_\rho$, the inner boundary term is bounded by the left-hand side above times $\sup|\phi|$, and therefore vanishes in the limit. This is the weak-derivative assertion on B_{R_0} , and $R_0 < R$ was arbitrary. $\square$

Lemma A.6 (Weak Cauchy–Riemann removability). *If $F \in W^{1,1}_{\text{loc}}(B_R; \mathbb{C}^N)$ satisfies $\partial_{\bar{z}} F = 0$ in the sense of distributions, then F agrees almost everywhere with a holomorphic function on B_R .*

Proof. Mollify on slightly smaller disks. The convolutions F_ε are smooth, satisfy the Cauchy–Riemann equation, and converge to F in L^1_{loc} . For a holomorphic H on a disk K' and $K \Subset K'$, the mean-value property on disks gives $\sup_K |H| \leq C_{K,K'} \|H\|_{L^1(K')}$. Applied to the differences $F_\varepsilon - F_{\varepsilon'}$, this makes the family uniformly Cauchy on compact subsets; the holomorphic limit agrees almost everywhere with F . $\square$

Remark A.7 (Where the hypothesis enters). The seam with Part VII is the potential identities $dC = \omega_g$ and $dV = \theta_g$ of Part II: the gradients of the eight real components of $C = B/y$ and $V = A - xB/y$ are *exactly* the coefficients of the two one-forms, hence are L^1 near p precisely when g is. So the hypothesis of Lemma A.5 is the local integrability hypothesis of the classification theorem, transported through the potentials with no loss; Lemma A.6 then extends the stem $F = A + \iota B$ holomorphically, and $\Delta_4 \mathcal{I}(F)$ is the regular extension of g . Neither lemma uses the structure of distributions supported at a point, and neither produces the surface source; the two routes are genuinely independent proofs of the same theorem.

A.3 Removability cutoffs and the planar elliptic estimate

Three cutoff schemes for L^2 removability occur in the source material. All are correct. Theorem 8.15 uses the first; the other two are recorded here so that the merged document keeps them without giving them a second proof of the theorem.

- (C1) *Logarithmic cutoff* (the proof adopted in Part VII, because it is shortest and needs no classification machinery, no growth bound and no axially). With $\chi_\varepsilon = 0$ for $\rho \leq \varepsilon^2$, $= 1$ for $\rho \geq \varepsilon$ and affine in $\log \rho$ between, $\|\nabla \chi_\varepsilon\|_{L^2}^2 \leq C/\log(1/\varepsilon) \rightarrow 0$, so the cutoff error itself vanishes.
- (C2) *Linear-shell cutoff*. With $\chi_\varepsilon = 0$ for $\rho < \varepsilon$, $= 1$ for $\rho > 2\varepsilon$ and $|\nabla \chi_\varepsilon| \leq C/\varepsilon$, a compact portion of a smooth codimension-two submanifold has shell volume $O(\varepsilon^2)$, so $\|\nabla \chi_\varepsilon\|_{L^2} \leq C$ — bounded, but *not* tending to zero. The cutoff error $C_\phi \|g\|_{L^2(\{\varepsilon < \rho < 2\varepsilon\})} \|\nabla \chi_\varepsilon\|_{L^2}$ still tends to zero, the decay now coming entirely from absolute continuity of the integral of $|g|^2$ rather than from the cutoff. This variant has the advantage of needing no distance function adapted to S_p beyond smoothness, and it applies verbatim to any smooth embedded two-sphere in a four-dimensional open set.
- (C3) *Planar reduction*. Writing $G = P + \iota Q$, the Vekua system is equivalent to $\partial_{\bar{z}} G = Q/y$, and four-dimensional local L^2 of g is equivalent to two-dimensional local L^2 of G (the spherical average being exact, and y bounded above and away from zero). The linear-shell cutoff in the plane then gives the equation distributionally across the puncture, and regularity follows from the elementary Fourier identity

$$\|\nabla w\|_{L^2} = 2\|\partial_{\bar{z}} w\|_{L^2} \tag{A.3}$$

for compactly supported smooth $\mathbb{C}^N$ -valued w , which holds distributionally whenever $w, \partial_{\bar{z}} w \in L^2$. Localizing G makes its $\bar{\partial}$ -derivative L^2 , so $G \in H^1_{\text{loc}}$; since $1/y$ is smooth, differentiating, relocalizing and reapplying (A.3) give $G \in H^k_{\text{loc}}$ for every k , and Sobolev embedding gives smoothness. This is the only one of the three that produces the regularity without invoking Lemma A.4, at the cost of being restricted to the axial class.

Remark A.8 (The fourth route, and the one thing it alone supplies). A fourth argument appears in the sources: derive $|g| = O(\rho^{-2})$ from the harmonic mean value property and Cauchy–Schwarz, apply the $N = 2$ case of the growth filtration, kill the surviving Laurent coefficient by the fact that a nonzero Laurent mode cannot have finite local L^2 energy, and only then invoke the energy law. It is correct, but it needs the whole $8N$ classification and is restricted to axial

fields, so it is not the primary proof; see Remark 8.16. Its one irreplaceable ingredient is the separate quantification of why the *meromorphic* part, as opposed to the logarithmic part, is excluded by finite energy, and that is kept in Part VIII next to the higher-order energy scale $16\pi^2 k \|c_k\|_{\mathbb{H}_\mathbb{C}}^2 \varepsilon^{-2k}$ of Proposition 9.15.

B Branch-free and polar formulas

The formulas an implementation should use are collected here. Throughout, $a, b \in \mathbb{H}$, $d = ua + b$, $\xi = x - u = \rho \cos \varphi$, $\eta = y - v = \rho \sin \varphi$, $\rho = |z - p|$, $\ell = \log \rho$, and $y = v + \eta$. The meridian angle is written φ , not θ , because θ_g denotes the second one-form.

B.1 The Cartesian branch-free form, and what it hides

The single-valued components of $G_{p;a,b}$ are (5.2)–(5.3),

$$P = \frac{1}{\pi y} \left[\frac{(xa + b)\xi + ya\eta}{\rho^2} + a \log \rho \right], \quad Q = \frac{ya\xi - (xa + b)\eta}{\pi y \rho^2} + \frac{(xa + b) \log \rho}{\pi y^2},$$

in which the multivalued real part of the logarithm has cancelled: only $\log \rho = \operatorname{Re}_\mathbb{C} \log(z - p)$ survives, because a branch change contributes $n(za + b) \in \ker \Delta_4$. Every one of the nine research manuscripts merged here derived this pair independently, in four mutually incompatible offset conventions; after translation into (ξ, η) they are literally the same function, and that agreement is one of the load-bearing facts of the merge.

What the Cartesian form hides is numerical: near S_p the bracket subtracts nearly equal quantities, and the critical pole is not separated from the logarithmic remainder. Both defects are repaired by passing to polar coordinates.

B.2 The polar form

Substituting $\xi = \rho \cos \varphi$, $\eta = \rho \sin \varphi$, and replacing the smooth coefficients $x, y, 1/y$ by their values at the centre inside the $O(\rho^{-1})$ terms, gives

$$P = \frac{1}{\pi y} \left[\frac{d \cos \varphi + v a \sin \varphi}{\rho} + a(1 + \ell) \right], \quad (\text{B.1})$$

$$Q = \frac{1}{\pi y} \left[\frac{v a \cos \varphi - d \sin \varphi}{\rho} + \frac{(d + \rho a \cos \varphi) \ell}{y} \right], \quad (\text{B.2})$$

algebraically identical to (5.2)–(5.3). The first bracketed term in each is the exact critical pole and the remainder is $O(1 + |\log \rho|)$, so the two scales never have to be recovered by cancellation. Letting $\rho \downarrow 0$ in (B.1)–(B.2) reproduces the uniform leading term (8.16) of Part VII, and taking the squared pair norm reproduces $\mathcal{R}_p(a, b)^2 / (\pi^2 v^2 \rho^2)$, which is the input to both leading-size laws and to the energy law. In practice this form is what makes the shell-energy and supremum tables of §12.3 stable down to $\rho = 10^{-5}$ and below; the Cartesian form loses several digits there.

B.3 The potentials as a second check of both normalizations

The potentials of the same mode are (5.4),

$$C = \frac{1}{2\pi} \left(a\varphi - \frac{(xa + b)\ell}{y} \right), \quad V = \frac{1}{2\pi} \left(b\varphi + \left(ya + \frac{x(xa + b)}{y} \right) \ell \right).$$

They deserve a place in an implementation appendix for a reason independent of their role in Part IV. Since $\ell = \log \rho$ is single valued and a positive circuit about p increases φ by 2π , these two formulas exhibit directly that C gains exactly a and V gains exactly b . Because $dC = \omega_g$

and $dV = \theta_g$, this is a *second and independent* verification of both period normalizations — one that never integrates either form numerically, and that fixes the sign conventions (slope period from ω_g , constant period from θ_g) without reference to any contour computation. A sign or factor error anywhere in the chain $\omega_g \rightarrow C \rightarrow a$ or $\theta_g \rightarrow V \rightarrow b$ shows up here as a mismatch of 2π or of sign, and is caught before any quadrature is run. The formulas also display, in one line, the mechanism of §11.1: for $a = 0$ the multivalued term in C disappears while the one in V does not.

C An audit of signs, dimensions and normalizations

Every constant in this document is traceable to the following seven conventions. A document assembled from sources using different ones would silently change the model fields, the residues and the energy coefficient.

- (A1) *The operator.* $\mathcal{D} = \partial_{x_0} + \mathbf{i}\partial_{x_1} + \mathbf{j}\partial_{x_2} + \mathbf{k}\partial_{x_3}$, with *no* factor $\frac{1}{2}$, and $\Delta_4 = \overline{\mathcal{D}}\mathcal{D} = \sum_{\mu=0}^3 \partial_{x_\mu}^2$ is the ordinary *positive* four-dimensional Laplacian. Diagnostic: $\Delta_4(q^2) = -4$, $\Delta_4(q^3) = -4(3x_0 + \text{Im } q)$, and $\Delta_4(q^{-1}) = -4\bar{q}/|q|^4 = -8\pi^2 E(q)$ with $E(q) = \bar{q}/(2\pi^2|q|^4)$. Using $-\Delta_4$ instead reverses the signs of the model fields and of both residues, but leaves the squared energy coefficient unchanged.
- (A2) *Sides.* All quaternionic coefficients multiply on the *right*, in $\text{Aff}_{\mathbb{H}}$, in the period pair, in the mode $G_{p;a,b}$, in the section S , and in every series. A right coefficient cannot be moved to the left without changing the problem.
- (A3) *The central unit.* ι is central in $\mathbb{H}_{\mathbb{C}} = \mathbb{H} \oplus \iota\mathbb{H}$, with $\kappa(A + \iota B) = A - \iota B$ and $|A + \iota B|_{\mathbb{H}_{\mathbb{C}}}^2 = |A|^2 + |B|^2$. It is never an element of $\mathbb{S}$, and it is never identified with the variable $I \in \mathbb{S}$.
- (A4) *Which form gives which period.* The slope period is $a_\gamma = \int_\gamma \omega_g$ with $\omega_g = \frac{1}{2}(-Pdx + Qdy)$; the constant period is $b_\gamma = \int_\gamma \theta_g$ with $\theta_g = \frac{1}{2}((xP + yQ)dx + (yP - xQ)dy)$. All loops are counterclockwise-positive. The logarithmic normalization is $1/(2\pi\iota)$.
- (A5) *Transverse geometry.* The normal circle $|z - p| = \rho$ has length $2\pi\rho$; $\mathbb{S}$ has area 4π ; the sphere S_p has area $4\pi v^2$; the tube boundary and volume elements are $dS = y^2\rho d\varphi d\sigma(I)$ and $dV = y^2\rho d\rho d\varphi d\sigma(I)$, so $dV = y^2 dx dy d\sigma(I)$ in planar coordinates.
- (A6) *The derived constants.* Everything else follows: $2/v$ in the source density $\sigma_p(s) = \frac{2}{v}(sa + b)$ (from (A.2) and $dS = v^2 d\sigma$); $8\pi v$ and $8\pi v^3$ in the two fluxes $\Phi_0 = 8\pi v(ua + b)$ and $\Phi_1 = -8\pi v^3 a$ (from $\int_{\mathbb{S}} d\sigma = 4\pi$ and the meridian integral 2π); the 8 in $\rho J_g(\rho) \rightarrow 8\mathcal{R}_p(a, b)^2$, which is $(2\pi)(4\pi)/\pi^2$; and the $16\pi^2 k$ in the higher-order energy scale.
- (A7) *Energy.* $\mathcal{E}_g(\varepsilon, R) = \int_{\varepsilon < \rho < R} |g|^2 dV$ is the squared L^2 norm, not a Dirichlet integral; $J_g(\rho) = \int_{\partial T_\rho} |g|^2 dS$ is the shell density, so that $\mathcal{E}_g(\varepsilon, R) = \int_\varepsilon^R J_g(s) ds$ and $E^{\text{ren}} = \int_0^R (J_g(s) - 8\mathcal{R}_p^2/s) ds$.

Remark C.1 (Two geometries, not one). The factor $2\pi^2$ of the point Cauchy–Fueter kernel E belongs to the three-sphere $|q| = r$ of area $2\pi^2 r^3$ and to the isolated-point geometry of Part I. It does not appear in any spherical constant above, whose geometry is codimension two: a normal circle times a two-sphere. The removability exponents follow the same split — $o(\rho^{-1})$ at a nonreal sphere, $o(r^{-3})$ at an isolated point — and mixing the two produces an exponent that is correct in neither.

D An implementation-oriented reconstruction procedure

On a finitely connected base the theory gives a direct procedure that solves no new boundary value problem. Given sampled or symbolic axial data:

1. *Recover P, Q .* From g on the circularization, $P = \frac{1}{4\pi} \int_{\mathbb{S}} g d\sigma$ and $Q = -\frac{1}{4\pi} \int_{\mathbb{S}} I g d\sigma$. These averages are exact and inexpensive; do not sample $\mathbb{S}$ where an algebraic identity is available.
2. *Check the equations.* Verify the Vekua residuals $P_x - Q_y - 2Q/y$ and $Q_x + P_y$. Arbitrary noisy pairs (P, Q) need not even define closed forms, and every statement below presupposes that they do.
3. *Form the two one-forms directly.* Assemble the coefficients of ω_g and θ_g from P and Q alone. Do *not* construct a first primitive first: the second obstruction is computable from the target, and the point of the elimination identity $\eta_C - d(xC) = \theta_g$ is that the two tests are simultaneous rather than sequential.
4. *Integrate around one loop per conjugate pair.* On a symmetric base, integrate only around the upper representative of each conjugation-exchanged hole pair; conjugation-fixed holes contribute nothing and need not be visited.
5. *Subtract the modes.* Form $Sg = \sum_j G_{p_j; a_j, b_j}$ using the branch-free formulas (5.2)–(5.3), or their polar version near a sphere, and replace g by $\Pi g = g - S\text{Per}(g)$. Moving a chosen point p_j inside its own hole changes the representative by a zero-period field and does not change Πg 's membership in the range.
6. *Integrate the now-exact forms along any convenient paths.* Compute C and V by path integration from a base point z_* and set $f(x + Iy) = (x + Iy)C + V$. Normalize by $C(z_*) = V(z_*) = 0$, equivalently $F(z_*) = 0$ in $\mathbb{H}_{\mathbb{C}}$, which makes the primitive vanish on the entire base sphere $x_* + y_*\mathbb{S}$ — eight real conditions, not the four conditions $f(x_* + I_*y_*) = 0$ at one quaternionic point.
7. *Use the quantitative bounds as diagnostics.* Before the subtraction, a nonzero computed period is a *certificate of impossibility*, not a failure of the path integrator, provided it exceeds the error budget: the bounds $|a_\gamma| \leq \frac{L_\gamma}{2} \sup_\gamma \|g\|_{\text{ax}}$ and $|b_\gamma| \leq \frac{L_\gamma R_\gamma}{2} \sup_\gamma \|g\|_{\text{ax}}$ convert data error into period error, and the lower bound $\sup_\gamma \|g - \Delta_4 f\|_{\text{ax}} \geq \max\{2|a_\gamma|/L_\gamma, 2|b_\gamma|/(L_\gamma R_\gamma)\}$, valid for *every* global slice-regular f , converts a period into a guaranteed error floor. After the subtraction, the inverse estimate $|f(q)| \leq LM(|q| + R)/2$ bounds the reconstruction.

A numerically small period is evidence only within specified integration and data error bounds; the theorem requires exact vanishing. Conversely, a period that is large compared with the bound in step 7 cannot be an artifact of the quadrature.

E Theorem audit

Statement	Essential hypothesis	What is <i>not</i> asserted
Global period criterion (Part II)	Smooth axial Vekua data on the circularization of a connected planar U ; no topology, no boundary regularity	Inversion of arbitrary nonaxial Cauchy–Fueter-regular fields

Statement		Essential hypothesis	What is <i>not</i> asserted
Affine (Part II)	monodromy	None beyond the above	A multiplicative or path-ordered holonomy; the group is translation by the fixed space $\text{Aff}_{\mathbb{H}}$
Range and kernel of $\mathcal{H}_g$ (Part III)		Axial monogenic g on a connected U	Injectivity of $\mathcal{H}_g$: its kernel is eight-dimensional on a product domain
Reality constraint $h_{-1} = (\iota/v) \text{Re}_{\mathbb{C}} h_{-2}$ (Part III)		An isolated <i>nonreal</i> puncture p , $v > 0$	Any analogue at a real point; the constraint uses $v > 0$
Cokernel $8m$, product domain (Part IV)		An explicit finite winding basis of size m	An $8m$ -dimensional function space; a finite cokernel for infinitely many holes
Cokernel $8m$, symmetric domain (Part V)		D connected, conjugation invariant, meeting $\mathbb{R}$; m exchanged hole pairs	A contribution from conjugation-fixed holes; the answer is not $8b_1(D) = 8(s + 2m)$
Closed range and stability (Part VI)		Connected base; compact-open topology	Boundedness in any weighted, Hardy, Bergman or Sobolev norm; orthogonality of the splitting
Period-corrected Runge (Part VI)		Stated pole-set hypothesis: one-sided on a product domain, conjugation-symmetric on a slice domain	That the approximants are rational with $\mathbb{H}$ coefficients in the <i>one-sided</i> case
Normal form, no hypothesis (Part VII)		Axial monogenic on a punctured tube	Finite order; the principal part may be an infinite series (Example 11.4)
L^1 (Part VII)	classification	Local integrability across S_p — formally weaker than $M_g^{\text{sph}} = O(\rho^{-1})$, which implies it, though the two classes coincide a posteriori	Completeness of (a, b) below L^1 ; see §11.3
Growth filtration (Part VII)	$8N$	$M_g^{\text{sph}}(\rho) = O(\rho^{-N})$	$4N$ or $8N + 8$; and no fractional orders occur
Sharp (Part VII)	removability	$M_g^{\text{sph}}(\rho) = o(\rho^{-1})$	That big- O suffices; $G_{p;a,b}$ is the counterexample
L^2 (Part VII)	removability	Finite local L^2 mass across the sphere; <i>no</i> axiality, <i>no</i> growth bound	A new general elliptic removability principle; the method is standard
Exact (Part VII)	leading size	Critical order, residue pair (a, b)	One constant: M^{ax} and M^{sph} have <i>different</i> limits, and the correction vanishes iff $\text{Im}_{\mathbb{H}}(b\bar{a}) = 0$, not iff $ab = ba$
Surface source and layer (Part VIII)		Locally integrable axial field; uniqueness step needs no axiality	A classification of arbitrary surface densities; only affine ones occur here
Two conserved fluxes (Part VIII)		Right-Fueter-regular weights 1 and $H_u = 3(\text{Re } q - u) + \text{Im } q$	That Φ_0 alone classifies, or that the weight $q - u$ works
Energy law (Part VIII)		Critical growth, fixed outer radius R	Uniformity as $v \downarrow 0$; E^{ren} depends on R and on the regular part
Growth-free lower bound (Part VIII)		Axial regularity off the sphere; <i>no</i> growth hypothesis	An optimal finite-radius correction term
Realization (Part IX)	theorem	Simply connected upper base; discrete locally finite family of spheres	Any compatibility condition between spheres — there is none

Statement			Essential hypothesis	What is <i>not</i> asserted
Global (Part IX)	normal	form	Prescribed transverse orders m_j and growth $O(q ^d)$	A classification including real punctures, whose principal parts are excluded

All statements listed are proved in the corresponding parts. The computational supplement of §12 checks identities, normalizations and examples only; it is not a proof-assistant certificate, and the qualified novelty assessment of §13.2 applies to the document as a whole, including every result labelled a contribution.

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This list is the union of the reference lists of the ten source manuscripts: 81 bibliography entries naming 26 distinct works. It is reproduced in full so that the merged document does not lose the literature those manuscripts consulted, and the comment on each entry records which of the ten cited it. Three works are cited by all ten. Only some are cited in the text above; the others were consulted by one or more of the sources for background, or for the priority audits reported in Part XI.

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